46

Primary and Secondary Hypertension

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CHAPTER OUTLINE

Hypertension has consistently been one of the major contributors to premature morbidity and mortality in the United States.[1] Hypertension was ranked first worldwide in an analysis of all risk factors for global disease burden in 2010[2] (Fig. 46.1) and again in 2015. By the year 2025, hypertension is expected to increase in prevalence worldwide by 60% and will affect 1.56 billion people.[3] Developing nations will experience an 80% increase (from 639 million to 1.15 billion afflicted persons). As emerging countries have improved sanitation and other basic public health measures, cardiovascular (CV) disease has or soon will become the most common cause of death, and hypertension will be its most common reversible risk factor, as it is already in the United States.

The major public health importance of hypertension can be amply demonstrated using data from the United States, but similar conclusions have been emerging as other industrialized countries analyze national health care databases.[4] Hypertension was listed as the principal or a contributing cause of death in 16.4% of Americans older than 45 years who died in 2013, a 48% increase since 2000.[1]

Hypertension is the most important modifiable risk factor for stroke,[1] which fell from the second most common cause of death in the United States in 1958, to third between 1959 and 2007, to fourth between 2008 and 2012, and to fifth in 2013 and 2014.[5] Current estimates are that 77% of those who have a first stroke have had a blood pressure (BP) above 140/90 mm Hg.

High BP is the leading antecedent condition for heart failure—regardless of whether left ventricular function is diminished or preserved—which is the most common reason for acute care hospitalization among Medicare beneficiaries (~1.023 million in 2010); approximately 74% of people experiencing an initial hospitalization for heart failure either had or have BP of 140/90 mm Hg or higher.[1]

Currently, end-stage kidney disease (ESKD) has the highest per-patient annualized cost to Medicare, amounting to about 20% of Medicare expenditures for Americans older than 65 years,[6] and hypertension is the second most common cause of ESKD. After tobacco use and diabetes, hypertension is the most important risk factor for peripheral vascular disease, the second leading cause of loss of limbs in the United States.[1] Hypertension is likely the most important treatable cause of (vascular) dementia, which ranked sixth among causes of death in the United States in 2014 and is the third leading cause of admission to nursing homes. Hypertension ranks first among the chronic conditions for which Americans visit a health care provider. This may be secondary to its high age-adjusted prevalence (33.5% of adults >18 years in the US National Health and Nutrition Examination Survey [NHANES], 2013–2014) and the fact that treatment clearly improves prognosis.

All health care providers routinely encounter people who are likely to benefit from lowered BP levels. In the future, more people will likely become candidates for antihypertensive therapy because the prevalence of hypertension is increasing secondary to the increasing prevalence of obesity and increased longevity of the general population.[7]

HYPERTENSION DEFINITIONS

Traditionally, high blood pressure has been defined as a persistent BP elevation in the office at or above 140/90 mm Hg.[8–11] Blood pressure is the phenotypic expression of the genetically predisposed disease hypertension. Blood pressure is a continuous variable. The threshold BP value to secure a diagnosis of hypertension has come from large epidemiologic studies demonstrating a higher mortality at levels above

1990

Mean rank (95% UI)	Risk factor
1.1 (1–2)	1 Childhood underweight
2.1 (1–4)	2 Household air pollution
2.9 (2–4)	3 Smoking (excluding SHS)
4.0 (3–5)	4 High blood pressure
5.4 (3–8)	5 Suboptimal breastfeeding
5.6 (5–6)	6 Alcohol use
7.4 (6–8)	7 Ambient PM pollution
7.4 (6–8)	8 Low fruit
9.7 (9–12)	9 High fasting plasma glucose
10.9 (9–14)	10 High body mass index
11.1 (9–15)	11 Iron deficiency
12.3 (9–17)	12 High sodium
13.9 (10–19)	13 Low nuts and seeds
14.1 (11–17)	14 High total cholesterol
16.2 (9–38)	15 Sanitation
16.7 (13–21)	16 Low vegetables
17.1 (10–23)	17 Vitamin A deficiency
17.3 (15–20)	18 Low whole grains
20.0 (13–29)	19 Zinc deficiency
20.6 (17–25)	20 Low omega-3
20.8 (18–24)	21 Occupational injury
21.7 (14–34)	22 Unimproved water
22.6 (19–26)	23 Occupational low back pain
23.2 (19–29)	24 High processed meat
24.2 (21–26)	25 Drug use
	26 Low fiber
	30 Lead

2010

Risk factor	Mean rank (95% UI)	% change (95% UI)
1 High blood pressure	1.1 (1–2)	27% (19 to 34)
2 Smoking (excluding SHS)	1.9 (1–2)	3% (−5 to 11)
3 Alcohol use	3.0 (2–4)	28% (17 to 39)
4 Household air pollution	4.7 (3–7)	−37% (−44 to −29)
5 Low fruit	5.0 (4–8)	29% (25 to 34)
6 High body mass index	6.1 (4–8)	82% (71 to 95)
7 High fasting plasma glucose	6.6 (5–8)	58% (43 to 73)
8 Childhood underweight	8.5 (6–11)	−61% (−66 to −55)
9 Ambient PM pollution	8.9 (7–11)	−7% (−13 to −1)
10 Physical inactivity	9.9 (8–12)	0% (0 to 0)
11 High sodium	11.2 (8–15)	33% (27 to 39)
12 Low nuts and seeds	12.9 (11–17)	27% (18 to 32)
13 Iron deficiency	13.5 (11–17)	−7% (−11 to −4)
14 Suboptimal breastfeeding	13.8 (10–18)	−57% (−63 to −51)
15 High total cholesterol	15.2 (12–17)	3% (−13 to 19)
16 Low whole grains	15.3 (13–17)	39% (32 to 45)
17 Low vegetables	15.8 (12–19)	22% (16 to 28)
18 Low omega-3	18.7 (17–23)	30% (21 to 35)
19 Drug use	20.2 (18–23)	57% (42 to 72)
20 Occupational injury	20.4 (18–23)	12% (−22 to 58)
21 Occupational low back pain	21.2(18–25)	22% (11 to 35)
22 High processed meat	22.0 (17–31)	22% (2 to 44)
23 Intimate partner violence	23.8 (20–28)	0% (0 to 0)
24 Low fiber	24.4 (19–32)	23% (13 to 33)
25 Lead	25.5 (25–29)	160% (143 to 176)
26 Sanitation		
29 Vitamin A deficiency		
31 Zinc deficiency		
33 Unimproved water		

——— Ascending order in rank
- - - - Descending order in rank

Fig. 46.1 Global risk factor ranks with 95% uncertainty interval for all ages and genders combined in 1990 and 2010 and percentage change. *PM,* Particulate matter; *SHS,* secondhand smoke. (From Lim SS, Vos T, Flaxman AD, et al: A comparative risk assessment of burden of disease and injury attributable to 67 risk factors and risk factor clusters in 21 regions, 1990-2010: a systematic analysis for the Global Burden of Disease Study 2010. *Lancet*. 2012;380:2224–2260.)

Table 46.1 Categorization of Blood Pressure for Hypertension and Its Related Diagnoses in the United States[a]

BP Category[b]	SBP		DBP
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120–129 mm Hg	and	<80 mm Hg
Hypertension			
• Stage 1	130–139 mm Hg	or	80–89 mm Hg
• Stage 2	140–159 mm Hg	or	90–99 mm Hg

[a]If the systolic and diastolic blood pressure levels fall into two different diagnostic categories, the higher category is used (e.g., 162/92 mm Hg is stage 3 hypertension; 134/72 mm Hg is stage 1).

[b]Definitions are from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324. Different classification schemes are used outside the United States.

BP, Blood pressure; *DBP*, diastolic blood pressure; *SBP*, systolic blood pressure.

Box 46.1 Essential Elements of Proper Blood Pressure Measurement in the Office

Allow patient to rest in the seated position for at least 3 to 5 minutes prior to measuring blood pressure (BP).

Neither patient nor examiner should talk during measurements.

The patient should have his or her legs uncrossed and should be seated comfortably on a chair with arm and back support.

Use the arm as the preferred site of measurement.

Make sure the measuring device is adequately maintained and calibrated.

Use a cuff that fits the arm circumference properly. The bladder length should cover at least 80% of the arm circumference. Recommended cuff sizes for adults are as follows:

- Small adult (12 × 22 cm): for arm circumferences between 22 and 26 cm
- Adult (16 × 30 cm): for arm circumferences between 27 and 34 cm
- Large adult (16 × 36 cm): for arm circumferences between 35 and 44 cm
- Adult thigh (16 × 42 cm): for arm circumferences between 45 and 52 cm

Place the lower end of the cuff approximately 2 to 3 cm above the antecubital fossa.

Have the arm positioned at the level of the heart.

Take at least two BP measurements and average them. Obtain more measurements if there is disparity between the first two values.

Measure BP in both arms to identify interarm differences (~20% of individuals may have a difference >10 mm Hg). If different, report the values obtained on the arm with higher BP.

If using the auscultatory method with a stethoscope, use Korotkoff phase I (appearance) and V (disappearance) to define systolic and diastolic BP, respectively.

If using an aneroid or mercury manometer, use a deflation rate of 2–3 mm Hg/sec. (Deflation rates with automated oscillometric devices vary substantially and are defined based on proprietary algorithms.)

Data from Whelton PK, Carey RM, Aronow WS, et al. ACC/AHA/AAPA/ABC/ACPM/ AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association and Task Force on Clinical Practice Guidelines. *Hypertension.* 2018;71(6):e13–e115.

140/90 mm Hg.[12] In the United States, national guidelines promulgated by the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC 7) in 2003 simplified the classification of hypertension and related conditions.[8] This classification has been endorsed and expanded by more recent guidelines (Table 46.1).[13] The five categories of BP—normal, elevated, stages 1, 2, and 3 hypertension—are associated with progressively increasing CV risk and are independent of any other risk factor (including age; see Table 46.1).[13] The most recent guidelines have defined the level of raised blood pressure that should be considered for beginning treatment based not only on BP level, but also on a more than 10% cardiovascular risk over the ensuing 10 years. Thus, in people with chronic kidney disease (CKD), all people with a BP more than 130/80 mm Hg not only need lifestyle intervention but treatment with at least one antihypertensive medication.[13]

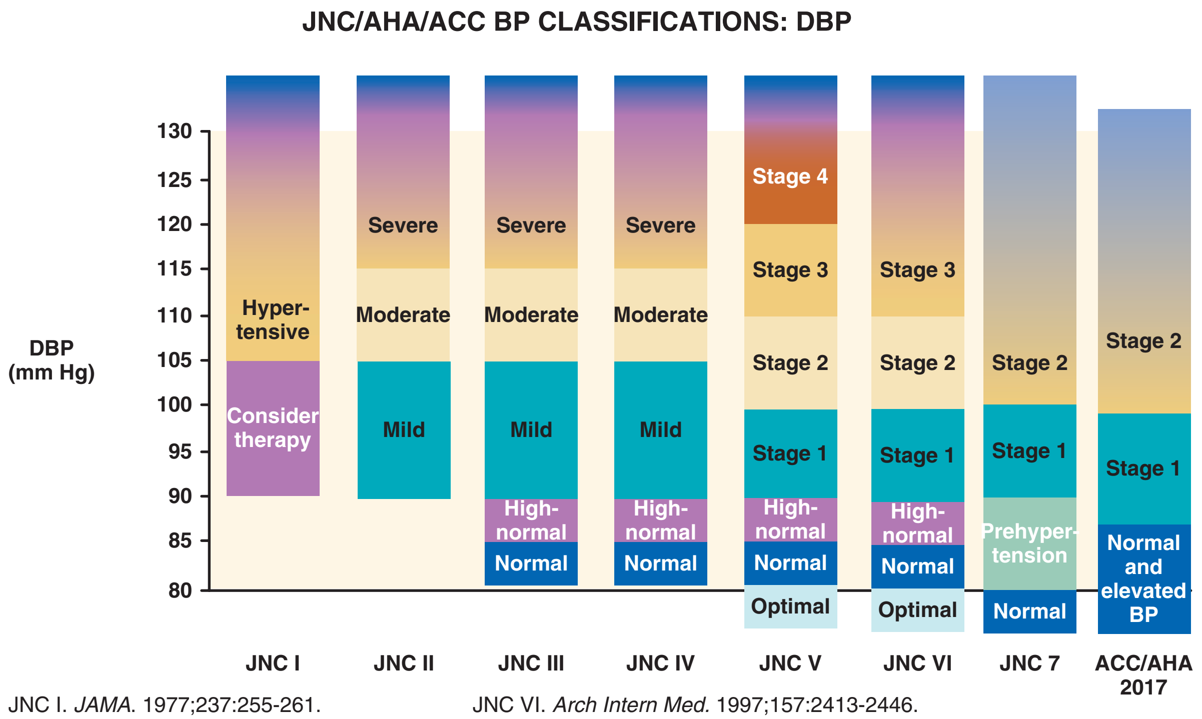
Office BP readings often display less accuracy and reproducibility than is desired, particularly if great care is not taken during the procedure (Box 46.1)[8,10] As a result, there is growing interest in other methods of measuring BP that either remove human error from the process or involve more expensive equipment (see later).

EVOLUTION OF BLOOD PRESSURE GOALS

As more data became available, the guideline-based definition of hypertension has evolved over the last 40 years (Fig. 46.2). The change of the diastolic threshold from 95 to 90 mm Hg occurred concomitantly with the recommendation to use Korotkoff phase V (disappearance of sound), rather than phase IV (muffling of sounds) in diagnosis. The inclusion of systolic BP in the definition of hypertension has been much more gradual (see Fig. 46.2).

In JNC 1 and 2 (1977–1982), the diagnosis of hypertension was made solely on diastolic BP values. In JNC 3 and 4, "isolated systolic hypertension" (systolic BP ≥160 mm Hg, but diastolic BP <90 mm Hg) was defined, but no recommendations about therapy were made (see Fig. 46.2).

Since 1993, many guidelines have recognized isolated systolic hypertension as worthy of treatment.[7,11] BP-lowering therapy in this subgroup of older adults has morbidity and mortality benefits.[14,15] A pooling of data from nearly 1 million people from 61 long-term epidemiologic studies has concluded that systolic BP predicts approximately 89% of age-stratified stroke deaths and 93% of coronary heart disease deaths, whereas diastolic BP is much less predictive (83% and 73%, respectively).[9]



JNC I. *JAMA.* 1977;237:255-261.
JNC II. *Arch Intern Med.* 1980;140:1280-1285.
JNC III. *Arch Intern Med.* 1984;144:1047-1057.
JNC IV. *Arch Intern Med.* 1988;148:1023-1038.
JNC V. *Arch Intern Med.* 1993;153:154-183.
JNC VI. *Arch Intern Med.* 1997;157:2413-2446.
JNC 7. *JAMA.* 2003;289:2560-2572.
Expert panel report-JAMA 2014;311:507-520.
ACC/AHA BP guidelines hypertension 2018.

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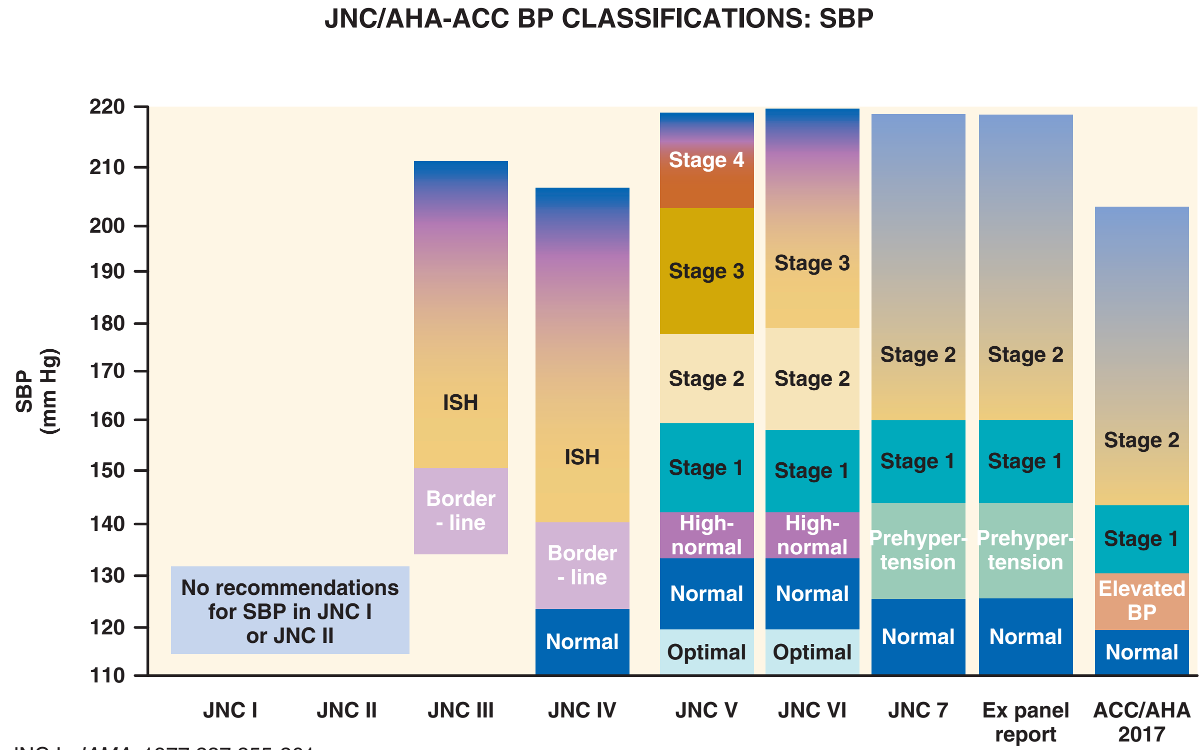

JNC I. *JAMA.* 1977;237:255-261.
JNC II. *Arch Intern Med.* 1980;140:1280-1285.
JNC III. *Arch Intern Med.* 1984;144:1047-1057.
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JNC 7. *JAMA.* 2003;289:2560-2572.
Expert panel report-JAMA 2014;311:507-520.
ACC/AHA BP guidelines hypertension 2018.

B

Fig. 46.2 (A) Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC)/American Heart Association (AHA)/American College of Cardiology (ACC) DBP classifications. (B) JNC/AHA/ACC SBP classifications. *DBP,* Diastolic blood pressure; *ISH,* isolated systolic hypertension; *SBP,* systolic blood pressure.

In addition to a large body of epidemiologic data linking elevated systolic BP with cardiovascular events,[2–4,9,16] 11 clinical trials have compared active antihypertensive drugs to either placebo or no treatment in 28,436 patients. These trials demonstrated morbidity and mortality benefits of treatment tied to reductions in systolic but not diastolic BP.[14]

EPIDEMIOLOGY

Hypertension is widely treated because of its increased risk for long-term CV and renal morbidity and mortality and the fact that antihypertensive treatment prevents some of these events. The risks attributable to elevated BP levels have been documented in numerous epidemiologic studies, beginning in 1948 with the Framingham Heart Study and extending to the present.[2–4,9,15,16] Meta-analyses of pooled data have confirmed the robust continuous relationship between BP level and cerebrovascular disease and coronary heart disease in both Western and Eastern populations.[2–4,9,15–17] In addition, BP has been linked directly in epidemiologic studies to incident left ventricular hypertrophy (LVH), heart failure, peripheral vascular disease, carotid atherosclerosis, ESKD, and subclinical CV disease. A natural history study that involved almost 12,000 veterans, followed over 15 years, has noted that the level of BP correlates with the risk for ESKD (Fig. 46.3).[18] Note that in this study, the highest risk for ESKD was found at levels above the renal autoregulatory range (i.e., a systolic BP >180 mm Hg).

CV risk factors tend to cluster; thus, hypertensive individuals are much more likely than normotensive people to have type 2 diabetes mellitus or dyslipidemia, especially elevated triglyceride levels and low high-density lipoprotein cholesterol levels. The common denominator may be insulin resistance, perhaps because of the frequent coexistence of hypertension and obesity.

AGE AND HYPERTENSION

Increasing age is a major risk factor for developing hypertension (Fig. 46.4), as well as a very strong confounder of its independent influence on CV and renal events. In the analysis

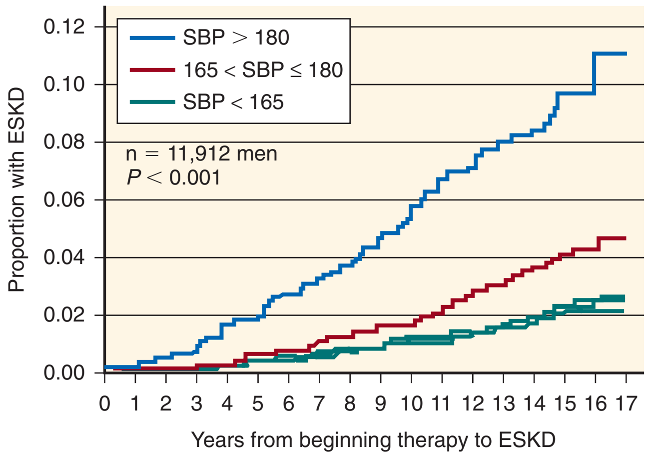


Fig. 46.3 A 17-year follow-up from Veterans Affairs hypertension clinics on end-stage kidney disease *(ESKD)*. *SBP,* Systolic blood pressure (in mm Hg).

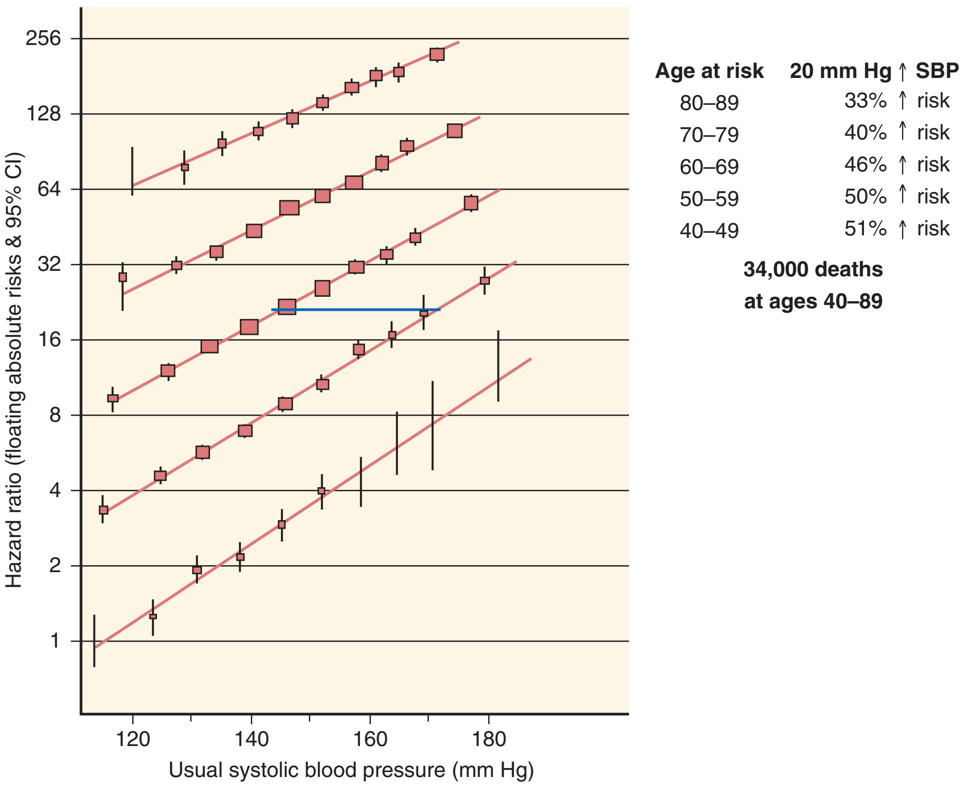


Fig. 46.4 Coronary heart disease mortality in each decade of age versus systolic blood pressure (SBP) at the start of that decade. *CI,* Confidence interval.

of nearly 1 million individuals in 61 epidemiologic studies followed for an average of 13.3 years, those with BP levels in the highest decile had roughly the same risk for death from ischemic heart disease or stroke as people who were 20 years older but had BP levels in the lowest decile.[9] In the Framingham study, the lifetime risk of 55- to 65-year-old men or women for developing hypertension was above 90%.[19] In this study, of those who survived to age 65 to 89 years, systolic BP elevations were found in 87% of the hypertensive men and 93% of the hypertensive women. In an analysis of the Framingham data set, classification of people with hypertension who were older than 60 years into the appropriate BP stages was done correctly in 99% of the cases using systolic rather than diastolic BP.[12]

These data highlight the great public health importance of systolic BP, particularly among people older than 50 years. In such individuals, systolic BP is a much better predictor of hypertensive target organ damage and future CV and renal events than diastolic BP.[9,20,21] Overall, each 20-mm Hg increase in systolic BP doubled the risk for CV death.[9] Systolic BP was less likely to be controlled to less than 140 mm Hg than diastolic to less than 90 mm Hg in the general US population, according to every NHANES data set from 1974 to 2014. Nearly 54% of people with uncontrolled hypertension in the United States in 2013 to 2014 were 60 years of age and older.[22] Yet, antihypertensive drug therapy reduces the risk for CV events across the full age spectrum and has its greatest absolute benefit in older people, including individuals older than 80 years.[23–25]

The diagnosis of hypertension in children and adolescents is becoming more important due to the epidemic of obesity in young Americans.[26] Current US guidelines recommend BP measurement in children at least annually, but normative values depend on gender, age, and height of the child.[27] As a result, interpretation of BP levels in children and adolescents usually involves comparison of a child's average BP (from three visits) to a comprehensive table that provides threshold values for elevated (traditionally, BP between the 90th and 95th percentiles), hypertension (BP between the 95th and 99th percentiles), and severe hypertension (99th percentile or higher).

GENDER AND HYPERTENSION

Hypertension is a major problem for both men and women, but men tend to develop it at an earlier age (Fig. 46.5), which is also true of the adverse clinical consequences of hypertension. Among individuals older than 70 years, women are more likely to have hypertension[1,22] and to have a CV event compared with men. Age and body mass index have been much stronger predictors of incident hypertension than gender in epidemiologic studies. Drug treatment of hypertension has roughly the same benefits for women and men.[28,29]

RACE AND ETHNICITY AND HYPERTENSION

Similar to previous surveys, NHANES 2011–2014 concluded that non-Hispanic blacks had approximately a 50% higher prevalence of hypertension than non-Hispanic whites, even after age adjustment (41.2% vs. 28.0%).[22] The prevalence of hypertension in either non-Hispanic Asians or Hispanics was slightly but not significantly lower (at 24.9% and 25.9%, respectively) than in non-Hispanic whites. The prevalence of hypertension is geographically heterogeneous, with the highest prevalence in both blacks and whites in the southeastern United States (the so-called stroke belt). Perhaps because of persistent public health initiatives, non-Hispanic blacks had the highest awareness (at 85.7%) and treatment (at 77.4%) of hypertension in NHANES 2011–2012, but their BP control rate in 2011–2014 lagged behind that of non-Hispanic whites (48.5% to 55.7%, age-adjusted). This pattern has been consistent over the last decade.

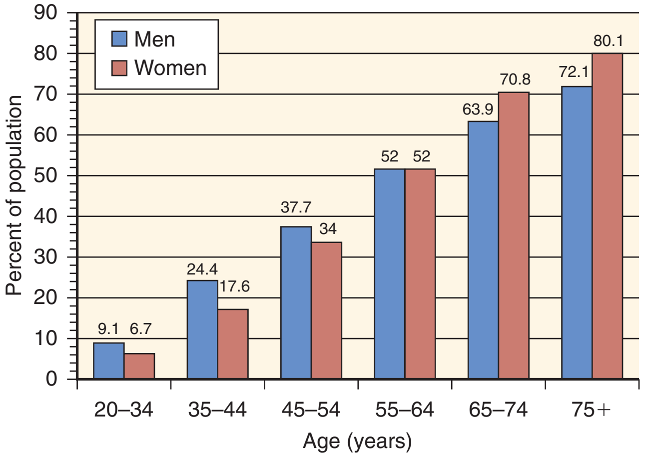


Fig. 46.5 Prevalence of high blood pressure in US adults (≥20 years of age) by age and gender, according to the National Health and Nutrition Examination Surveys 2007–2010.

Although the prevalence of hypertension has increased between NHANES 1988–1991 and 2011–2014, the increase was greatest for non-Hispanic blacks compared with Mexican Americans or non-Hispanic whites. In contrast to non-Hispanic whites and Mexican Americans, a slightly higher prevalence of hypertension was observed in NHANES 2011–2014 for non-Hispanic black women, compared with men (41.5% vs. 40.8%). In all three racial and ethnic groups, women had higher rates of awareness, treatment, and control of BP than men in NHANES 2011–2014.

Perhaps because of the persistent historical difference in BP control rates, the adverse long-term consequences of hypertension are still more common in blacks than whites, but disparities have decreased in the last decade.[1] In 2014, the age-adjusted death rate from heart disease was 24% higher in blacks, as was stroke (by 41%) and hypertension or hypertensive renal disease (by 111%).[5] Incident ESKD was 3.1 or 1.2 times more common in 2014 in blacks or Native Americans, compared with whites.[30] Although diuretics or calcium channel blockers (CCBs) may have an advantage as initial therapy for reducing CV events in blacks, an angiotensin-converting enzyme (ACE) inhibitor was better than either a CCB or a beta-blocker in preventing the decline of renal function in African-Americans with hypertensive nephrosclerosis.[1,29,31] Most current hypertension guidelines therefore recommend controlling BP with multidrug regimens in all racial and ethnic groups.[11,13,32,33]

Although the prevalence of hypertension in Hispanics is lower than in non-Hispanic blacks and whites, hypertension is a concern. Mexican Americans continue to have the lowest age-adjusted prevalence of controlled hypertension in both men and women (25.6% and 31.9%, respectively, in NHANES 1999–2006, compared with 37.0% and 49.2% in NHANES 2007–2014).[1]

BLOOD PRESSURE CONTROL RATES

Despite major progress in identifying the risks associated with elevated BP and demonstration that reducing BP to within a certain range reduces the risk for death from CV disease and stroke, as well as kidney disease progression, improved methods of achieving and sustaining BP control are needed. We have more than 125 different medications, most of which are generically available, from 8 different antihypertensive drug classes, to help lower BP, as well as more than 20 fixed-dose combination single-pill agents. In spite of this, BP control remains suboptimal in many parts of the world.[2,3,16,22,34,35]

BP control rates (to <140/90 mm Hg) have improved substantially in the United States since 1974 (Fig. 46.6) and have stabilized at just over 50% in the last four biennial NHANES reports.[1] Successful national efforts to increase hypertension treatment and control rates have been associated with significant reductions in CV hospitalizations or death in both Canada[35] and the United Kingdom.[36]

The prevalence of uncontrolled hypertension is greater for undiagnosed, untreated, or older individuals and for systolic (rather than diastolic) BP. Some health care delivery organizations have reported BP control rates in excess of 60% to 80%.[37] These improvements in BP control have been attributed to system improvements that routinely call the health care provider's attention to the uncontrolled BP at every clinical encounter, development of a registry of hypertensive patients, increasing convenience for BP measurements, and more widespread use of fixed-dose combination single-pill therapy.[37]

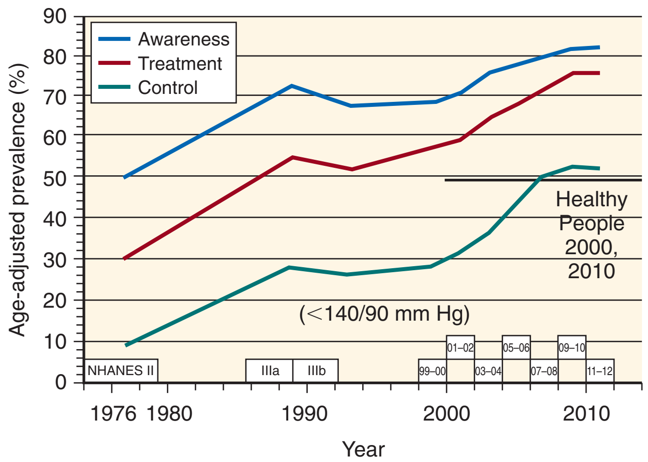


Fig. 46.6 Awareness, treatment, and control of hypertension to less than 140/90 mm Hg in US National Health and Nutrition Examination Surveys (NHANES), from 1973 to 2012. The *horizontal line* at 50%, starting at the year 2000, corresponds to the national target for hypertension control promulgated by Healthy People 2000 and 2010, which has been increased to 61.2% by Healthy People 2020. The *small numbers* (e.g., IIIa, IIIb, 99-00) in boxes just above the *x*-axis of the figure reflect the nomenclature of the NHANES data collection period.

The wisdom of controlling BP over a relatively short time course after its discovery, rather than taking months to do so, has been most clearly demonstrated in the Valsartan Antihypertensive Long-term Use Evaluation (VALUE) trial.[38] Although the randomized comparison was between high-risk patients with hypertension who received valsartan or amlodipine initially, prevention of CV events was clearly better among individuals who achieved their goal BP during the first 6 months of treatment, regardless of initial randomized therapy. Similar long-term benefits of early control of BP have been seen for stroke or CV events in the Systolic Hypertension in Europe trial[39] and for death in the Systolic Hypertension in the Elderly Program (SHEP).[40]

ECONOMICS OF HYPERTENSION

Although greater availability of generic antihypertensive medications (including every class except renin inhibitors) has greatly reduced the cost of treatment in the United States, limited financial resources and drug shortages have always been a major concern in the rest of the world. Even in the United States, restricted formularies, prior authorization processes, "step-edits" (which start with the most cost-effective drug therapy and progress to more costly therapies as necessary), and other barriers to prescribing and dispensing limit overall BP control rates. Currently, many of the single-pill combinations are priced generically, and the cost is often less than what would be paid for the individual components, purchased separately. Many single-pill combinations that include a thiazide diuretic cost no more than the nondiuretic component alone.

A proper analysis of the economics of hypertension and its treatment should include more than what is spent on drugs, patient visits, and/or laboratory tests.[41] For many high-risk patients, the expensive complications of untreated hypertension far outweigh the inconvenience and costs associated with effective treatment. For the United States in 2012–2013, hypertension was expected to cost approximately $51.2 billion (related to CV disease)[1] and roughly another $58 billion related to CKD in 2014,[30] although there is an undefined overlap (e.g., a hypertensive patient with CKD who is hospitalized for heart failure). Antihypertensive drugs cost approximately $19.7 billion or 18% of the total expenditures for hypertension. This proportion of expenditures has decreased steadily over the last decade due to wider use of generics, which recently constituted about 90% of the dispensed antihypertensive medications in the United States.

PATHOPHYSIOLOGY

The physiology that generates BP involves the integration of cardiac output (CO) and systemic vascular resistance (SVR):

$$BP = CO \times SVR$$

Each of these parameters has its own determinants:

$$CO = \text{heart rate} \times \text{stroke volume}$$
$$SVR = 80 \times (\text{mean arterial pressure} - \text{central venous pressure})/CO$$

This view is simplified, but it provides a framework to define the relevant factors in BP regulation. Changes in CO typically produce short-lived BP changes (hours to days) as adaptive mechanisms adjust SVR to normalize BP. However, changes in SVR are able to produce sustained increases in BP. The following sections summarize relevant mechanisms leading to BP regulation.

PRESSURE NATRIURESIS AND SALT SENSITIVITY

A key factor in the regulation of BP as a factor of CO and SVR is the phenomenon of pressure natriuresis. Pressure natriuresis is defined as the increase in renal sodium excretion because of mild increases in BP, typically due to extracellular fluid volume expansion, allowing BP to remain in the normal range.[42–44] This concept is essential to the understanding of the sustainability of hypertension. If one understands the set point BP as the BP at the point when extracellular volume and pressure natriuresis are in equilibrium, it necessarily follows that a change in BP can only be sustained if pressure natriureis is abnormal. Pressure natriuresis occurs over hours to days and is modulated by biophysical and humoral factors.

In the normal state, increased sodium intake causes an increase in extracellular volume and BP. Because of the steep relationship between volume and pressure, small increases in BP produce natriuresis that restores sodium balance and returns BP to normal (Fig. 46.7). Expansion of extracellular fluid volume and increased BP result in a rise in blood flow through the vasa recta, which stimulates the production of paracrine factors such as nitric oxide (NO) and adenosine triphosphate (ATP), which can inhibit tubular sodium reabsorption at multiple sites of the nephron.[45] Nitric oxide also blunts the myogenic response of arteriolar autoregulation, thus allowing increased blood flow, which is necessary to increase renal blood flow and interstitial pressure (see later).[44,45]

The ability of the kidneys to adjust to sodium loading is remarkable, adapting to fluctuations in sodium intake as high as 50-fold,[43] but this response is markedly blunted in the setting of chronic hypertension, resulting in a need for much higher BP levels to promote natriuresis. These states of abnormal sodium handling lead to sodium-sensitive hypertension, such as in conditions of a reduced glomerular filtration rate (GFR) or high levels of Ang II (see Fig. 46.7). In such situations, the change in extracellular fluid volume is relatively small (3%–5%), but a state of chronic high BP develops, resulting from increased SVR. The mechanisms responsible for this vascular effect are not completely understood but likely involve increased activity of the renin-angiotensin-aldosterone system (RAAS; high Ang II level) and several other vasoconstricting substances,[44] as will be discussed later in this section.

The pressure natriuresis process is also mediated by biophysical factors. Increased renal interstitial hydrostatic pressure is an important factor. Sodium loading results in increased pressure in the vasa recta, which have noticeably poor autoregulation, whereas pressure in cortical peritubular capillaries remains normal.[46,47] Vasa recta blood flow approximates 10% of total renal blood flow. This increase in interstitial pressure inhibits sodium transport largely by increasing 20-hydroxyeicosatetraenoic acid, a lipoxygenase product of arachidonic acid and an inhibitor of the sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase), whose inhibition causes decreased activity of the Na^+-H^+-exchanger isoform 3 (NHE3).[46] In addition, increased interstitial pressure limits proximal tubular paracellular pathways, thus maximizing natriuresis.

Because abnormalities in pressure-sodium relationships are essential to maintaining chronic elevations in BP, they represent a fundamental step in the pathogenesis of any type of hypertension—not only primary, but also in the maintenance phase of most secondary causes, such as renal and renovascular hypertension, hyperaldosteronism, glucocorticoid excess, coarctation of the aorta, and pheochromocytoma.

The interplay between renal sodium retention and hypertension involves changes in sodium handling throughout the nephron. A theory with substantial experimental support has proposed that increased renal vasoconstriction due to a variety of possible mechanisms (e.g., increased levels of angiotensin II, catecholamines, or uric acid, progressive aging) induces a preglomerular (afferent) arteriolopathy that results in impaired sodium filtration.[48,49] In addition, renal vasoconstriction results in tubular ischemia, another mediator of increased sodium avidity.

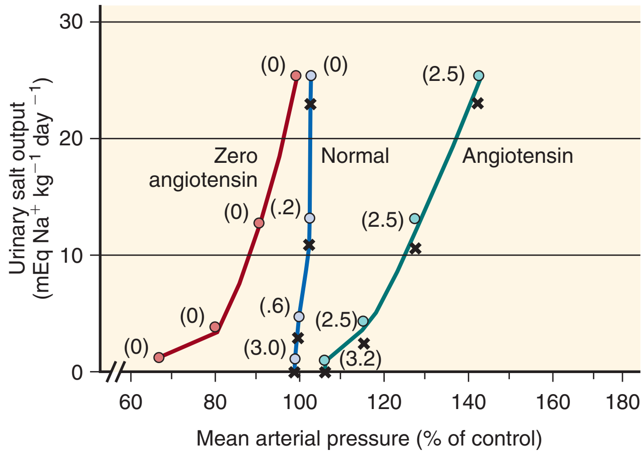


Fig. 46.7 Pressure natriuresis curves in dogs at different levels of sodium intake (reflected in urinary salt output, *y*-axis). In the normal state, massive fluctuations in sodium intake produce minimal changes in blood pressure (BP). In states of high angiotensin II (Ang II) levels (Ang II infusion in this model), BP increases significantly, even with modest increases in sodium intake. Conversely, in states where Ang II is absent or low (administration of captopril in this model), the increased sodium results in increased BP only at low BP levels; once BP becomes normal, the relationship is similar to that of normal kidneys. Values in *parentheses* are the relative estimated concentrations of Ang II (1 being the reference). (From Elliott W. Economic considerations in hypertension management. In: Izzo JL J, Black HR, eds. Hypertension Primer. Dallas: American Heart Association; 2008; 331–334.)

GENETICS OF HYPERTENSION

Hypertension clusters in families; an individual with a family history of hypertension has a fourfold greater chance of developing hypertension,[50] and it has been estimated that the heritability of hypertension ranges from 31% to 68%. Genome-wide association studies (GWAS) in several multinational cohorts have identified a large number of single-nucleotide polymorphisms (SNPs) associated with hypertension.[51] However, these individual SNPs are responsible for only minor BP effects (0.5–1 mm Hg), and the overall impact of these identified SNPs on the overall BP variance is only approximately 1% to 2%.[50] Among these genes, *FOS* (fos protooncogene) and *PTGS2* (cyclooxygenase-2 [COX-2]) have been replicated in a number of studies. There are many shortcomings of the use of GWAS and other large population approaches.[52] For example, the SNP platforms used for testing are not hypothesis-driven; they simply include common genetic variants for exploratory analyses that might provide clues for molecular pathways leading to better understanding of disease or new targets for therapy.

To advance this use in the progress toward personalized medicine in hypertension, a recent GWAS based on the large United Kingdom Biobank Study cohort also performed functional and transcript expression analyses of candidate genes in target tissues in vitro (vascular smooth muscle cells, aortic fibroblasts, and endothelial cells), with the goal of identifying potential therapeutic targets.[51] The study also developed and validated an unbiased genetic risk score that included clinical and genotype information, including data on 107 independent risk loci to generate estimates of the risk of hypertension and the risk of specific hypertension-related outcomes (e.g., stroke, coronary disease, any cardiovascular outcome). These analyses showed a gender-adjusted systolic BP difference of 9.3 mm Hg between the lowest and highest risk quintile (higher in the high-risk group, which also had 2.3-fold greater odds of hypertension and a 1.35-fold increase in the odds of any cardiovascular outcome).[51] These results indicate the potential value for genetic risk-based clinical scoring.

BP measurement is another important concern, as it is not uniform in these very large, population-based GWAS. In addition, large numbers of patients are receiving treatment at the time of testing, thus limiting the strength of any associations. Finally, hypertensive phenotypes are not well defined, so patients with very different phenotypes (e.g., isolated diastolic hypertension, isolated systolic hypertension of the young, isolated systolic hypertension of older adults) are all lumped together.[53] We now understand that each of these phenotypes likely has different underlying pathophysiologic mechanisms, and that even within each group there is substantial variability in a hemodynamic profile.[54,55]

With the improvement in techniques that allow expeditious and cheaper whole-exome or whole-genome analyses and the expansion of precision medicine, it is possible that greater mechanistic insights on the genetics of hypertension will become available. Unfortunately, compared with other clinical phenotypes, the heavy influence of lifestyle and environmental factors in hypertension makes it unlikely that the simple analysis of genome sequence will have a great impact on hypertension.[56] Instead, the systematic evaluation of epigenetic modifications associated with detailed functional phenotyping is more likely to allow us to move forward in the understanding of this complex interplay in patients with primary hypertension.[56,57]

Whereas the attempts at using current genetic approaches to understand essential hypertension in general have not been very fruitful, the study of monogenic hypertension has. Monogenic causes of hypertension, although rare, have provided substantial insight into the pathogenesis of hypertension. Of the monogenic forms of hypertension with well-described molecular mechanisms, all have one thing in common—a defect in renal sodium handling. This commonality points to the primacy of the kidney in regulating BP by way of sodium balance.

In Liddle syndrome, mutations in the epithelial sodium channel (ENaC) lead to increased ENaC expression and decreased channel removal from the luminal membrane, both of which contribute to persistent channel activation. This leads to sodium avidity, volume expansion, hypertension, hypokalemia, and metabolic alkalosis with suppressed aldosterone levels.[58]

In Gordon syndrome (pseudohypoaldosteronism type 2), a variety of mutations have been described leading to changes in the function of the thiazide-sensitive sodium-chloride cotransporter (NCC).[58] These mutations were initially mapped to the with-no-lysine (WNK) kinases 1 and 4, which regulate NCC phosphorylation and activity. Mutations in two E3 ubiquitin ligase complex proteins (kelchlike 3 and cullin 3) were discovered later.[59] These mutations are responsible for most cases of the syndrome. The presumed mechanism is related to decreased channel ubiquitination—and therefore persistent presence in the luminal membrane and augmented sodium removal from the tubular fluid—leading to the clinical phenotype of hypertension, hyperkalemia, and metabolic acidosis.

Mutations in the mineralocorticoid receptor can also produce hypertensive syndromes, such as hypertension exacerbated by pregnancy (Geller syndrome), in which there is constitutive activity of the receptor in addition to marked sensitivity to progesterone, leading to hypertension during pregnancy in addition to chronic severe hypertension with hypokalemia.[60] Likewise, increases in the aldosterone level due to a chimeric gene duplication involving the 11β-hydroxylase and the aldosterone synthase genes result in control of aldosterone synthase by adrenocorticotropic hormone (ACTH), independently of sodium balance or angiotensin II (Ang II) or serum potassium levels. Such patients have hyperaldosteronism that is only blunted by ACTH suppression—thus, the term "glucocorticoid-remediable hyperaldosteronism."[61]

Other patients may have apparent mineralocorticoid excess due to mutations in the 11β-hydroxysteroid dehydrogenase type 2 gene. This enzyme is responsible for the conversion of cortisol to the inactive cortisone in target epithelia, including the kidneys. As a result, excess cortisol, which is 1000-fold more abundant in blood than aldosterone, is available to activate the mineralocorticoid receptor, leading to a state of apparent mineralocorticoid excess (salt-sensitive hypertension, hypokalemia, metabolic alkalosis) in the absence of aldosterone.

Similarly, patients with congenital adrenal hyperplasia due to 11β-hydroxylase or 17α-hydroxylase deficiency have an excess production of 21-hydroxylated steroids such as deoxycorticosterone and corticosterone, which are potent activators of the mineralocorticoid receptor. Thus, also the syndrome of apparent mineralocorticoid excess is also produced, in addition to the well-known sexual developmental abnormalities of the syndromes.[62] Taken together, this is strong evidence of the importance of renal sodium handling in the genesis of hypertension.

It must be noted, however, that the recent elucidation of the molecular genetics of the rare syndrome of autosomal hypertension with brachydactyly points to a vascular cause that does not involve the kidney.[63] This is a monogenic form of salt-independent and age-related hypertension transmitted via gain-of-function mutations in the phosphodiasterase 3A gene (*PDE3A*), which results in increased protein kinase A–mediated PDE3A phosphorylation and increased cyclic adenosine monophosphate (cAMP) hydrolytic activity. These cellular actions result in enhanced cell proliferation in fibroblasts and vascular smooth muscle and dysregulated parathyroid hormone–related peptide physiology, accounting for the hypertensive and skeletal phenotypes.[63]

NONOSMOTIC SODIUM STORAGE

The paradigm of sodium balance described earlier assumes that sodium and its accompanying anion are osmotically active and therefore retained isosmotically with water. However, this paradigm cannot explain the observation that acute sodium loading in humans and animals results in positive sodium balance, without the expected water (weight) gain. Consequently, sodium may accumulate without water, most prominently in the skin,[64] where negatively charged glycosaminoglycans bind sodium.[65] This system of interstitial sodium buffering adds to the classical Guytonian approach, wherein nonosmotic accumulation occurs acutely and is presumably followed by increased removal from skin (via an enhanced lymphatic network) for ultimate renal excretion.

The mechanisms explaining isosmotic sodium storage have been under intense investigation.[66] Mice and rats receiving a high-salt diet develop hypertonicity of the skin interstitium, which triggers a series of mechanisms to keep interstitial volume constant.[67] The hypertonic sodium content activates the tonicity-responsive enhancer-binding protein (TonEBP) present in mononuclear cells infiltrating the skin. Consequently, these skin macrophages act as local osmosensors and secrete vascular endothelial growth factor type C, resulting in increased density and hyperplasia of the skin lymphocapillary network and increased endothelial nitric oxide synthase (eNOS). If these responses are blocked, salt-sensitive hypertension develops.[67,68] These findings link the mononuclear phagocyte system to extracellular fluid volume control.

The translation of these findings into clinical implications was addressed in a study evaluating sodium balance in cosmonauts undergoing prolonged training (up to 205 days) in a facility simulating life in space.[69] By carefully monitoring water and electrolyte intake and excretion, as well as factors regulating sodium balance, these individuals exhibited a large variability in sodium excretion on a daily basis, despite relatively stable diets. In the long term, approximately 95% (70%–103%) of ingested sodium was recovered, but daily sodium excretion during stable sodium intake varied considerably and was independent of BP and sodium intake. Instead, urine sodium excretion varied as a function of circaseptan fluctuations (6–9 days in this case) in levels of aldosterone, cortisol, and cortisone. Moreover, total body sodium stores had even longer infradian variations (averaging several weeks).

The factors regulating these intriguing changes are still unknown. These observations have clinical implications for the use of urine sodium excretion to assess sodium intake because they suggest wide daily variations that cannot be captured in a single 24-hour urine collection.[69]

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The RAAS has wide-ranging effects on BP regulation. Fig. 46.8 summarizes the most relevant elements of the RAAS and its role in the pathogenesis of hypertension and its complications (see Chapter 11 for greater detail). The different elements of the RAAS have key roles in mediating sodium retention, pressure natriuresis, salt sensitivity, vasoconstriction, endothelium dysfunction, and vascular injury, and the use of RAAS blockers is an effective means of treating hypertension. Taken together, the RAAS has an important role in the pathogenesis of hypertension. However, there are some still unanswered issues about this relationship. For example, in a very large GWAS of 2.5 million genotyped or imputed SNPs in 69,395 individuals of European ancestry from 29 studies,[70] the meta-analysis showed that most SNPs involved issues with natriuretic peptide abnormalities. Thus, these hormones play a prominent role in the pathogenesis of hypertension and may be more important than the RAAS system, which did not have prominent SNPs associated with this analysis. Another meta-analysis evaluating the relationship between polymorphisms in key RAAS genes (ACE, AGT, and CYP11B2) and salt sensitivity also found no significant role for these polymorphisms.[71]

Despite these genetic inconsistencies, there is a wealth of experimental evidence linking the RAAS to hypertension, which we review concisely later. In addition, a classic paradigm has been to highlight the role of the circulating (systemic) RAAS. However, it is known that tissue expression of different elements of the RAAS is also important, including the protection of animals from the development of hypertension after targeted elimination of renal ACE activity.[72] Interestingly, these very same experiements have also indicated that the absence of ACE in other tissues may also be protective from hypertension caused by triggers that do not involve the RAAS (e.g., nitric oxide inhibition), therefore raising the possibility that other (extrarenal) sites may also be relevant.[73]

Renin and prorenin are synthesized and stored in the juxtaglomerular cell apparatus and released in response to decreased renal afferent perfusion pressure, decreased sodium delivery to the macula densa, activation of renal nerves (via β_1-adrenergic receptor stimulation), and a variety of metabolic products, including prostaglandin E2 and several others. The main function of renin is to cleave angiotensinogen into angiotensin I. Prorenin, previously regarded as an inactive substrate for renin production, is known to also stimulate the (pro)renin receptor (PRR). This receptor leads to more efficient cleavage of angiotensinogen and activates downstream intracellular signaling through the mitogen-activated protein (MAP) kinases extracellular signal–regulated kinases 1 and 2 (ERK1/2) pathways that have been associated with profibrotic effects in some, but not all, experimental models.[74–76] It is still uncertain if the PRR is involved in the genesis or complications of hypertension in a manner that is independent of the effects of Ang II .[74]

Angiotensin II, formed by the cleavage of angiotensin I by ACE, is at the center of the pathogenetic role of the RAAS in hypertension. Primarily through its actions mediated by the Ang II type 1 receptor (AT1R), Ang II is a potent vasoconstrictor of vascular smooth muscle, causing systemic vasoconstriction and increased renovascular resistance and decreased medullary flow, which is a mediator of salt sensitivity.

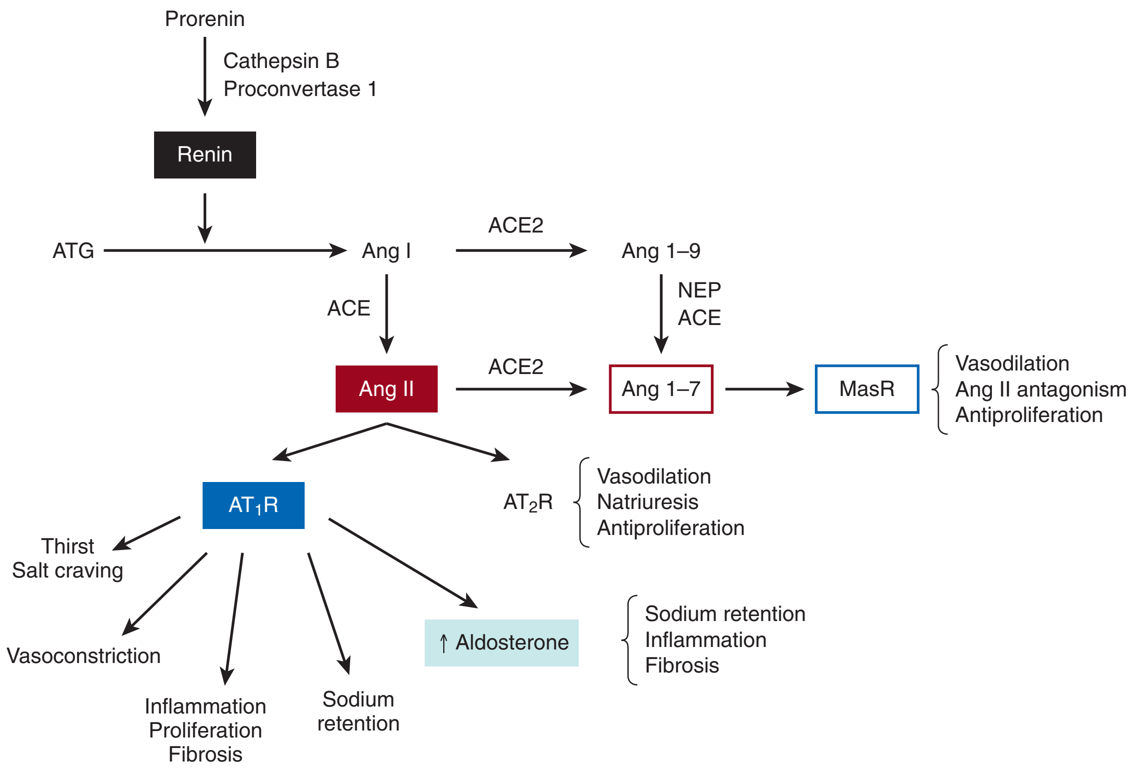


Fig. 46.8 Key elements of the renin-angiotensin-aldosterone system. *ACE,* Angiotensin-converting enzyme; *ACE2,* Angiotensin-converting enzyme type 2; *Ang,* angiotensin; *ATG,* angiotensinogen; *AT$_1$R,* Ang II type 1 receptor; *AT$_2$R,* Ang II type 2 receptor; *MasR,* Mas receptor; *NEP,* neutral endopeptidase.

It produces increased sodium reabsorption in the proximal tubule by increasing the activity of NHE3, the sodium-bicarbonate exchanger, and Na^+-K^+-ATPase and by inducing aldosterone synthesis and release from the adrenal zona glomerulosa. In addition, it is associated with endothelial cell dysfunction and produces extensive profibrotic and proinflammatory changes, largely mediated by increased oxidative stress, resulting in renal, cardiac, and vascular injury, thus giving Ang II a tight link to target organ injury in hypertension. Conversely, stimulation of the Ang II type 2 receptor (AT2R) is associated with opposite effects, resulting in vasodilation, natriuresis, and antiproliferative effects.

The relative importance of the renal and vascular effects of Ang II has been evaluated in classic cross-transplantation studies using both wild-type mice and mice lacking the AT1R.[77,78] By cross-transplanting the kidneys of wild-type mice into AT1R knockout mice and vice versa, investigators were able to generate animals that were selective renal AT1R knockouts or selective systemic (nonrenal) AT1R knockouts. In physiologic conditions, renal, systemic, and total knockout animals had lower BP than wild-type animals, indicating a role of both renal and extrarenal AT1R in BP regulation.[78] The systemic AT1R absence was associated with approximately 50% lower aldosterone levels, but the lower BP observed in this group was independent of this lower aldosterone production. This was because BP remained low, despite aldosterone infusions to supraphysiologic levels following adrenalectomy in the systemic knockout animals. In addition, the BP reduction in kidney knockout animals occurred despite normal aldosterone excretion, again confirming the importance of aldosterone-independent renal Ang II effects.

In the hypertensive environment, it is the presence of renal AT1R that mediates hypertension and organ injury[78] (Fig. 46.9). When animals were infused with Ang II for 4 weeks, animals lacking renal AT1R did not develop sustained hypertension, whereas wild-type and systemic knockout mice had a significant increase in BP. Additionally, only animals with elevated BP developed cardiac hypertrophy and fibrosis. This indicates that cardiac injury is largely dependent on hypertension and not on the presence of AT1R in the heart, because the (hypertensive) systemic knockout animals developed significant cardiac abnormalities, despite the absence of AT1R in the heart.[77] In summary, these experiments have indicated that both systemic and renal actions of Ang II are relevant to physiologic BP regulation but, in hypertension, the detrimental effects of Ang II are mediated via its renal effects.

It must be understood that there may be a role for extrarenal RAAS activation in hypertension, albeit likely less important than its renal role.[72,73] Because Ang II is necessary for normal renal development, experimental manipulation of the intrarenal RAAS demands the presence of some level of activity of the system. In experiments where the ACE gene was knocked out systemically then reintroduced in myelomonocytic cells, there was no demonstrable renal ACE activity, but the systemic levels generated from the myelomonocytic cells were enough to allow normal renal development.[72] These animals, with absent renal ACE activity, have markedly blunted hypertensive responses to the infusion of Ang II or N(ω)-nitro-L-arginine methyl ester (L-NAME, a low-renin, Ang II–independent model of hypertension). At face value, this indicates that the absence of renal ACE protects against

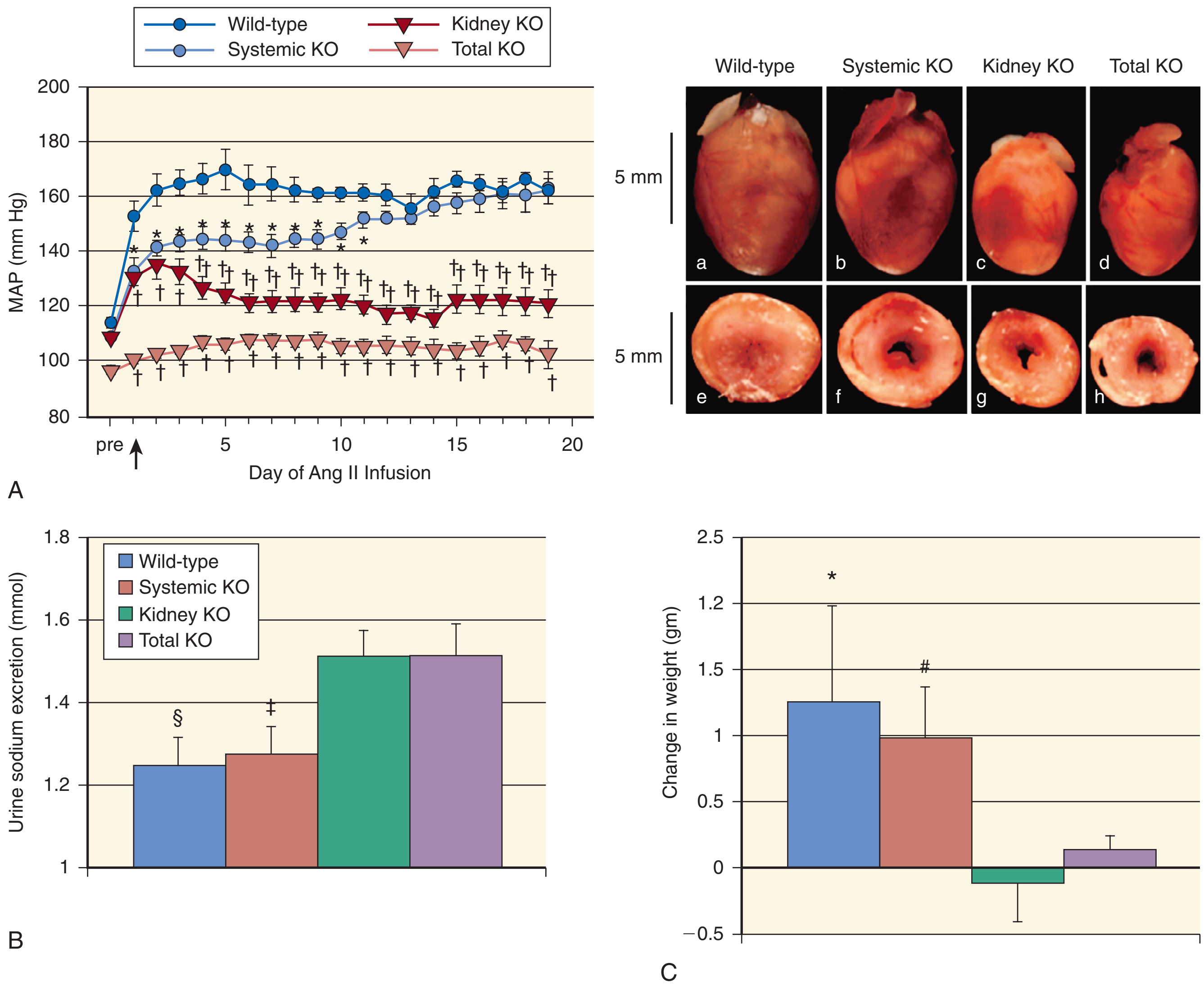


Fig. 46.9 Effects of angiotensin II *(Ang II)* infusion on blood pressure (A), urinary sodium excretion (B), body weight (C), and cardiac hypertrophy (photos) according to renal and extrarenal presence of Ang II type 1 receptor. See text for details. *KO,* Knockout; *MAP,* mean arterial pressure. (From Crowley SD, Gurley SB, Herrera MJ, et al. Ang II causes hypertension and cardiac hypertrophy through its receptors in the kidney. *Proc Natl Acad Sci U S A.* 2006;103:17985–17990.)

hypertension, regardless of the plasma Ang II levels. This is an unexpected finding given the traditional view of the RAAS, in which an infusion of Ang II bypasses ACE and produces sustained hypertension, regardless of ACE activity,[79] but well in line with the growing evidence for the importance of ACE-induced intrarenal tubular generation of Ang II in the mediation of hypertension. However, because ACE was not expressed in other organs, it is also possible that the hypotensive effects of the absence of ACE activity was the result of the absence of an effect in other locations, such as the vasculature.[73,80] This question remains unanswered while developments continue to occur in the understanding of the function of tissue RAAS (especially intrarenal). This system has different regulators from the systemic RAAS and, differently from the endocrine RAAS, has feed-forward mechanisms that generate further augmentation of local responses on activation by the circulating RAAS.[80]

Aldosterone, the adrenocortical hormone synthesized in the zona glomerulosa, plays a critical role in hypertension through its well-known effects on sodium reabsorption that are largely mediated by transcriptional effects via activation of the mineralocorticoid receptor, leading to increased expression of ENaC. An extensive body of literature has identified other genomic and nongenomic effects of aldosterone with relevance to hypertension. Extensive nonepithelial effects include vascular smooth muscle cell proliferation, vascular extracellular matrix deposition, vascular remodeling and fibrosis, and increased oxidative stress, leading to endothelial dysfunction and vasoconstriction.[81]

Several other elements of the RAAS have been identified as having potentially important roles in hypertension. The importance of ACE2 and Ang–1-7 to BP regulation and Ang II–associated target organ injury has become apparent. ACE2 is expressed largely in heart, kidney, and endothelium; it has partial homology to ACE and is unaffected directly by ACE inhibitors.[82] It has a variety of substrates, but its most important action is the conversion of Ang II to Ang–1-7. Ang–1-7 is formed primarily though the hydrolysis of Ang

II by ACE2. Its actions are opposite to those of Ang II, including vasodilatory and antiproliferative properties mediated by the Mas receptor, a G protein–coupled receptor that on activation, forms complexes with the AT1R, thus antagonizing the effects of Ang II. The vasodilatory effects are mediated by increased cyclic guanosine monophosphate (cGMP), decreased norepinephrine release, and amplification of bradykinin effects. Studies have identified ACE2 and Ang–1-7 as protective factors in the development of atherosclerosis and cardiac and renal injury.[82,83] The administration of recombinant ACE2 or its activator, xanthenone, has resulted in improved endothelial function, decreased BP, and improved renal, cardiac, and perivascular fibrosis in hypertensive animals.[84–86] However, a phase 1 study of recombinant ACE2 in healthy humans did not show any BP-lowering effects, despite appropriate modulation of the RAAS, including a sustained increase in Ang–1-7 levels.[87] Therefore, the clinical value of the manipulation of any of the elements of this vasodepressor component of the RAAS remains to be determined.[88]

SYMPATHETIC NERVOUS SYSTEM

The sympathetic nervous system (SNS) is activated consistently in patients with hypertension compared with normotensive individuals, particularly in those who are obese (Fig. 46.10). Many patients with hypertension are in a state of autonomic imbalance that encompasses increased sympathetic and decreased parasympathetic activity.[89,90] SNS hyperactivity is relevant to the generation and maintenance of hypertension and is observed in human hypertensives from the very earliest stages. For example, studies in humans have identified markers of sympathetic overactivity in normotensive individuals with a family history of hypertension.[89] Among patients with hypertension, increasing severity of hypertension is associated with increasing levels of sympathetic activity as measured by microneurography.[91,92] In human hypertension, plasma catecholamine levels, microneurographic recordings, and systemic catecholamine spillover studies have consistently found elevation of these markers in obesity, the metabolic syndrome, and hypertension complicated by heart failure or kidney disease.[89] In addition, SNS hyperactivity is observed in most hypertensive subgroups, although it appears more pronounced in men than in women and in younger than in older patients.

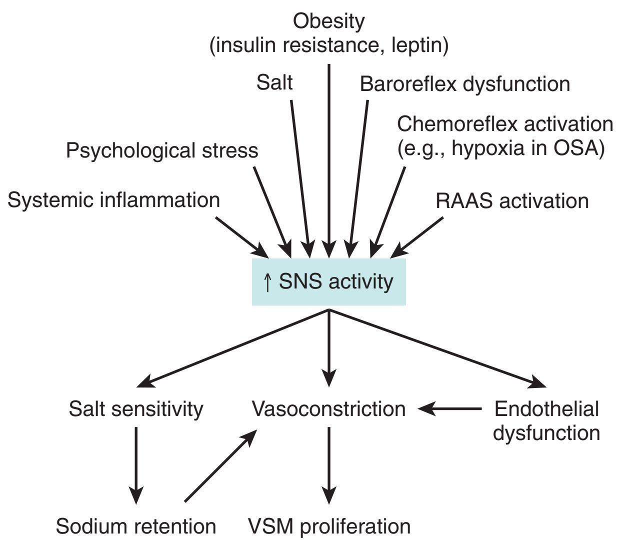


Fig. 46.10 Causes and consequences of sympathetic nervous system activation in the pathogenesis of hypertension. *OSA,* Obstructive sleep apnea; *RAAS,* renin-angiotensin- aldosterone system; *SNS,* sympathetic nervous system; *VSM,* vascular smooth muscle.

Several experimental models have outlined the importance of the SNS in generating hypertension. Different models of obesity-related hypertension have indicated that the SNS is activated early in the development of increased adiposity,[90] and the key factor in the maintenance of sustained hypertension is increased renal sympathetic nerve activity and its attendant sodium avidity.[90]

SNS-mediated induction of salt sensitivity is a key element for sustaining high BP in other models of hypertension as well. For example, rats receiving daily infusions of phenylephrine for 8 weeks developed hypertension during the infusions, but BP normalized under a low-salt diet after discontinuation of phenylephrine.[93] However, once exposed to a high-salt diet, the animals again became hypertensive. The degree of BP elevation on a high-salt diet was directly related to the degree of renal tubulointerstitial fibrosis and decrement of GFR. These findings can be interpreted within the paradigm that catecholamine-induced hypertension causes renal interstitial injury that is associated with a salt-sensitive phenotype, even after sympathetic overactivity is no longer present.[93] In addition, enhanced SNS activity results in α_1 receptor–mediated endothelial dysfunction, vasoconstriction, vascular smooth muscle proliferation, and arterial stiffness, all of which contribute to the development of hypertension. Finally, studies have shown that sympathetic overactivity results in salt sensitivity due to a reduction in the activity of WNK4, which causes increased sodium avidity through the thiazide-sensitive NCC.[94] Fig. 46.10 summarizes the causes and consequences of SNS activation in the genesis of hypertension.

Renalase is a flavoprotein highly expressed in kidney and heart that metabolizes catecholamines and catecholamine-like substances to aminochrome.[95] Tissue and plasma renalase is decreased in experimental models with renal mass reduction, and renalase knockout mice have increased BP and elevated circulating catecholamine levels. A normal phenotype is restored by the administration of recombinant renalase. Also of relevance to catecholamine metabolism is catestatin, a product of the proteolysis of the neuroendocrine peptide chromogranin A.[96] Catestatin acts at nicotinic cholinergic receptors in adrenal chromaffin cells as an inhibitor of catecholamine release. Chromogranin A knockout mice are hypertensive and have elevated catecholamine levels, both of which are normalized by the administration of catestatin. Moreover, serum catestatin levels are decreased in patients with hypertension and their normotensive offspring, raising the possibility of a regulatory role in the development of hypertension. The role of renalase and catestatin in the modulation of SNS-mediated hypertension, as well as their possible value in the treatment of hypertension in humans, remains uncertain.

Because increased SNS activity is associated with vascular smooth muscle proliferation, LVH, large artery stiffness, myocardial ischemia, and arrhythmogenesis, there is also a mechanistic role for the SNS in the complications of hypertension. In support of this concept, several cohort studies have reported an association between physiologic or biochemical markers of SNS activation and adverse outcomes in heart failure, stroke, and ESKD.[89,97] However, there are no such studies among patients with hypertension, and the indirect evaluation of the impact of treatment-induced heart rate reduction in hypertension has yielded paradoxic results.

In a meta-analysis of hypertension trials, heart rate reduction during treatment with beta-blockers was associated with increased risk for death and CV events in patients with hypertension.[98] In contrast, in a very large (n = 10,000) patient outcome trial, a post hoc analysis of heart rate at baseline has demonstrated that those with a resting heart rate above 80 beats/min, even with a BP below 140/90 mm Hg, had a higher mortality rate.[99] Therefore, although it is apparent that SNS activation is deleterious to patients with CV disease, and presumably with hypertension, a cause for the overactivity should be sought and an attempt made to affect that mechanism.

OBESITY

Obesity-related hypertension is characterized primarily by impaired sodium excretion and endothelial dysfunction, both of which are dependent on SNS overactivity, activation of the RAAS, and increased oxidative stress.[89,100] Fat tissue in obesity is hypertrophied, meaning larger cells versus more cells, and marked by increased macrophage infiltration in the adipose tissue.[101] It is now well described; adipose tissue is not inert and secretes a wide range of cytokines and chemokines whose profile is abnormal in obesity, marked by increased levels of leptin, resistin, interleukin-6, and tumor necrosis factor-α secretion; elevated free fatty acid release; and reduced adiponectin levels. Decreased adiponectin level results in insulin resistance, decreased induction of eNOS, and possibly increased sympathetic activity.

Resistin impairs nitric oxide (NO) synthesis (eNOS inhibition) and enhances endothelin-1 (ET-1) production, shifting the vasodilation-vasoconstriction balance toward vasoconstriction. Hyperleptinemia directly stimulates the SNS through complex mechanisms that involve central leptin receptors, as well as activation of the pro-opiomelanocortin system (via the melanocortin 4 receptor).[100]

Finally, visceral adipocyte mass is directly correlated with aldosterone secretion by the zona glomerulosa, a process mediated by angiotensinogen production by adipocytes, as well as increased secretion of Wnt-signaling molecules that modulate steroidogenesis.[102–104] In addition, despite yet unclear mechanisms, there is consistent evidence that obese individuals tend to have lower natriuretic peptide levels than lean individuals,[100] and this relative deficiency is amplified in obese hypertensives.[105] All these factors compound the tendency toward sodium retention and shifting the pressure-natriuresis curve to the right. Activation of these same systems leads to a proinflammatory state related to increased reactive oxygen species, factors directly associated with endothelial dysfunction and vascular proliferation. Therefore, multiple mechanisms contribute to the development and maintenance of hypertension in obese individuals.

NATRIURETIC PEPTIDES

Natriuretic peptides—atrial natriuretic peptide (ANP), brain natriuretic peptide (BNP), and urodilatin—play an important role in salt sensitivity, heart failure, and hypertension.[106] These peptides have important natriuretic and vasodilatory properties that allow maintenance of sodium balance and BP during sodium loading. On administration of a sodium load, atrial and ventricular stretch lead to the release of ANP and BNP, respectively, which result in immediate BP lowering due to systemic vasodilation and decreased plasma volume. The latter is caused by fluid shifts from the intravascular to the interstitial compartment.[107] All natriuretic peptides directly increase GFR, which in volume-expanded states is mediated by an increase in efferent arteriolar tone and increased K_f. They also inhibit renal sodium reabsorption through direct and indirect effects. Direct effects include decreased activity of Na^+-ATPase and the sodium-glucose cotransporter in the proximal tubule and inhibition of the ENaC in the distal nephron.[106] The inhibitory effects of natriuretic peptides on renin and aldosterone release mediate indirect effects. Unfortunately, understanding the contribution of natriuretic peptides to the development of hypertension in humans is complicated by the elevation of their levels in association with increased BP (due to increased afterload) and hypertensive heart disease.

Some studies have tested whether polymorphisms in ANP or BNP genes resulting in higher levels of these peptides would associate with lower BP; results of these studies have been inconsistent, and effects have been small.[108–110] There have been no published studies evaluating sequential changes in natriuretic peptides and risk for incident hypertension.

Attention has been given to corin, the serine protease that is largely expressed in the heart and converts pro-ANP and pro-BNP to their active forms. Corin activates ANP. Genetic variants and defects of the enzyme that activates corin have been found in people with hypertension, heart failure, and preeclampsia. Corin intereacts with natriuretic peptides and hence impaired corin intracellular trafficking, cell surface expression, and zymogen activation will result in fluid retention.[111,112] Experiments have suggested that states of corin deficiency are associated with sodium overload, heart failure, and salt-sensitive hypertension,[111,112] and recent clinical studies have observed an association between low corin levels and increased risk of adverse cardiovascular outcomes in patients with coronary disease, heart failure, and stroke.[111]

THE ENDOTHELIUM

The endothelium is a major regulator of vascular tone and thus plays a key role in BP regulation. Endothelial cells produce a host of vasoactive substances, of which NO is the most important to BP regulation. NO is continuously released by endothelial cells, especially in response to flow-induced shear stress in arteries and arterioles, leading to vascular smooth muscle relaxation through the activation of guanylate cyclase and generation of intracellular cGMP.[113] Interruption of NO production via inhibition of the constitutively expressed eNOS causes BP elevation and development of hypertension in animals and humans. Using brachial artery flow-mediated vasodilation and measurement of urinary excretion of NO metabolites as methods to evaluate NO activity in humans, several studies have demonstrated decreased whole-body production of NO in patients with hypertension compared with normotensive controls.

Several elements are responsible for endothelial dysfunction in hypertension. Normotensive offspring of patients with hypertension have impaired endothelium-dependent vasodilation, despite normal endothelium-independent responses, thus suggesting a genetic component to the development of endothelial dysfunction.[114] Besides direct pressure-induced injury in the setting of chronically elevated BP, a mechanism of major importance is increased oxidative stress. Reactive oxygen species are generated from enhanced activity of several enzyme systems—reduced nicotinamide adenine dinucleotide phosphate oxidase (NADPH oxidase), xanthine oxidase, and cyclooxygenase in particular—and decreased activity of the oxygen free radical detoxifying enzyme superoxide dismutase.[113,115] Excess availability of superoxide anions leads to their binding to NO, resulting in decreased NO bioavailability, in addition to generating the oxidant, proinflammatory peroxynitrite. It is the decreased NO bioavailability that links oxidative stress to endothelial dysfunction and hypertension.[114] Ang II is a major enhancer of vascular NADPH-oxidase activity and plays a central role in the generation of oxidative stress in hypertension, although several other factors are also involved, including cyclic vascular stretch, ET-1, uric acid, systemic inflammation, norepinephrine, free fatty acids, and tobacco smoking.[116]

ET-1 is the endothelial cell product that counteracts NO to maintain the balance between vasodilation and vasoconstriction. ET-1 expression is increased by shear stress, catecholamines, Ang II, hypoxia, and several proinflammatory cytokines, such as tumor necrosis factor-α, interleukins 1 and 2, and transforming growth factor-β.[113] ET-1 is a potent vasoconstrictor through stimulation of ET-A receptors in vascular smooth muscle.[117] In hypertension, increased ET-1 levels are not consistently found. However, there is a trend of increased sensitivity to the vasoconstrictor effects of ET-1. ET-1, therefore, is considered a relevant mediator of BP elevation—ET-A and ET-B receptor antagonists attenuate or abolish hypertension in several experimental models of hypertension (e.g., Ang II–mediated models, deoxycorticosterone acetate–salt hypertension, and Dahl salt-sensitive rats) and are effective in lowering BP in humans.[118]

Endothelial cells also secrete a variety of other vasoregulatory substances. These include the vasodilating prostaglandin prostacyclin (PGI2), CNP, and several vasodilating endothelium-derived hyperpolarizing factors, the identity of which remains uncertain. There are also endothelium-derived contracting factors in addition to ET-1, such as locally generated Ang II and vasoconstricting prostanoids such as thromboxane A2 and prostaglandin A2. The balance of these factors, along with NO and ET-1, determine the final impact of the endothelium on vascular tone.

Other non–endothelium-derived factors may be of relevance to the genesis of hypertension via endothelium dysfunction. Much attention has been given to uric acid, which can induce endothelial dysfunction and produce salt-sensitive hypertension through mechanisms that involve renal microvascular injury.[119,120] These changes can be abrogated by therapies that lower uric acid in animals and may be of value in lowering BP and limiting renal injury in humans with hyperuricemia. Current evidence from randomized trials, however, have only supported a modest BP-lowering effect of allopurinol in hyperuricemic hypertensive adolescents,[121] with no observable effect of allopurinol or probenecid on flow-mediated vasodilation in obese nonhypertensive adults with mild hyperuricemia.[122] This implies an age-dependent effect or an effect of duration of exposure to the hyperuricemia, even if mild. Also of relevance is the possible role of high dietary fructose consumption in intracellular adenosine triphosphate (ATP) depletion, increased oxidative stress, increased uric acid production, and endothelial dysfunction.[123,124]

Nitric oxide is not the only gasotransmitter relevant to vascular biology. Hydrogen sulfide (H_2S) is another compound that has received recent attention due to potential treatment relevance.[125] H_2S is a vasodilating gas produced from sulfated amino acids by the action of one of two key enzymes, cystathione gamma-lyase (CSE) and cystathione beta-synthase. CSE homozygous knockout mice have around 80% decreased H_2S expression in the heart and aorta and around 60% reduction in serum, and both homozygous and heterozygous animals demonstrate impaired endothelial function and develop age-dependent hypertension.[126] The mechanisms underlying the BP effects of H_2S are multiple, including enhanced NO-mediated vasodilation, activation of potassium ATP channels, activation of protein kinase G1α, inhibition of phosphodiesterase type 5, and inhibition of SNS activity.[125] Modulation of H_2S levels with the administration of gaseous H_2S or H_2S donors (sodium hydrosulfide or sodium thiosulfate) has resulted in lower BP and decreased cardiovascular and renal injury in several experiemental models.[125] If delivery systems permit, and successful oral use of these agents is developed, it is possible that H_2S may become a therapeutic target in hypertension and vascular disease.

Taken together, the net result observed in patients with hypertension is one of endothelial dysfunction. In cross-sectional analyses, the lower the degree of forearm flow-mediated vasodilation, the greater the prevalence of hypertension.[115,127] Prospective cohort studies have used flow-mediated vasodilation as a measure of endothelial dysfunction, regardless of a specific mechanism, to evaluate its relationship with hypertension and test whether endothelial dysfunction is a cause or consequence of hypertension, or both.[128] These studies have shown conflicting results, but the larger of them was unable to demonstrate an association between endothelial dysfunction and incident hypertension among 3500 patients followed for 4.8 years.[128]

Furthermore, endothelial dysfunction carries a genetic predisposition that is independent of BP and may be improved by agents that have little or no impact on BP (e.g., some antioxidants).[129] Therefore, as it stands, the evidence is stronger for endothelial dysfunction as a consequence, not a cause, of hypertension.[127,129]

ARTERIAL STIFFNESS IN HYPERTENSION

Arterial stiffness is an important factor in the pathogenesis of hypertension, particularly the syndrome of isolated systolic hypertension, because it is a common accompaniment of elevated systolic BP and pulse pressure. Arterial stiffness develops as a result of structural changes in large arteries, particularly elastic arteries.[130] These include loss of elastic fibers and substitution with less distensible collagen fibers.

Factors strongly associated with arterial stiffening include aging, hypertension, diabetes mellitus, CKD, smoking, and high-sodium intake.[131]

The most commonly used measure to assess arterial stiffness in humans is carotid-femoral pulse wave velocity (cf-PWV). The traditional view linking arterial stiffness (measured as increased cf-PWV) to hypertension noted that faster PWV produced faster reflection of the incident pulse wave, which resulted in an earlier reflected wave that returned to the central circulation before the end of systole, resulting in increased systolic BP.[132] Although these mechanisms still hold true, later data have indicated the importance of two other factors, increased amplitude of the forward wave and increased characteristic impedance of the proximal aorta.[132] When these specific factors are taken into account, the relative contribution of wave reflection to the observed age-dependent change in pulse pressure is only 4% to 11%.

Arterial stiffness was previously thought to be a consequence of hypertension. Cyclic pulsatile load is associated with fracture of elastin fibers and wall stiffening, and increased distending pressure demands recruitment of the less distensible collagen fibers, thus making vessels stiffer.[130] Evidence from several studies, however, have indicated that arterial stiffness may precede and predispose to hypertension.[132] For example, in the Framingham Heart Study, markers of arterial stiffness (cf-PWV and amplitude of the forward pressure wave) were associated with a 30% to 60% increased risk for incident hypertension (per standard deviation of each variable) during 7 years of follow-up in a cohort with a baseline mean age of 60 years.[133] Conversely, baseline BP levels were not associated with future changes in arterial stiffness. Some studies have corroborated these findings, but other studies have also suggested a bidirectional relationship, so that arterial stiffness is also a consequence of chronic hypertension.[132] In contraposition, a recent cohort study of younger adults (baseline age, 36 years) has indicated that higher BP is associated with higher large artery stiffness, not the opposite.[134] These differences in results between younger and older adult populations may indicate that earlier in life, hypertension is mediated by factors that are largely independent of large vessel stiffness, whereas later in life, arterial stiffness has a more important causal role in the development of hypertension.

Arterial stiffening is relevant to target organ damage in hypertension. Increased PWV is associated with increased mortality and CV events,[135] as well as with a variety of subclinical CV injury markers, such as coronary calcification, cerebral white matter lesions, ankle-brachial index, and albuminuria. A relationship with cardiac complications has been suggested—increased impedance to left ventricular ejection results in LVH, diastolic dysfunction, and subendocardial myocardial ischemia.

The relationship to brain and renal complications is more complex. It is apparent that the mechanism of damage of these organs, which are characterized by vasculatures with high flow and low impedance, is mediated by the increased transmission of increased pulsatile pressure to the brain and renal parenchyma. The reason for this is related to the abnormal process of impedance matching. For individuals with normal vessels, the elastic arteries are much less stiff than muscular arteries, thus creating an impedance mismatch. This mismatch provokes wave reflection, thereby protecting the tissue located distally to this reflection point from injury from the traveling pulse wave. In states of increased arterial stiffness, the stiffening of elastic arteries approximates the stiffness of muscular arteries, thus eliminating the protective impedance mismatch. Once impedances become matched, there is less reflection and greater tissue injury, as supported by a growing body of clinical and experimental literature.[132] Therefore, it is clear that the relationship between hemodynamic forces and blood vessel pathology warrants greater attention in the study of hypertension.

ROLE OF THE IMMUNE SYSTEM IN HYPERTENSION

Immune responses, both innate and adaptive, participate in several of the mechanisms discussed earlier, including the generation of reactive oxygen species, mediation of the afferent arteriolopathy thought to be important to maintain salt sensitivity, and participation in the inflammatory changes noted in the kidneys, vessels, and brain in hypertension.[136,137] Innate responses, especially those mediated by macrophages, have been linked to hypertension induced by Ang II, aldosterone, and NO antagonism. Reductions in macrophage infiltration of the kidney or periadventitial space of the aorta and medium-sized vessels have led to improvements in BP and salt sensitivity in several experimental models.[137,138] Adaptive responses via T cells have been linked to the genesis and complications of hypertension. T cells express AT1R and mediate Ang II–dependent hypertension, as demonstrated by the observations that adoptive transfer of T cells restores the hypertensive phenotype in response to Ang II infusion that was absent in mice without lymphocytes.[138] Abnormalities in both proinflammatory T cells and regulatory T cells are implicated in complications of hypertension because they appear to regulate vascular and renal inflammation that underlies target organ injury.

Suppression of these inflammatory responses can improve BP control.[136,137,139] B lymphocytes may also play a causative role in hypertension. This has been suggested by reports of the effects of several autoantibodies, including agonistic antibodies against adrenergic receptors, vascular calcium channels, and AT1R, and antibodies against endothelial cells causing endothelial dysfunction, or heat shock proteins (hsp70) causing salt-sensitive hypertension. Further research will determine if manipulation of immune targets is of value in the prevention and treatment of hypertension.

OTHER METABOLIC PEPTIDES AND HYPERTENSION

Several vasodilating substances act as compensatory vasodilators to balance the heavily provasoconstrictive milieu in hypertension. Some of these vasodilators act primarily through an increase in NO release from endothelial cells, such as calcitonin gene–related peptide, adrenomedullin, and substance P. The glucose-regulating, gut hormone glucagon-like peptide-1 (GLP-1) has vasodilating properties, and its administration to Dahl salt-sensitive rats improves endothelial function, induces natriuresis, and lowers BP.[140] Additionally, the use of recombinant GLP-1 to treat diabetes in humans (exenatide) results in significant BP reduction, especially in those with high BP at baseline.[141]

GUT MICROBIOME

The gut microbiome may have a relationship with several humoral and hemodynamic aspects of hypertension, including neuroimmunologic aspects of BP regulation, such as activation of the SNS and regulation of the RAAS.[142] Bacteria that produce short-chain fatty acids (especially acetate, propionate, and butyrate) are seen as favorable to producing an environment of lower BP.[142] A proof of concept study has demonstrated an unfavorable gut microbiome profile (i.e., one that leads to less available short-chain fatty acids) in spontaneous hypertensive rats as compared with normotensive Wistar Kyoto rats and has been observed in animals exposed to Ang II infusions.[143] In addition, analyses of stool samples of 10 normotensive and seven hypertensive patients have shown a similar pattern.[143] The use of probiotics, typically as yogurt or other dairy products, is the most commonly used approach to alter the gut microbiome. A meta-analysis of nine randomized clinical trials investigating the use of probiotics to lower BP has shown an average BP reduction of 3.6/2.4 mm Hg.[144] Only one study included hypertensive patients (average BP reduction, 9.7/4.4 mm Hg), but an analysis of BP lowering according to baseline BP above or below 130/85 mm Hg has shown no differences in systolic BP and a small difference in diastolic BP reduction (2.7 vs. 0.9 mm Hg). These early results indicate that modification of the gut microbiome may alter hemodynamics and that the use of probiotics, presumably through such microbiome changes, may have salutary effects on BP, although the magnitude and clinical relevance of this effect remain to be determined.

CLINICAL EVALUATION

The evaluation of patients with hypertension focuses on six key components:

1. The confirmation that the patient is indeed hypertensive through careful measurements of BP
2. An assessment of clinical features that might suggest specific remediable causes of hypertension
3. The identification of comorbid conditions that confer additional CV risk or that may affect treatment decisions
4. The discussion of patient-related lifestyle factors and preferences that will affect management
5. The systematic evaluation of hypertensive target organ damage
6. Shared decision making about the treatment plan

To accomplish this, the clinician often needs multiple visits, a targeted clinical examination, and selected laboratory and imaging tests.

HISTORY AND PHYSICAL EXAMINATION

The medical history and physical examination are essential to uncovering possible secondary causes of hypertension, identifying symptoms suggestive of hypertensive target organ damage, and diagnosing comorbid conditions that may affect treatment decisions. Although the focus is traditionally on the CV, neurologic, and renal systems, a complete review of systems is indicated when the patient is first evaluated. This is because some patients will present with hypertension because of sleep apnea (snoring, witnessed apneas or gasping), hyperthyroidism or hypothyroidism (each with their litany of possible symptoms), hyperparathyroidism (symptoms of hypercalcemia), Cushing syndrome (symptoms of cortisol excess), pheochromocytoma or paraganglioma (symptoms of catecholamine excess), or acromegaly with its distinctive physical findings. These conditions are discussed in detail later in this chapter.

High BP is typically asymptomatic, but some symptoms are common among patients with very high BP levels, such as headaches, epistaxis, dyspnea, chest pain, and faintness, all of which were present in more than 10% of patients presenting with diastolic BP levels above 120 mm Hg.[145] Other common symptoms are nocturia and unsteady gait; treated patients often complain of fatigue in addition to symptoms of overtreatment and those related to specific side effects of medications. In patients with lower BP levels, the occurrence of symptoms is often difficult to tie to observed BP, as demonstrated in a study evaluating the relationship between headaches and BP levels, where the frequently observed headaches in patients with hypertension did not correlate well with office or ambulatory BP levels.[146]

When searching for target organ damage, one looks for symptoms to suggest a previous stroke or transient ischemic attack, previous or ongoing coronary ischemia, heart failure, peripheral arterial disease, or a past history of kidney disease or current symptoms, such as hematuria or flank pain.

Obtaining a detailed family history as it pertains to hypertension is essential. Focus should be on the development of hypertension at a young age or clustering of endocrine problems (e.g., pheochromocytoma, multiple endocrine neoplasia, primary aldosteronism) or renal problems (e.g., polycystic kidney disease or any inherited form of kidney disease). The young patient with hypertension and a family history of hypertension poses a particular challenge and should be evaluated in detail. Table 46.2 provides a guide to possible causes to be considered.[147]

Knowledge of conditions with potential relevance to treatment is important. For example, issues related to CV risk management, such as diabetes mellitus, hypercholesterolemia, inflammation (C-reactive protein [CRP]), obesity, and tobacco smoking need to be evaluated. Patients with established CV disease will need some treatment for their hypertension and underlying disorder (e.g., beta-blockers for angina pectoris), so knowledge of specific CV diagnoses is essential. Finally, some non-CV conditions may have an impact on treatment options. For example, patients with reactive airways disease (asthma) probably should not receive beta-blockers, patients with prostatic hyperplasia may benefit from a regimen that includes an alpha-blocker, and patients with attention-deficit/hyperactivity disorder or anxiety may benefit from a central sympatholytic (e.g., guanfacine), but those with major depression should probably not be treated with this drug class.

It is also important to recognize that it is while obtaining the history that the clinician has the opportunity to explore issues related to lifestyle, cultural beliefs, and patient preferences that will be essential in designing an effective treatment plan. It is important to define dietary and physical activity patterns and, when problems are identified, to determine if the patient is willing and/or able to modify them. Cultural

Table 46.2 Clinical Clues to Guide the Investigation in Young Patients with Hypertension with a Potential Hereditary Cause

Specific Conditions	Possible Causes of Familial Hypertension	Clinical Clues
Catecholamine-Producing Tumors		
Pheochromocytoma, paraganglioma	Familial cases are responsible for ~30% of cases, including MEN2A and MEN2B, von Hippel–Lindau disease, neurofibromatosis, and familial paraganglioma syndromes (SDH complex mutations)	Paroxysmal palpitations, headaches, diaphoresis, pale flushing; syndromic features of any of the associated disorders
Neuroblastomas (adrenal); aortic or renovascular lesions	1%–2% of neuroblastomas are familial	
Coarctation of the aorta	Overrepresented in families but no familial distribution	Asymmetry between upper- and lower-extremity BP, radial-formal pulse delay; associated with Turner syndrome, Williams' syndrome, and bicuspid aortic valve
Renal artery stenosis caused by fibromuscular dysplasia or inherited arterial wall lesions	<10% familial with AD pattern	Abnormal renal vascular imaging results; vascular disease in the carotid territory at an early age; common in neurofibromatosis and Williams' syndrome; also present in tuberous sclerosis, Ehlers-Danlos syndrome, and Marfan syndrome
Parenchymal kidney disease GN	Alport disease (X-linked, AR, or AD), familial IgA nephropathy (AD with incomplete penetrance)	Proteinuria, hematuria, low eGFR
PKD	ADPKD type 1 or 2, ARPKD	Multiple renal cysts (as few as three in patients <30 years)
Adrenocortical disease; glucocorticoid-remediable aldosteronism (familial hyperaldosteronism type I)	AD chimeric fusion of the 11β-hydroxylase and aldosterone synthase genes	Cerebral hemorrhages at young age, cerebral aneurysms; mild hypokalemia; high plasma aldosterone, low renin
Familial hyperaldosteronism	AD; unknown defect	Severe type 2 hypertension in early adulthood; high plasma aldosterone, low renin; no response to glucocorticoid treatment
Familial hyperaldosteronism type III	AD; unknown defect	Severe hypertension in childhood with extensive target organ damage; high plasma aldosterone, low renin; marked bilateral adrenal enlargement
Congenital adrenal hyperplasia	AR mutations in 11β-hydroxylase or 21-hydroxylase	Hirsutism, virilization; hypokalemia and metabolic alkalosis; low plasma aldosterone and renin
Monogenic Primary Renal Tubular Defects		
Gordon syndrome	AD mutations of *KLHL3*, *CUL3*, *WNK1*, and *WNK4;* AR mutations of *KLHL3*	Hyperkalemia and metabolic acidosis with normal renal function
Liddle syndrome	AD mutations of the epithelial sodium channel	Hypokalemia and metabolic alkalosis; low plasma aldosterone and renin
Apparent mineralocorticoid excess	AD mutation in 11β-hydroxysteroid dehydrogenase type 2	Hypokalemia and metabolic alkalosis; low plasma aldosterone and renin
Geller syndrome; hypertension-brachydactyly syndrome	AD mutation in the mineralocorticoid receptor AD mutations in the phosphodiesterase E3A enzyme	Hypokalemia and metabolic alkalosis; low plasma aldosterone and renin; increased BP during pregnancy or exposure to spironolactone; short fingers (small phalanges) and short stature; brain stem compression from vascular tortuosity in the posterior fossa
Essential hypertension	Polygenic	When obesity or metabolic syndrome is present, likelihood of essential hypertension is higher

AD, Autosomal dominant; *ADPKD,* autosomal dominant polycystic kidney disease; *AR,* autosomal recessive; *ARPKD,* autosomal recessive polycystic kidney disease; *BP,* blood pressure; *eGFR,* estimated glomerular filtration rate; *GN,* glomerulonephritis; *IgA,* immunoglobulin A; *MEN,* multiple endocrine neoplasia; *PKD,* polycystic kidney disease; *SDH,* succinate dehydrogenase.

beliefs related to the treatment of hypertension, health illiteracy, and mistrust in physicians and the pharmaceutical industry are several of the factors that can affect the relationship with the patient and should be discussed openly. It is only then that patients will be able to participate in shared decision making about their treatment, an essential tenet of patient-centered care.

The physical examination is designed to complement the items discussed in the history. One should pay attention to syndromic features of cortisol excess (e.g., moon face, central obesity, frontal balding, cervical and supraclavicular fat deposits, skin thinning, abdominal striae), hyperthyroidism (e.g., tachycardia, anxiety, lid lag or proptosis, hypertelorism, pretibial myxedema), hypothyroidism (e.g., bradycardia, coarse facial features, macroglossia, myxedema, hyporeflexia), acromegaly (e.g., frontal bossing, widened nose, enlarged jaw, dental separation, acral enlargement, carpal tunnel syndrome), neurofibromatosis (e.g., neurofibromas, café au lait spots, because neurofibromatosis is associated with pheochromocytoma and renal artery stenosis), or tuberous sclerosis (e.g., hypopigmented ash leaf patches, facial angiofibromas, because tuberous sclerosis is associated with renal hypertension, usually related to angiomyolipomas). Many other even rarer associations exist but are beyond the scope of this chapter.

In younger patients or patients with unexplained, difficult to treat hypertension, it is worth exploring the possibility of coarctation of the aorta by measuring BP in both arms and in one thigh. If present, there will be a significantly lower BP in the thigh (typically by >30 mm Hg). Sometimes, in case of a lesion proximal to the left subclavian, there may be a significant interarm difference, lower on the left. In addition, there is a significant decrease in intensity of the femoral pulses and a palpable radial-femoral pulse delay.

All patients should have a funduscopic examination to evaluate vascular changes associated with hypertension. The retinal changes are associated with severity of acute and chronic BP elevation. Acute changes can happen quite abruptly (hours to days); these range from arteriolar spasm in most patients with uncontrolled BP to retinal infarcts (exudates) and microvascular rupture (flame hemorrhages) to papilledema once the protection afforded by vasoconstriction is overcome. Chronic changes take much longer to develop and include vascular tortuosity (arteriovenous nicking) due to perivascular fibrosis, followed by progressive arteriolar wall thickening that prevents visualization of the blood column, thus leading to the appearance of copper wiring and then silver wiring. Several studies have demonstrated a relationship between severity of hypertensive retinopathy and risk for LVH and stroke.

An important recent development is the availability of smartphone-based technology that allows the use of a condensing lens coupled with the smartphone's own camera for video and photography of the retina in place of a conventional ophthalmoscope.[148,149] In a study of hypertensive patients seen in the emergency room, use of this technology has resulted in improved identification of abnormal retinal findings (e.g., exudates, hemorrhages, papilledema) by an observer with very little clinical experience compared with conventional bedside opththalmoscopy, while requiring about half the time to complete the examination (74 vs. 130 seconds).[148] The price for these devices (~$450 for the D-EYE camera, D-EYE, Truckee, CA) is higher than conventional ophthalmoscopes (~US$150–US$300) but competitive with that of panoptic hand-held scopes (~US$500).

The CV examination focuses on the identification of volume overload (e.g., jugular venous distention, lung crackles, edema), cardiac enlargement (e.g., deviated cardiac impulse), and the presence of a third or fourth heart sound as markers of impaired left ventricular compliance. We also routinely look for bruits over the carotid arteries because the prevalence of carotid atherosclerosis is increased in patients with hypertension, as well in the abdomen, primarily looking for renal arterial bruits heard over the epigastrium and/or flanks. These bruits are of greater significance if they occur on both systole and diastole. Finally, detailed palpation of the peripheral pulses of the arms and legs is important to look for signs of peripheral arterial disease.

To finish the examination, a focused neurologic examination looks for obvious cranial nerve abnormalities, motor deficits, or speech or gait abnormalities. Any further testing is based on specific symptoms or on focal findings on the screening examination.

BLOOD PRESSURE MEASUREMENT

Because treatment decisions are based largely on BP levels, accurate BP measurement is essential. Cuff-based brachial BP is the method most used to measure BP, typically in the office setting. However, a rapidly growing body of evidence has indicated the value of out-of-office BP methods, such as 24-hour ambulatory BP monitoring (ABPM) and home BP monitoring, as superior methods to evaluate BP burden and evaluate BP-related risk in patients with hypertension.[32,150] Additionally, the most recent guidelines point to much more careful assessment of blood pressure in the office setting.[13]

OFFICE BLOOD PRESSURE MEASUREMENT

Office BP measurement is the time-honored method for the diagnosis and management of hypertension. It is strongly associated with hypertension-related outcomes based on more than 50 years of observational and clinical trial data. Accordingly, guidance provided to clinicians for the diagnosis and treatment of hypertension by most major guidelines is based on office BP values (see Box 46.1).[11] The British National Institute for Health and Care Excellence (NICE)[151] and US Preventive Services Taskforce (https://www.uspreventiveservicestaskforce.org/ Page/Document/UpdateSummaryFinal/high-blood-pressure-in-adults-screening) guidelines recommend ABPM or home BP monitoring as preferred methods for the initial diagnosis of hypertension but still recommend office BP for monitoring treatment. Additionally, the most recent American Heart Association (AHA)– American College of Cardiology (ACC) guidelines on hypertension assessment strongly recommend the following approach for blood pressure assessment in the office setting, as shown in Box 46.1. Attention to measurement technique is essential; Box 46.1 summarizes essential elements of the proper BP measurement technique. Most patients should have their BP measured in the arm in the seated position. In some situations, such as malformations, injuries, or extensive vascular disease of the upper extremities, or when comparing BP levels in the upper and lower extremities, it may be necessary to use thigh measurements with an appropriately sized thigh cuff, which should be obtained with the patient in the prone position to allow the cuff to be at the level of the heart. Mercury sphygmomanometers are now seldom available in clinical practice because of environmental concerns. Aneroid and electronic oscillometric manometers are accurate but should have periodic maintenance (every 12 months) to ensure that they are properly calibrated, as well as whenever poor function is suspected.

A recent development in office BP measurement is the use of an electronic oscillometric device for multiple

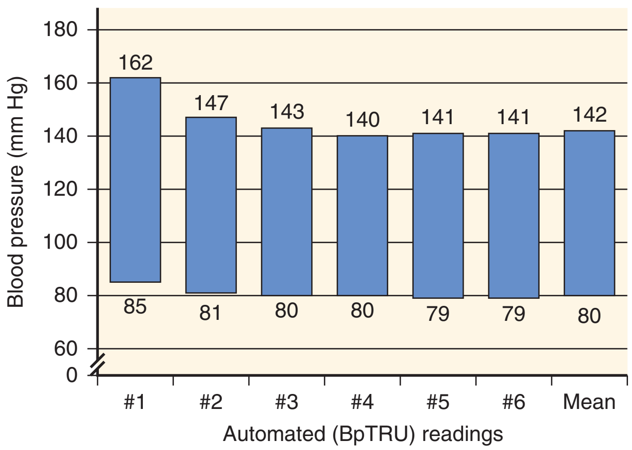


Fig. 46.11 Blood pressure *(BP)* behavior with multiple measurements in the office in 50 patients with hypertension. The first BP reading was taken by the physician. The following five readings were taken by the patient, with only the patient in the examination room. *BpTRU,* Automated blood pressure monitor. (From Myers MG. The great myth of office blood pressure measurement. *J Hypertens.* 2012;30:1894–1898.)

measurements, but with the patient alone in the room for five automated readings. Using this method, the "white coat effect" can be largely eliminated (Fig. 46.11).[152] In addition, this automated approach results in better correlations with ambulatory BP averages and with left ventricular mass than routine office BP,[153,154] but there is a risk of underestimating masked hypertension, which is potentially problematic given the increased risk associated with this condition. The phenomenon of masked hypertension is defined as a clinical condition in which a patient's office BP level is normal but ambulatory or home BP readings are in the hypertensive range. This phenomenon, the opposite of white coat hypertension (WCH), would suggest the necessity for measuring out-of-office BP in those with apparently normal or well-controlled office BP.

ORTHOSTATIC BLOOD PRESSURE MEASUREMENT

Orthostatic hypotension commonly accompanies uncontrolled hypertension, especially among older patients, where it occurs in 8% to 34% of patients.[155] Some guidelines now provide specific recommendations for the measurement of standing BP to screen for orthostatic hypotension in older patients with hypertension, as well as in patients at increased risk for autonomic dysfunction, such as those with diabetes and kidney disease.[156,157]

The frequency of orthostatic hypotension increases with increasing age, the presence of hypertension, and the number of antihypertensive drugs, but whether the type of antihypertensive drug has any specific impact is unclear. The development of orthostatic hypotension is important because it is a risk factor for syncope and falls. Therefore, evaluation for the presence of orthostatic hypotension should be part of the assessment of risks and benefits of drug treatment.

Orthostatic vital signs (heart rate and BP) are best obtained after at least 5 minutes in the supine position followed by immediate assumption of the patient in the standing position, when sequential measurements are taken for up to 3 minutes. The difficulties of following this protocol in a busy clinical practice are recognized, so it is acceptable to compare values in the seated position with those after standing for 1 minute; this approach results in decreased sensitivity for the detection of orthostatic hypotension but is better than no measurement at all.[158] To account for this, it has been proposed that a fall of 15/7 mm Hg be used for the definition of orthostatic hypotension when the test is performed using the seated BP as baseline, as compared with the the generally accepted definition of orthostatic hypotension as a drop in BP of more than 20/10 mm Hg that occurs after 3 minutes of standing.[159] Among patients with supine hypertension, it is recommended that the definition of systolic fall in BP for the diagnosis is a decrease of more than 30 mm Hg because the level of baseline BP is directly proportional to the orthostatic BP fall.

Integration of the heart rate response to changes in BP during orthostasis is important to guide the differential diagnosis and further evaluation of orthostatic hypotension. In the absence of medications with a negative chronotropic effect, the lack of a tachycardic response to orthostatic hypotension is indicative of baroreflex or sympathetic autonomic dysfunction. Patients with a robust tachycardic response, on the other hand, likely have effective arterial volume depletion or excessive vasodilation.

Management of orthostasis is dependent on the pathophysiology of the cause. If a baroreceptor abnormality or autonomic dysfunction is a cause, a general approach using nocturnal clonidine or nitrates to mitigate the hypertension—while requiring compression garments and increased morning sodium intake—has been shown to be beneficial to reduce the symptoms and close the pressure gap.[160]

OFFICE VERSUS OUT-OF-OFFICE BLOOD PRESSURE

Office BP measurement is the time-honored method to evaluate hypertension. It is easy to perform and is widely available at low cost. Home BP is also widely available, although accessibility to low-income patients is still a problem, despite the availability of low-cost devices. ABPM, on the other hand, is less widely available due to costs and limited reimbursement by third-party payers in the United States. Both home BP monitoring protocols and ABPM include larger numbers of readings, thus decreasing variability and improving reproducibility.[161]

In the last 30 years, ABPM and home BP have become accepted as better markers of hypertensive target organ damage and adverse clinical outcomes. ABPM has stronger associations with several measures of LVH, albuminuria, kidney dysfunction, retinal damage, carotid atherosclerosis, and aortic stiffness than office BP, although this is not consistent among studies.[162] Likewise, home BP is a better marker than office BP for LVH and proteinuria, although it is not consistently superior for other measures of target organ damage.

In the assessment of hard CV endpoints, out-of-office BP has consistently outperformed office BP in studies that account for the values observed in the office. In other words, no matter what the office BP, it is the out-of-office BP that decisively drives outcomes. In a systematic review by the NICE clinical guidelines group in the United Kingdom of nine cohort studies comparing ABPM with office BP, ABPM was superior in eight and equal to office BP in one.[163] For home BP, they identified three studies compared with office BP; home BP was superior in two and equal in one. Finally, two studies compared ABPM, home BP, and office BP; of these, one showed superiority of both ABPM and home BP, whereas the other study did not show differences among any of the three methods.

In meta-analyses of studies that evaluated both office and ABPM on outcomes, only ABPM values retained significance.[164,165] Likewise, in the largest home BP cohort study that included simultaneous use of office and home BP to predict CV events and mortality, only home BP remained significantly associated with these adverse outcomes.[166] Similar observations of the superior prognostic performance of out-of-office methods have been made for patients with resistant hypertension, CKD, hemodialysis, and the general population.

In summary, evidence from prospective cohort studies has convincingly demonstrated the superiority of out-of-office BP measurements as predictors of hypertension outcomes. There are several possible explanations for the superiority of out-of-office measurements in outcomes assessment:

1. There is lower variability and better reproducibility afforded by the larger number of readings across a longer period of observation, thus making ABPM/home BP better reflections of so-called BP burden.
2. Home BP and ABPM can detect white coat and masked hypertension (see Fig. 46.12).

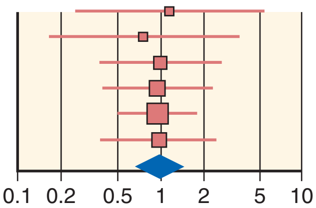
White coat hypertension, or isolated office hypertension, is the occurrence of high BP in the office and normal BP values in the out-of-office environment. It occurs in 20% to 30% of patients with a diagnosis of office hypertension.[167] It has generally been noted that patients with WCH have similar CV outcomes as normotensive individuals.[168] However, data from the large International Database of Home Blood Pressure in Relation to Cardiovascular Outcome have shown a significant increase in fatal and nonfatal CV events among untreated WCH patients diagnosed based on home BP compared with untreated normotensive persons (hazard ratio [HR], 1.42; $P = .02$).[169] Interestingly, treated patients with hypertension who retained a white coat effect had the same overall risk as treated patients whose BP was controlled both in the office and at home (HR, 1.16; $P = .45$). Moreover, an updated meta-analysis of 14 observational ABPM studies revealed an increase in risk of CV events (HR, 1.7) and CV mortality (HR, 2.8) among WCH patients compared with normotensive controls, but no statistically significant increase in stroke or all-cause death. As in previous analyses, sustained normotension was associated with a much larger risk.[170]

In addition, WCH has been associated with an intermediate phenotype between normotension and hypertension as it pertains to left ventricular mass, carotid intima-media thickness, aortic pulse wave velocity, and albuminuria.[169] However, there are no data available to demonstrate that patients with WCH benefit from drug therapy. Therefore, it appears that WCH may not be as benign as previously considered, and patients should be advised on general lifestyle changes to improve BP levels and overall vascular risk, especially as their risk for progressing to sustained hypertension is approximately 40% to 50% after 10 to 11 years of follow-up.[171,172] Masked hypertension, conversely, consists of normal BP in the office but high BP in the ambulatory setting, with an estimated prevalence of 10% to 15% in population studies. It has been

White coat hypertension vs. normotension

Study name	Hazard ratio	Lower limit	Upper limit	Z value	P value
Verdecchia 1994	1.170	0.253	5.402	0.201	0.841
Kario 2001	0.760	0.164	3.529	−0.350	0.726
Fagard 2005	1.000	0.372	2.686	0.000	1.000
Ohkubo 2005	0.950	0.389	2.322	−0.112	0.910
Hansen 2006	0.960	0.500	1.842	−0.123	0.902
Pierdomenico 2008	0.970	0.381	2.468	−0.064	0.949
Summary	0.964	0.654	1.421	−0.186	0.852

Hazard ratio and 95% CI



Masked hypertension vs. normotension

Study name	Hazard ratio	Lower limit	Upper limit	Z value	P value
Bjorklund 2003	2.770	1.149	6.676	2.270	0.023
Fagard 2005	1.650	0.526	5.172	0.859	0.390
Ohkubo 2005	2.560	1.410	4.649	3.088	0.002
Hansen 2006	1.660	1.056	2.610	2.195	0.028
Pierdomenico 2008	2.650	1.177	5.966	2.354	0.019
Summary	2.088	1.550	2.812	4.844	0.000

Hazard ratio and 95% CI

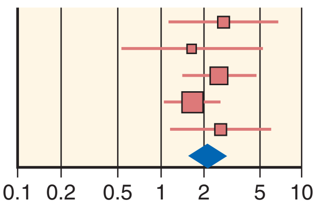


Fig. 46.12 Results of a meta-analysis of the effects of white coat hypertension *(top)* and masked hypertension *(bottom)* on the occurrence of fatal and nonfatal cardiovascular events. There is no statistical difference between white coat hypertension and normotension (hazard ratio, 0.96; $P = .85$), whereas masked hypertension is associated with a 2.09-fold increase in risk ($P < .0001$). *CI,* Confidence interval. (From Pierdomenico SD, Cuccurullo F. Prognostic value of white-coat and masked hypertension diagnosed by ambulatory monitoring in initially untreated subjects: an updated meta-analysis. *Am J Hypertens.* 2011;24:52–58.)

consistently and strongly associated with increased risk for adverse CV endpoints and mortality to a level indistinguishable from that associated with sustained hypertension.[168] The very existence of masked hypertension is troublesome because its identification is only possible with BP measurement in the out-of-office environment. This finding has important public policy implications related to screening, which remain unresolved at this time. However, in a first step to address this issue, the ACC/AHA guidelines have indicated that it may be reasonable to use out-of-office BP to screen for masked hypertension in adults with office BP readings consistently between 120 to 129/80 to 84 mm Hg.

Because WCH and masked hypertension afflict such a substantial number of patients and have a diametrically different impact on outcomes, their identification improves outcome prediction in patients with hypertension.

3. The ability to evaluate BP during sleep was a feature until now restricted to ABPM, although newer home BP monitors can be programmed for activation during sleep. In some, but not all, studies, nighttime BP is a better marker of CV disease than daytime or 24-hour average BP.[173–175] The importance of nighttime BP (compared with daytime levels) appears greater among treated patients, perhaps because antihypertensive treatment, often taken in the morning, might result in better BP control during the day than during the night.[176]

The pattern of BP fluctuation between day and night is also associated with prognosis. The normal circadian BP pattern includes a decrease in BP of approximately 15% to 20% during sleep. Patients who lack this normal BP dip during sleep are called *nondippers* (arbitrarily defined as a sleep BP that falls by less than 10% compared with awake levels); they have increased target organ damage and overall CV risk. In large observational studies, patients whose systolic BP falls by 20% or more during the night have lower fatal and nonfatal CV event rates than those whose BP decreases by less than 20%, and those whose BP does not decrease at all during the night have significantly worse CV outcomes than all other patients.[177]

4. ABPM also provides information on BP variability throughout the day, which may add further prognostic information. Increased BP variability (measured as the standard deviation of BP) has been associated with increased event rates, although these findings are of small magnitude when taken independently from BP values.[178]

Despite these observations, objective evidence demonstrating that outcomes are better when patients are managed using an out-of-office method is lacking. Three randomized clinical trials have compared management of hypertension with office or out-of-office BP, one using 24-hour ABPM[179] and two using home BP.[39,180,181] All these studies have shown that more patients managed with out-of-office methods could have treatment stopped or deescalated, thus resulting in marginal cost savings. However, none of them could demonstrate the superiority of ABPM or home BP in achieving better BP control (the primary outcome of all three trials) or less LVH (evaluated in all studies as a secondary outcome).

CLINICAL USE OF AMBULATORY AND HOME BLOOD PRESSURE MONITORING

ABPM has been in clinical use for almost 50 years. In the United States, problems related to limited reimbursement have significantly limited its expansion compared with other parts of the world. Despite this limitation, there is general agreement on its value in several clinical circumstances, as outlined in Table 46.3.

ABPM is performed, typically, over a period of 24 hours, although it can be extended for longer periods (e.g., 48 hours) to provide information covering more than one wake-sleep cycle or to cover a specific period in detail, such as a 2-day interdialytic period for a patient undergoing hemodialysis. Clinicians should use an independently validated monitor (for a list, refer to www.dableducational.org). A typical measurement interval is every 20 minutes during the daytime (7 AM to 11 PM) and every 30 minutes at night (11 PM to 7 AM), although the frequency and time windows can be adjusted based on clinical needs, such as the need to identify factors such as frequent BP swings and atypical sleep patterns. Patients should keep a log of activities during

Table 46.3 Indications for 24-Hour Ambulatory and Home Blood Pressure Monitoring

Indication	Home BP Monitoring	ABPM	Comment
Identify white coat hypertension.	++	+++	ABPM still the gold standard when patients have home BP values that are borderline (125–135/80–85 mm Hg)
Identify masked hypertension.	++	+++	
Identify true resistant hypertension.	++	+++	
Evaluate borderline office BP values without target organ damage.	++	+++	
Evaluate nocturnal hypertension.	—	+++	
Evaluate labile hypertension.	++	++	Home BP better for infrequent symptoms or paroxysms; ABPM better if frequent within a 24-hr period
Evaluate hypotensive symptoms.	+++	++	
Evaluate autonomic dysfunction.	+	++	Home BP useful to monitor orthostatic hypotension; ABPM useful to quantify supine hypertension and determine overall (average) BP levels
Clinical research (treatment, prognosis)	++	+++	

ABPM, Ambulatory blood pressure monitoring; *BP,* blood pressure.

the day, the time of retiring to bed and waking up, and the time of taking vasoactive medications (if applicable). It is preferred that the periods designated as "night" and "day" reflect the actual periods of sleep and wakefulness obtained from the patient's diary. Most patients tolerate the procedure well, although sometimes sleep is compromised (<10% of cases) and, rarely, patients have excessive bruising or discomfort from the frequent cuff inflations. Up-to-date instructions on how to perform and interpret ABPM studies are available in guideline format from the European Society of Hypertension.

Home BP is performed by the patient in the home (or sometimes work) environment. It is used commonly in clinical practice and is associated with improved adherence to therapy. It has also been used successfully for the treatment self-titration of BP medications and is amenable to telemedicine approaches, in which the patient can upload BP values via telephone or direct entry to an Internet server so that clinicians can inspect the BP logs and make treatment decisions remotely.

Just as with office BP, it is important that the equipment fits the patient well and that measurements are obtained using the same technique as outlined earlier for office BP. Independently validated devices are listed at www.dableducational.org; unfortunately, many of the marketed devices have not been independently validated. The preferred devices use arm cuffs.[182] Finger cuffs are inaccurate and should not be used. Wrist cuffs often provide incorrect readings because of inappropriate technique but, if used correctly, can be convenient and accurate and particularly useful in obese patients.

Smartphone applications are often marketed to obtain biologic information from users, including BP, and it is likely that in the near future their use will become important in the care of hypertensive patients. However, at the present time, none of the available technologies has been adequately validated, and a clinical validation study of the best-sold BP application (Instant Blood Pressure [IBP]) has shown wildly unreliable performance, leading to immediate removal of the application from the market on publication of the article.[183]

To allow management decisions, home BP monitoring is best performed using specific periods of monitoring. For most patients, a BP log obtained over 7 days before each office visit suffices because it has excellent reproducibility.[182] We recommended that the patient obtain readings in duplicate (~1 minute apart), twice daily (in the morning before taking medications and in the evening before dinner). In some situations, more frequent or more prolonged monitoring may be needed. For example, patients with hypotensive symptoms may benefit from BP measurements during the peak action of medications, such as in the mid to late morning or late evening, depending on when medications are taken. Likewise, patients with a labile BP can be monitored more often to capture the overall BP variability better, although we prefer to use ABPM in such patients. As for ABPM, detailed home BP guidelines are available from the European Society of Hypertension[150] and the ACC/AHA.[10]

Normative values for the interpretation of ABPM and home BP results are available based on observed outcomes in longitudinal studies[184] (Table 46.4). For ease of use, these thresholds were matched to specific office BP levels at which the observed rate of CV events was the same, thus allowing clinicians to relate to office values that have historically driven clinical decisions. For ABPM, other measures, such as the nocturnal dip, early morning surge (magnitude of BP rise during the first hours postawakening), BP load (percentage of time BP remains above a certain threshold, such as 140/90 mm Hg during the day and 120/80 mm Hg during the night), and overall BP variability (standard deviation of the 24-hour BP or awake BP), were not studied in relationship to hard outcomes for precise normative results. Table 46.5 also presents general home and ambulatory BP values that mirror office BP values; these represent consensus-based values and are provided to allow the clinician to relate values seen out of the office with the heretofore more commonly used office readings.

Table 46.4 Normative Values for Ambulatory Blood Pressure Monitoring and Home Blood Pressure Monitoring[a]

	Blood Pressure Equivalent to Office Blood Pressure of	
Parameter	**120/80 mm Hg**	**140/90 mm Hg**
24-Hour Ambulatory BP Monitoring		
24-hour BP	117/74 mm Hg	131/79 mm Hg
Awake BP	122/79 mm Hg	138/86 mm Hg
Sleep BP	101/65 mm Hg	120/71 mm Hg
Home BP Monitoring		
Average BP[b]	121/78 mm Hg	133/82 mm Hg

[a]Based on clinical outcomes.
[b]Average of all values during the monitoring period, usually 7 days.
Normative data based on equivalent of cardiovascular event rates observed at each level of office BP.
ABPM, Ambulatory blood pressure monitoring; *BP*, blood pressure.
Data from Richards EM, Pepine CJ, Raizada MK, Kim S. The gut, its microbiome, and hypertension. *Curr Hypertens Rep.* 2017;19(4):36; and Kikuya M, Hansen TW, Thijs L, et al. Diagnostic thresholds for ambulatory blood pressure monitoring based on 10-year cardiovascular risk. *Circulation.* 2007;115;2145–2152.

INTEGRATING OUT-OF-OFFICE BLOOD PRESSURE INTO CLINICAL PRACTICE

All current hypertension guidelines recognize the value of out-of-office BP in the diagnosis of hypertension; the UK NICE guidelines, the US Preventive Services Taskforce, and the ACC/AHA guidelines formally recommend its use to confirm the diagnosis of hypertension in patients with elevated office BP prior to initiating treatment. The ACC/AHA guidelines also recommend the use of out-of-office BP to evaluate patients who are receiving treatment for hypertension but remain above goal in the office, with the explicit caveat that the recommendation is on the basis of expert opinion. A finding to support this recommendation is the high prevalence (~40%–51%) of a white coat effect in patients with

Table 46.5 Consensus-Based Home and Ambulatory Blood Pressure Values[a]

Clinic	Home BP	Daytime BP	Nighttime	24-hour BP
120/80	120/80	120/80	100/65	115/75
130/80	130/80	130/80	110/65	125/75
140/90	135/85	135/85	120/70	130/80
160/100	145/90	145/90	140/85	145/90

[a]And their equivalent office values.
BP, Blood pressure.
Data from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324.

Box 46.2 Proper Method of Measurement and Integration of Out-of-Office Blood Pressure

Method

1. Make sure BP device is validated by medical personnel (nurse)-annually
2. Measure BP in morning before breakfast (maximal risk for CV events between 4-10 AM)
3. Method of measurement: (1) sit in chair with back firmly supported; (2) place appropriate sized BP cuff on arm and rest arm on comfortable surface at heart level; (3) rest for 5 minutes with no distractions (e.g., TV, radio, computer, phone; (4) obtain three readings 1 minute apart and average the last two readings.
4. Once BP is controlled and stabilized, do this on average four or five times a month.

Integration

1. Use the average range of BPs at home and compare with office readings. If home BPs are at least 10 or more mm Hg higher than office BPs, then a white coat effect is present.
2. Tailoring therapy to home BPs rather than office BPs has resulted in fewer medication side effects, especially hypotension.

BP, Blood pressure.

resistant hypertension. Another caveat is that patients with an office BP above 160/100 mm Hg do not need confirmation and should be treated.

We are also in agreement with the recommendation to use out-of-office BP to guide treatment, which reflects our clinical practice of systematically using this approach. Box 46.2 summarizes the steps involved in the integration of out-of-office BP in the diagnosis and management of hypertension using the proposed ACC/AHA BP thresholds for diagnosis and treatment.

LABORATORY AND OTHER COMPLEMENTARY TESTS

Similar to the history and physical examination, laboratory tests, imaging, and other complementary tests also focus on the evaluation of comorbid conditions, established target organ damage, and possible secondary causes. In the absence of worrisome signs or symptoms during the initial evaluation, the clinician should obtain a basic set of tests, including renal function, levels of electrolytes, calcium, glucose, and hemoglobin, lipid profile, urinalysis, and electrocardiogram (Table 46.6).

Further testing may be required in case any of these initial test results are abnormal or there are specific symptoms or physical findings suggesting a diagnosis (see "Secondary Hypertension"). Likewise, patients who are resistant to treatment during follow-up have higher rates of secondary causes of hypertension—in particular, sleep apnea, hyperaldosteronism, and renovascular disease—thus deserving a more dedicated search for secondary causes in their evaluation.

ECHOCARDIOGRAPHY

LVH is the most common target organ damage in hypertension and is independently associated with a worse prognosis, marked by increased risk for CV events (coronary, cerebrovascular), heart failure, and death.[185] The electrocardiogram is very specific but insensitive for the detection of LVH. Not surprisingly, the prevalence of LVH among patients with hypertension is only approximately 18% based on electrocardiographic criteria, whereas this number increases to approximately 40% when more sensitive echocardiographic criteria are used. The echocardiogram also provides information on left ventricular diastolic function, which is often impaired early in the course of hypertensive heart disease and does not require the presence of LVH. Finally, it allows the assessment of left ventricular systolic dysfunction, which is uncommonly present in hypertension (~4%) but is associated with a worse prognosis. Even though echocardiography is not recommended as a routine test in patients with hypertension, it often provides important information to help guide treatment, such as defining the need to initiate or escalate treatment in patients with borderline office or ambulatory BP levels.

EVALUATION OF SODIUM AND POTASSIUM INTAKE

Because of the importance of sodium and potassium as dietary interventions in hypertension, it is often useful to quantify intake objectively. Dietary recall is the method used most often in clinical practice; however, it is often problematic because many patients have difficulty defining portions. In situations where detailed knowledge of sodium and potassium intake is important to management, our practice is to obtain a 24-hour urine collection to evaluate sodium and potassium on a stable diet. These ions are measured in milliequivalents per day and then converted to dietary target in milligrams per day (1 mEq of sodium = 23 mg of sodium or 58 mg of salt as NaCl; 1 mEq of potassium = 39 mg of potassium). A diuretic can be maintained as long as the dose has been stable over time. One must recognize that sodium excretion may follow a circaseptan rhythm[60] and may therefore be imprecise on a single 24-hour collection, but it is still

Table 46.6 Initial Laboratory Evaluation of the Hypertensive Patient[a]

Test	Clinical Usefulness
Serum creatinine (and estimated glomerular filtration rate)	Assessment of renal function. Identifies parenchymal kidney disease as a possible secondary cause, as well as established TOD.
Serum potassium	Low potassium (of renal origin) suggests mineralocorticoid excess (primary or secondary), glucocorticoid excess, Liddle syndrome; high potassium with normal renal function suggests Gordon syndrome; low levels raise caution about the use of thiazides and loop diuretics; high levels preclude the use of ACE inhibitors, ARBs, renin inhibitors, and potassium-sparing diuretics.
Serum sodium	If high, suggests primary aldosteronism; if low, alerts to the need to avoid thiazide diuretics.
Serum bicarbonate	If high, suggests aldosterone excess (primary or secondary); if low with normal renal function, suggests Gordon syndrome (with high potassium) or primary hyperparathyroidism (with high calcium).
Serum calcium	If high, suggests primary hyperparathyroidism.
Serum glucose	Identifies prediabetes or diabetes; in the appropriate setting, suggests glucocorticoid excess, pheochromocytoma, or acromegaly.
Lipid profile	Identifies hyperlipidemia.
Hemoglobin, hematocrit	If high, in the absence of other hematologic abnormalities or underlying lung disease, suggests sleep apnea.
Urinalysis[b]	Proteinuria and hematuria identify a possible secondary cause (glomerulonephritis); proteinuria can also be a marker of TOD.
Electrocardiography	Identifies left ventricular hypertrophy, old myocardial infarction, or other ischemic changes; identifies conduction abnormalities that may preclude the use of beta-blockers or nondihydropyridine CCBs.

[a]To investigate the presence of comorbid conditions, secondary causes, or established target organ damage.

[b]Some organizations recommend screening microalbuminuria as a more sensitive tool to identify early renal injury. The most recent guidelines do not recommend BUN measurement alone, whereas some recommend the measurement of uric acid (as a marker of cardiovascular risk) and thyroid-stimulating hormone (to more specifically screen for hypo- and hyperthyroidism).

ACE, Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *BUN,* blood urea nitrogen; *CCB,* calcium channel blocker; *TOD,* target organ damage.

valuable as a general guide to allow more precise dietary advice to patients.

RENIN PROFILING

The evaluation of plasma renin activity has been proposed by Laragh as an empiric method for the evaluation and treatment of hypertension.[186] The premise for this approach is mechanistic; patients with high plasma renin activity levels (>0.65 ng/mL/h, and particularly >6.5 ng/mL/h) have vasoconstriction mediated by the RAAS as the primary operative mechanism of hypertension, whereas those with suppressed plasma renin activity levels (<0.65 ng/mL/h) are volume-overloaded. Accordingly, patients with high levels of plasma renin activity are treated with blockers of the RAAS (ACE inhibitors, angiotensin receptor antagonists, renin inhibitors, beta-blockers), and those with low levels of renin are treated with diuretics (including aldosterone antagonists), CCBs, or alpha-blockers. The approach not only includes using drugs that address the underlying pathophysiology directly but also proposes removal of drugs from the opposite group because there have been reports of paradoxic BP elevations in such cases.[187] A case series has reported streamlined drug regimens and improved BP control in patients with resistant hypertension, and a small randomized trial of renin-guided therapy versus conventional therapy has yielded greater systolic BP lowering with the renin-guided system (−29 vs. −19 mm Hg; $P = .03$).[188] It is reasonable to entertain renin profiling, especially in patients who do not respond to initial therapy. In such cases, renin measurement, along with plasma aldosterone measurement, will also be useful to rule out primary hyperaldosteronism.

SYSTEMIC HEMODYNAMICS AND EXTRACELLULAR FLUID VOLUME

An alternative to the renin-profiling approach is to measure systemic hemodynamics and extracellular fluid volume noninvasively. Such measurements can be achieved with several methodologies, but impedance cardiography has the advantage of simultaneously obtaining both volume (thoracic fluid content) and hemodynamic (CO, SVR) data. This approach has been used in patients with resistant hypertension, with some success in two randomized trials.[189,190] In one study, patients managed using the hemodynamic approach achieved better BP control while receiving more diuretics, whereas in the other, control was also better but was not associated with the specific extra use of any particular drug class. Direct measurement of volume excess and hemodynamics and availability of the information at the point of care make this methodology more attractive than renin profiling. However, relatively high costs associated with the technology and lack of reimbursement in the current health care environment make the wide use of this method out of reach for most physicians and their patients.

SECONDARY HYPERTENSION

Secondary (or remediable) hypertension is elevated BP due to a specific cause.[8,32] For several reasons, it should always be considered in every newly diagnosed or referred patient with hypertension, especially those with a history of hypokalemia. First, the proportion of patients with secondary hypertension who can eventually be cured is far greater than those with primary hypertension. This is particularly important in young hypertensive people, especially in children and adolescents,[27] who are more likely to harbor nearly all types of secondary hypertension,with the important exception of atherosclerotic renovascular hypertension. For such people, and those with an otherwise long life expectancy, the cost of diagnosis and treatment (and even cure) of secondary hypertension may be less than the cost of chronic medical therapy, including drugs, office visits, and laboratory monitoring. Finally, it is intellectually appealing to consider the possibility of cure for patients with hypertension, which not only keeps the health care provider mentally awake, but also may help avoid failure to diagnose disease in patients who actually harbor a secondary cause, but its presence is never contemplated or formally evaluated. The most common cause of secondary hypertension is primary hyperaldosteronism.[191,192]

RISK FACTORS AND EPIDEMIOLOGY

In general clinical practice, there is a higher probability of secondary hypertension in patients: (1) with higher levels of untreated blood pressure (except in some cases of primary hyperaldosteronism); (2) with characteristic physical signs (for details, see later); (3) with refractory hypertension (now more commonly called "resistant hypertension," defined as a patient who has had all secondary causes of hypertension excluded, as well as white coat and masked hypertension, and is confirmed to be adherent with their medications (despite proper doses of three appropriately chosen antihypertensive drugs, one of which is a diuretic, and has a persistent office blood pressure ≥140/90 mm Hg); or (4) seen in tertiary referral centers (largely caused by referral bias). Because of these factors, the prevalence of secondary hypertension varies widely.

The largest prospective study reported from a primary care setting evaluated 1020 consecutive patients with hypertension in Yokohama, Japan. The authors found that 9.1% of patients had a secondary cause, which was primary hyperaldosteronism in 6%, Cushing syndrome (full-blown in 1% and preclinical in 1%), pheochromocytoma (0.6%), and renovascular hypertension (0.5%).[193]

The largest series of consecutive patients evaluated by a whole-day protocol from 1976 to 1994 in a tertiary referral center was from Syracuse, New York. In this study, 10.1% of 4429 patients with hypertension had secondary hypertension—3.1% with renovascular hypertension, 1.4% with primary hyperaldosteronism, 0.5% with Cushing syndrome, 0.3% with pheochromocytoma, 3% with primary hypothyroidism, and 1.8% with hypertension attributed to CKD.[194] Later series from worldwide studies have suggested that primary hyperaldosteronism (especially due to sleep-disordered breathing and obstructive sleep apnea) is far more common than before the year 2000, with an average prevalence of approximately 10% to 11.2% in population-based studies; its prevalence is about doubled in resistant hypertension.[149] This change in the most common cause of secondary hypertension represents a marked reduction in smoking and much higher use of cholesterol-lowering medications—hence, a major reduction in the incidence of atherosclerotic arterial disease.

EVALUATION OF SECONDARY HYPERTENSION

Hypertension guidelines and a great deal of clinical experience have suggested a more detailed evaluation for secondary causes in patients with hypertension younger than 30 years who have no family history of hypertension. Additionally, those older than 55 years, with new-onset hypertension, sudden worsening of BP control (despite years of previously controlled BP levels), recurrent flash pulmonary edema, an abdominal bruit (especially with a louder diastolic component), and sudden increases of the serum creatinine level by 30% or more after an RAAS blocker, should prompt evaluation for secondary causes of hypertension. Such patients have a higher pretest probability of renovascular hypertension.

The initial set of laboratory tests recommended for newly diagnosed patients with hypertension includes serum levels of urea and creatinine and a urinalysis, which is generally sufficient for identifying patients with underlying intrinsic renal disease, even if the 3-month criterion for CKD is not yet met (see Table 46.6). Hyperthyroidism or hypothyroidism can be a cause of hypertension, but the presence of either is commonly detected by a serum ultrasensitive thyroid-stimulating hormone (thyrotropin) level.

Sometimes the demographic and clinical features of the patient help direct the search for a secondary cause. Fibromuscular dysplasia is much more common in young white women, whereas atherosclerotic renovascular disease is more common in older smokers (both current and former). Some symptoms, when elicited by a careful history, are also quite suggestive, although incompletely sensitive and not very specific.

Typically, paroxysmal spells occur in approximately 25% to 30% of patients with pheochromocytoma; the associated symptoms are variable across patients but are commonly experienced repetitively in a given patient. Sadly, the specificity of these paroxysms, even when they include headache, sweating, and elevated BP levels, is less than 5% in most large series. Similarly, the classic symptoms of large muscle weakness (particularly when rising from a chair or climbing stairs) reported with Cushing syndrome, or lower extremity weakness and leg cramps reported in primary hyperaldosteronism and attributed to hypokalemia, are uncommon in today's literature and patients. Given the relatively low prevalence of secondary hypertension, the decision to undertake a formal evaluation for specific causes can (and should) be individualized.

EXACERBATION OF PREEXISTING HYPERTENSION ATTRIBUTED TO LIFESTYLE FACTORS

Although generally not considered in most discussions of secondary (or remediable) hypertension, BP can be influenced by prescription or nonprescription medications or both,[195] excessive dietary sodium intake,[196] body weight and obesity, and excessive alcohol intake.[8] The health care provider or patient may not immediately appreciate some of these factors. Appropriate attention to these issues can result in improved BP profiles and a better prognosis.[197]

Modification of these factors is the cornerstone of therapy for primary hypertension and can mimic secondary hypertension. Of these factors, the most common issues relate to excessive sodium intake, poor sleep hygiene (i.e., getting <6 hours of uninterrupted sleep a night),[198] excessive caffeine or other stimulants, and use of nonsteroidal antiinflammatory drugs (NSAIDs).

INTRINSIC KIDNEY DISEASE

Hypertension can be both a cause and consequence of CKD—that is, estimated GFR (eGFR) is less than 60 mL/min/1.73 m^2. It is often difficult to discern which occurred first when a patient presents initially with both, but the screening and diagnostic processes are identical to those used for each individually. CKD is currently diagnosed using the 2012 Kidney Disease: Improving Global Outcomes (KDIGO) criteria from the National Kidney Foundation: persistent (≥3 months) evidence of kidney damage (e.g., proteinuria, abnormal urinary sediment, abnormal blood or urine chemistry levels, imaging studies, or biopsy), but primarily based on the eGFR.[199] Although it is possible to have CKD with eGFR greater than 60 mL/min/1.73 m^2, most authorities recommend this threshold for common use. Management strategies for hypertension due to CKD are also identical to those used in primary hypertension, except that doses and frequency of antihypertensive (and other) medications normally cleared by the kidney are decreased inversely to the eGFR. Although most antihypertensive drugs do not need dose adjustments in stage 3b or higher CKD, some agents that affect the RAAS mechanistically should theoretically be reduced. Renally excreted beta-blockers (e.g., atenolol, metoprolol, bisoprolol, nadolol, acebutolol) and all ACE inhibitors, except fosinopril and trandolapril, are reduced in dose or dosing frequency, but no serious adverse effects (other than possibly hyperkalemia) have been reported, even if no adjustment of dosing is performed. Also note that losartan and valsartan, normally twice-daily agents from the angiotensin receptor blocker (ARB) class when kidney function is normal, should be dosed once daily in those with an eGFR less than 60 mL/min/1.73 m^2.

Restriction of dietary protein intake had been recommended in the distant past, based on several small trials (primarily in Australia), but had marginal success in the Modification of Diet in Renal Disease trial.[200] It is usually challenging to carry out effectively, even in tertiary centers with a dedicated renal nutritionist. Dietary sodium restriction, although a somewhat lesser challenge, has benefits in patients with CKD and hypertension, not only to lower BP, but also to reduce urinary protein (and albumin) excretion.[201]

PRIMARY HYPERALDOSTERONISM

This form of secondary hypertension has been increasing in prevalence worldwide over the last 25 years[195,202] and is generally due to one of six subtypes: (1) an aldosterone-producing (Conn) adenoma, nearly always in one adrenal gland (~35% of cases); (2) bilateral adrenal hyperplasia (also known as *idiopathic primary hyperaldosteronism*, ~60% of cases); (3) primary (or unilateral) adrenal hyperplasia (~2% of cases); (4) aldosterone-producing adrenal carcinoma (~35 cases in the world literature); (5) familial hyperaldosteronism, which takes one of two forms—glucocorticoid-suppressible hyperaldosteronism, due to a chimeric chromosome 8, in which the 5′-regulatory sequence for corticotropin responsiveness of 11β-hydroxylase is fused to the enzyme coding sequence for aldosterone synthase (<1% of cases), or familial occurrences of either an aldosterone-producing adenoma or bilateral adrenal hyperplasia (<2% of cases); or (6) ectopic production of aldosterone by an adenoma or carcinoma outside the adrenal gland (<0.1% of cases). In addition, obstructive sleep apnea and sleep-disordered breathing also cause hyperaldosteronism. This is classically described as secondary hyperaldosteronism, but its evaluation and medical treatment are often quite similar to that of bilateral adrenal hyperplasia.

The prevalence of primary hyperaldosteronism depends on where and how one looks and is controversial. Some referral centers have reported a prevalence of hyperaldosteronism related to sleep apnea at approximately 20%, similar to the original prevalence of aldosterone-secreting adenomas estimated by Conn in the 1950s. In large population-based studies, the prevalence of primary hyperaldosteronism has been estimated at approximately 10% to 11.2% of hypertensives. The condition appears to be more common in people with higher levels of BP (2% for BP levels of 140–159/90–99 mm Hg, 8% for BP levels of 160–179/100–109 mm Hg, and 13% for BP levels > 180/110 mm Hg), treatment-resistant hypertension (17%–23% in several series), patients with hypertension with spontaneous or diuretic-associated hypokalemia, and hypertension and a serendipitously discovered adrenal mass (1%–10%).

In the last millennium, hypokalemia was thought to be very common—if not nearly universal—among patients with primary hyperaldosteronism, particularly if provoked by diuretic therapy. Today, however, more afflicted patients have eukalemia than hypokalemia, although sometimes more severe cases have weakness, muscle cramps, and even periodic paralysis. Patients with primary hyperaldosteronism experience higher CV morbidity and mortality than age-, gender-, and BP-matched patients with primary hypertension.[203]

Screening for primary hyperaldosteronism is most efficiently performed in potassium-repleted patients using the ratio of plasma aldosterone concentration to plasma renin activity (ARR; Table 46.7). The ARR can be affected by many factors, including antihypertensive drug therapy, dietary sodium restriction, posture, time of day, and sample handling (see Table 46.7). Most authorities recommend sustained-release verapamil, hydralazine, and peripheral α_1-adrenoceptor antagonists as medications that have little, if any, effect on the ARR. The likelihood of a false-positive ARR is increased by a low plasma renin activity (e.g., <0.5 ng of Ang II/mL per hour), so some investigators require the plasma aldosterone concentration to be above a given threshold (e.g., >15 ng/dL), for the screening to be considered positive, but levels between 12 and 15 ng/dL need to be considered individually, because some patients with proven aldosteronism have values in this range. Confirmation of primary aldosteronism in patients with aldosterone levels below 10 ng/dL, on the other hand, are rare. The most common cutoff value for an ARR that usually leads to further investigation is 30 (when the aldosterone level is measured in nanograms per deciliter and plasma renin activity in nanograms of Ang II per milliliter per hour), but higher thresholds lead to more falsely negative tests.

Clinical practice guidelines from the Endocrine Society have recommended one of four confirmatory tests before proceeding to an imaging study because of the expense and radiation involved in the latter. There have been only a few comparative studies of these four tests; they seem to have similar performance characteristics (75%–90% sensitivity,

Table 46.7 Factors That May Cause False-Positive or False-Negative Results of the Aldosterone/Renin Ratio

Factor	False-Positives	False-Negatives
Aldosterone relatively high	Potassium loading	
Renin relatively low	Beta-blockers; central alpha-adrenoceptor agonists; direct renin inhibitors; nonsteroidal antiinflammatory drugs; chronic kidney disease; sodium loading	
Aldosterone relatively low		Hypokalemia
Renin relatively high		Diuretics; ACE inhibitors, angiotensin receptor blockers; calcium channel blockers (dihydropyridines); acute sodium depletion

ACE, Angiotensin-converting enzyme.

80%–100% specificity). Cost, patient preference, local experience, local laboratory methods, and insurance reimbursement all factor into which confirmatory test is chosen.

The traditional saline-loading test (2 L infused over 4 hours) is confirmatory if the postinfusion plasma aldosterone concentration is greater than 10 ng/mL. Patients with aldoserone concentrations from 5 to 10 ng/mL are considered indeterminate and should be restested. Needless to say, intravenous saline is not often recommended for patients with heart failure, CKD, or uncontrolled hypertension.

Many centers have reported success with an oral sodium-loading protocol, which involves liberalizing sodium intake to approximately 6 g/day for 3 to 5 days and then assaying 24-hour urine collections for sodium (to ensure loading) and aldosterone content. The test is considered positive if the urinary aldosterone excretion is greater than 12 to 14 μg/day, but oral sodium loading can be as problematic in some patients as intravenous saline.

The fludrocortisone suppression test involves giving 0.1 mg of fludrocortisone every 6 hours for 4 days and then assaying the plasma aldosterone concentration when the patient is standing upright. It is considered confirmatory if the concentration is greater than 6 ng/dL, and plasma renin activity and serum cortisol levels are low. Execution of the test may be difficult for patients who have a long journey to the office or who are nonadherent.

Finally, the captopril challenge test is performed by assaying the plasma aldosterone concentration before and 1 and 2 hours after administration of 25 to 50 mg of oral captopril. It is considered confirmatory if the plasma aldosterone concentration remains elevated (and unchanged from baseline); many false-negative and equivocal captopril challenge test results have been reported, although several Japanese series have shown excellent results with this method.

After the diagnosis of primary aldosteronism is confirmed, a computed tomography (CT) scan of the adrenals is undertaken, which is useful in detecting large masses that might be adrenal carcinomas. Adrenal carcinomas typically are larger (>4 cm in diameter), an inhomogeneous character (often with internal hemorrhage), internal calcifications (in ~40%), and irregular borders (often because of micrometastases) and show enhancement after intravenous contrast medium is administered. Aldosterone-producing adenomas are usually small (<2 cm in diameter), hypodense, unilateral nodules. Idiopathic hyperaldosteronism usually has normal-appearing adrenal glands, but sometimes nodular changes and/or general enlargement are visible in one or both adrenals. Magnetic resonance imaging (MRI) is no better at detecting these abnormalities than CT, which is usually less expensive. Both techniques often detect nonfunctioning nodules, especially in older patients. At some centers, patients with hypertension with proven primary hyperaldosteronism who are younger than 40 years with a single typical hypodense nodule in one adrenal gland are offered an adrenalectomy directly.

Because CT scans identify unilateral adrenal disease with a sensitivity of only 78% and specificity of only 75%, the Endocrine Society has recommended adrenal venous sampling for most surgical candidates.[204] Despite being invasive, expensive, technically challenging, and potentially dangerous, and requiring an experienced and well-coordinated team, it has sensitivity and specificity of 95% and 100%, respectively, for detecting unilateral aldosterone production. It is commonly performed at 8 AM, with continuous cosyntropin administration, and simultaneous adrenal vein cortisol level measurement. Most centers use a 4 : 1 cutoff value of the cortisol-corrected aldosterone ratio (i.e., the ratio between the aldosterone/cortisol ratios on each side) to define a positive lateralization.

Several older (some prefer classic) tests remain available for the presumably rare circumstance in which adrenal venous sampling was technically unsuccessful or nondiagnostic. These are usually much less expensive and therefore can sometimes be preauthorized when financial and other roadblocks can affect adrenal venous sampling. The postural stimulation test, developed in the 1970s, depends on the propensity of patients with an aldosterone-producing adenoma to retain diurnal variation in the plasma aldosterone concentration, whereas those with idiopathic hyperaldosteronism often show an increased sensitivity to small increases in Ang II levels (e.g., after standing). In some centers, this test is performed by drawing two additional blood samples (before and after standing) at the conclusion of the intravenous saline infusion test. In a review of the Mayo Clinic experience from the last millennium in 246 patients with surgically proven adenomas, the test was only 85% accurate. There are also other tests. For example, iodocholesterol scintigraphy was developed at the University of Michigan in the 1970s; its sensitivity correlates directly with tumor size, so it is not very helpful in discerning microadenomas from bilateral hyperplasia. Also, serum 18-hydroxycorticosterone levels, typically measured with the patient in the recumbent position at 8 AM, are often higher than 100 ng/dL in patients with an adenoma, but the opposite is true in patients with bilateral hyperplasia, so blood was traditionally taken for this analyte before the 2-L saline infusion test. However, the accuracy of this test has been found to be even lower than the postural stimulation test.

Genetic testing for familial forms of primary hyperaldosteronism is recommended for those who are younger than 20 years at diagnosis and in those with a family history of primary aldosteronism or stroke at an early age (typically, <30 years). This strategy was successful in approximately half of large, qualifying, unrelated cohorts. Genetic testing by a Southern blot or long polymerase chain reaction assay is both sensitive and specific for glucocorticoid-remediable hyperaldosteronism (familial hyperaldosteronism, type I, the most common monogenetic cause of hypertension). Such testing is expensive, so many managed care organizations will not pay for the test unless an appropriate response (in both BP and plasma aldosterone concentration) is seen after weeks of empiric glucocorticoid administration. Familial hyperaldosteronism type II is genetically heterogeneous, despite being autosomal dominant in most affected cases. Genetic testing is not yet available for this more common type of familial hyperaldosteronism, so the diagnosis is primarily clinical, based on the biochemical findings and the pedigree.

Laparoscopic procedures for unilateral adrenalectomy have improved to the point that most patients with adrenal venous sampling–proven hyperaldosteronism have shorter hospital stays, fewer complications, and lower costs than those who have undergone an open procedure. Although nearly all return to eukalemia, hypertension is cured (i.e., follow-up BP levels of <140/90 mm Hg without antihypertensive drug therapy) in only approximately 50%. Cure is more likely in younger people, those with a short duration of hypertension, prior BP control with only one or two agents, and a pedigree that includes fewer than two first-degree relatives with hypertension. Typically, plasma aldosterone concentration and plasma renin activity are measured shortly after successful surgery, and potassium supplementation and aldosterone antagonists are discontinued. Intravenous saline is often required because the remaining adrenal gland needs to recover its normal function, which may take a few weeks. The nonsurgical option for patients with idiopathic hyperaldosteronism is spironolactone, which has had significantly better efficacy than its successor, eplerenone, in an international randomized clinical trial in hypertensive subjects with primary aldosteronism. Most physicians use dexamethasone or prednisone at bedtime (over twice-daily hydrocortisone) for glucocorticoid-remediable hyperaldosteronism, but the doses are kept low to avoid iatrogenic Cushing syndrome (see later).

HYPERALDOSTERONISM ASSOCIATED WITH SLEEP-DISORDERED BREATHING

Several reports have highlighted this association, which is thought to account for approximately 20% of resistant hypertension and typically responds very nicely to selective aldosterone antagonists. Polysomnography is the gold standard test for obstructive sleep apnea diagnosis, but requires overnight evaluation and is expensive. The Berlin questionnaire, which includes questions about snoring, daytime somnolence, body mass index, and hypertension, is a brief and validated screening tool that identifies persons in the community who are at high risk for obstructive sleep apnea. The ratio of serum aldosterone to plasma renin activity is used most often to diagnose the condition. A therapeutic trial of spironolactone may be warranted, especially if the Berlin questionnaire or the sleep study is sufficiently suggestive.

APPARENT MINERALOCORTICOID EXCESS STATES

Several unusual diagnoses manifest with a similar set of signs and symptoms (especially hypokalemia and often other Cushingoid features) that result from an apparent mineralocorticoid excess. These are most easily distinguished from the several types of hyperaldosteronism by the suppressed plasma aldosterone level.

The most common type, classic congenital adrenal hyperplasia, is due to one of several autosomal recessive genetic deficiencies in enzymes involved in adrenal steroidogenesis, usually resulting in mineralocorticoid and androgen excess. Although most common in infants and children, some patients (especially those with milder loss-of-function mutations) go undiagnosed until adulthood. The most common deficiency, 21-hydroxylase, accounts for approximately 95% of cases and is often discovered by universal screening programs, particularly for female newborns with ambiguous genitalia. If left untreated, approximately 75% of such babies suffer salt wasting, failure to thrive, hyponatremia, hypovolemia, shock, and death.

Hypertension occurs in approximately two-thirds of patients with congenital adrenal hyperplasia due to either 11β-hydroxylase deficiency (~5% of cases of congenital adrenal hyperplasia and a prevalence of approximately 1 : 100,000 in whites) or 17α-hydroxylase deficiency, which is rare. More than 40 mutations have been identified that lead to 11β-hydroxylase deficiency, which results in high circulating levels of deoxycorticosterone and 11-deoxycortisol, with increased production of adrenal androgens. Thus girls present in infancy or childhood with hypertension, hypokalemia, acne, hirsutism, and virilization, whereas boys present with pseudoprecocious puberty. Because 17α-hydroxylase is required for the production of both cortisol and sex steroids, its deficiency can delay puberty and present as pseudohermaphroditism or phenotypic female features (in genetic 45,XY boys), or as primary amenorrhea in girls. After appropriate diagnosis, typically by determination of levels of steroid precursors in serum, patients with congenital adrenal hyperplasia receive supplementation with glucocorticoids, which suppress corticotropin secretion and reduce the signs and symptoms of mineralocorticoid excess. Long-term care of adults with congenital adrenal hyperplasia is challenging.

Very rare causes of apparent mineralocorticoid excess include deoxycorticosterone-producing tumors, which are usually quite large and often malignant, primary cortisol resistance, or 11β-hydroxysteroid dehydrogenase deficiency, of which approximately 50 cases worldwide are congenital. Most are acquired and associated with imported licorice or licorice-flavored chewing tobacco. Some would also include Liddle syndrome—hypertension, hypokalemia, and inappropriate kaliuresis, associated with low plasma aldosterone and renin activity—due to an autosomal dominant condition that results in mutations in the beta or gamma subunits of the renal amiloride-sensitive ENaC.

RENOVASCULAR HYPERTENSION

Hypertension due to renal artery stenosis (fibromuscular dysplasia or atherosclerotic disease) has been studied extensively since the pioneering work of Harry Goldblatt (see Chapter 47).

The probability of renovascular hypertension can be calculated based on clinical characteristics in a given patient, which eliminates the need for a screening test in most cases. The choice among the several screening tests for renovascular hypertension is usually based on patient and physician preference, local expertise, and a favorable decision about prior authorization for the test from insurance companies. This has become less common since the publication of several outcome studies that showed no significant benefit to renal angioplasty (usually with stenting) over medical management in atherosclerotic renal artery disease.[160]

PHEOCHROMOCYTOMA

Although chromaffin tumors (pheochromocytomas and paragangliomas) that secrete catecholamines are rare (estimated incidence, two to eight cases/million per year), their diagnosis and management are important because of the following: (1) they might cause fatal hypertensive crises, despite otherwise appropriate treatment; (2) probably more than 10% are metastatic at diagnosis; (3) specific therapy can be curative; and (4) a much higher proportion of cases are more familial than once thought.[205]

Catecholamine-secreting tumors are found in 0.2% to 0.6% of patients with hypertension, may be more common in hypertensive children (1.7%), and are too often found incidentally, or worse, only at autopsy. The clinical presentation of patients with these tumors is variable because symptoms may occur constantly or in paroxysms. The classic triad of headache, sweating attacks, and hypertension was said to be present in 95% of patients in one large French series, but most centers reported having to see more than 100 such patients on referral before one is positively diagnosed.

The differential diagnosis includes many disorders, some of which are functional or factitious, so a high index of suspicion is required, even when faced with common admitting conditions (e.g., heart failure, which can be precipitated, if not exacerbated, by a functioning tumor). Some familial conditions that include pheochromocytoma have characteristic physical signs (e.g., café au lait spots and neurofibromas, retinal hemangiomas, port wine stains, subungual fibromas, ash leaf or shagreen patches, adenoma sebaceum, marfanoid body habitus) that provide clues to the underlying syndromes.[205]

There seem to be more exceptions to pheochromocytoma than with many other conditions: approximately 10% of such tumors are extraadrenal, multiple or bilateral, recurrent (after surgical extirpation), discovered as incidentalomas, or in children. More than 10% are likely familial or metastatic at presentation, and both of these have increased in the last 30 years. Although approximately 90% of pheochromocytomas are found in or in close proximity to the adrenal gland, paragangliomas can occur anywhere along the sympathetic ganglia, but most commonly in or near the organ of Zuckerkandl (at the aortic bifurcation) or near the bladder, which gives rise to rather unusual symptoms of so-called micturition headache, syncope, or similar problems.

Pheochromocytomas play important parts in the multiple endocrine neoplasia (MEN) syndromes, especially MEN2A (pheochromocytoma in ~50%, usually bilateral, medullary carcinoma of the thyroid, parathyroid adenomas, and cutaneous lichen amyloidosis, associated with the *RET* proto-oncogene) and MEN2B (usually bilateral pheochromocytomas, medullary carcinoma of the thyroid, submucosal neuromas, hyperplastic corneal nerves, joint laxity, Hirschsprung disease, and sometimes marfanoid body habitus). Pheochromocytomas are also found in patients with phakomatoses.

Approximately 20% of patients with von Hippel-Lindau disease type 2 (retinal and/or cerebellar hemangioblastomas, occasionally with clear cell renal carcinoma, pancreatic neuroendocrine tumors, retinal angiomas or hemangioblastomas, mediated by the *VHL* tumor suppressor gene, located on chromosome 3p25-26) will have pheochromocytomas or paragangliomas. Approximately 2% of patients with neurofibromatosis type 1 (autosomal dominant von Recklinghausen disease; neurofibromas, with café au lait spots, axillary and/or inguinal freckling, hamartomas of the iris—Lisch nodules, bony abnormalities, central nervous system gliomas, and sometimes macrocephaly, or cognitive deficits, mediated by the *NF1* tumor suppressor gene on chromosome 17q11.2) will develop a catecholamine-secreting tumor, usually an adrenal pheochromocytoma. Both these conditions can be diagnosed using genetic screening, although this is often more fruitful for screening family members after an index case has been identified.

Neither the prevalence nor genetics of pheochromocytoma in Sturge-Weber syndrome (choroidal and leptomeningeal angiomas, port wine stain in the trigeminal distribution) or tuberous sclerosis (sometimes called Bourneville or Pringle disease—adenoma sebaceum, subungual fibromas, and occasionally mental retardation) are as well understood. Familial paraganglioma is an autosomal dominant syndrome with paragangliomas in the skull base and neck, thorax, abdomen, pelvis, or urinary bladder wall. A number of studies in the last 2 decades have characterized mutations in one of several genes that code for components of the mitochondrial succinate dehydrogenase complex—*SDHD,* located on chromosome 11q23, or *SDHB,* located on chromosome 1p35-36. Availability of genetic testing for these mutations has made disease surveillance for those who carry these genes (typically relatives of index cases) much simpler.

The process of case finding for catecholamine-secreting tumors typically begins with biochemical testing for catecholamine metabolites. Measurements of plasma-free or urinary fractionated metanephrines are usually recommended, but a wide variety of factors are known to produce both false-positive and false-negative results. In some centers, plasma-free metanephrines (which provide a very brief picture of catecholamine production and metabolism) can be assayed quickly and accurately; in others, integration of catecholamine production and metabolism over a longer time period is less expensively performed using urinary collections. False-negative urinary collections are common in patients with pheochromocytomas that are familial, normotensive, dopamine β-hydroxylase deficient, or intermittently secreting. Pharmacologic testing for pheochromocytoma is occasionally used in equivocal cases; clonidine suppression testing is usually preferred over glucagon stimulation testing, although most managed care organizations recommend a repeat plasma or urinary collection 6 months or more after the initial evaluation.

To improve cost-effectiveness and reduce radiation exposure, imaging studies for pheochromocytoma and related tumors are generally not ordered until after biochemical evidence of catecholamine overproduction has been obtained. In this setting, the higher resolution available from CT scans, with thin cuts of the adrenals, outweighs the more specific finding of a T2-weighted bright spot via MRI. For some patients (e.g., those with metastatic disease, intraabdominal surgical clips, allergies to radiocontrast media, pregnant), MRI may be a more suitable option. Positron emission tomography scans using fluorine 18-labeled fludeoxyglucose or nuclear medicine scans using Iodine[123]-labeled *m*-iodobenzylguanidine (^{123}I-MIBG) can localize and define the extent of metastatic disease, especially if delivery of a ^{131}I-MIBG scan is a viable therapeutic option. Prior to obtaining this scan, certain antihypertensive medications should be discontinued, if possible, for at least 7 to 10 days before the test. These classes include calcium antagonists (e.g., amlodipine), alpha- and beta-blockers (e.g., labetolol), and many other non–BP-lowering agents.[206]

The role of genetic testing in the routine care of patients with pheochromocytoma is evolving; current guidelines recommend a shared decision-making process, often involving more family members than just the index patient.[207] Eight studies have shown a high prevalence of germline mutations in patients with presumed sporadic pheochromocytoma or ganglioma, so some authorities recommend genetic screening for all afflicted patients; others base the decision on the pedigree, syndromic features, and/or extent of disease (e.g., multifocal, bilateral, or metastatic tumors at diagnosis).

Proper pharmacologic preparation of the patient with pheochromocytoma is critical for successful extirpation of the tumor; alpha-blockers (e.g., phentolamine intravenously or phenoxybenzamine orally) are given first, followed by beta-blockers if needed to control tachycardia. Most experts use a calcium antagonist before adding a beta-blocker and recommend delaying the operation for 7 to 14 days after localization of the tumor to normalize BP, heart rate, and intravascular volume, which helps minimize hypotension after removal of the tumor. Metyrosine is also used occasionally in people with very large tumors or nonsurgical candidates.

Most surgeons favor laparoscopic procedures for small adrenal pheochromocytomas or paragangliomas in accessible locations. Postoperatively, vigilant monitoring may reduce the risk for severe hypotension, hypoglycemia, or adrenal insufficiency. The diagnostic technique that originally demonstrated the overproduction of catecholamines in a given patient is usually repeated 4 to 6 weeks postoperatively to document successful tumor removal and occasionally (often annually) during long-term follow-up. The specific frequency is best individualized, based on the pedigree, results of genetic testing, and risk factors that predict recurrence.

HYPERCORTISOLISM

Most cases of Cushing syndrome today are iatrogenic due to prescribed oral corticosteroids, but occasional sporadic cases are still seen and were found in 0.5% to 1.0% of hypertensive subjects in two large series. The pathophysiology of hypertension in Cushing syndrome overlaps somewhat with mineralocorticoid excess states. This is because excess cortisol often overwhelms the capacity of 11β-hydroxysteroid dehydrogenase type 2 to degrade cortisol to cortisone selectively in the aldosterone-producing cells of the adrenal cortex and can increase circulating levels of deoxycorticosterone, which has only mineralocorticoid activity.

The full-blown syndrome of hypertension, truncal obesity with striae, diabetes, hirsutism, acne, hyperglycemia, hypokalemia, and muscular weakness is less common today than in Cushing's era, and the recommended diagnostic sequence is also much shorter. After an appropriate screening test (urinary-free cortisol, late night salivary cortisol, or overnight dexamethasone suppression test) has positive results, an endocrine referral for a second test is recommended before imaging studies are ordered. In some centers, plasma corticotropin levels are used to discriminate between corticotropin-dependent Cushing syndrome (>15 pg/mL, probably 85%–90% of cases) and corticotropin-independent Cushing syndrome (<5 pg/mL). In most cases, dynamic testing of the hypothalamic-pituitary-adrenal axis is performed next, with a corticotropin-releasing hormone test, which assays plasma cortisol and corticotropin levels before and after intravenous releasing hormone, or a high-dose dexamethasone (2 mg every 6 hours) suppression test, which assays the serum cortisol level.

Most expert centers have reported that this localizes the tumor to the pituitary in 60% to 75% of cases, a single adrenal gland in approximately 20% (split ~60 : 40 between adenomas and carcinomas), or ectopic production of corticotropin (10%–12%, usually by small cell lung cancers), with less than 1% due to ectopic production of corticotropin-releasing hormone, typically by bronchial carcinoid tumors. Petrosal venous sinus sampling is not needed very often today. The anatomic site of hormonal overproduction is then usually approached surgically, although other modalities (e.g., radiation of the sella turcica) can be used in special circumstances. Medical therapy is also possible in some cases, especially those patients for whom surgery is not feasible.

THYROID DYSFUNCTION

The literature about thyroid dysfunction being associated with hypertension is inconsistent. Many patients with hyperthyroidism have wide pulse pressures (and therefore elevated systolic BP levels) and high pulse rates, but this is seldom missed, especially in younger patients. The ultrasensitive serum thyroid-stimulating hormone (TSH) level is widely available and is most commonly used for screening. After diagnosis, a nonselective beta-blocker such as propranolol may be specifically useful because it treats the tachycardia and hypertension and allegedly inhibits peripheral conversion of thyroxine to triiodothyronine. However, some now question these "classic" clinical pharmacologic reports.

The role of hypothyroidism as a potential cause of hypertension, especially isolated diastolic, is less clear, despite the experience in upstate New York, in which 3% of patients with hypertension reverted to normotension after treatment of hypothyroidism.[151] The hypertension in hypothyroidism is predominantly diastolic and usually in the range of stage 1 (i.e., diastolic BP < 99 mm Hg). In children and adolescents, especially in areas of iodide deficiency, a positive association between serum TSH and BP has been noted. A pooling of seven population-based European data sets has shown similar findings in adults, but there was no consistent relationship with either a 5-year change in BP or incident hypertension. It is likely that targeted screening of patients with hypertension for hypothyroidism with determination of the serum TSH level may be useful, but routine screening for all newly diagnosed patients with hypertension is not currently recommended by any set of national or international guidelines.

HYPERPARATHYROIDISM, CALCIUM INTAKE, VITAMIN D, AND HYPERTENSION

Although hypercalcemia and hypertension associated with hyperparathyroidism often improve after appropriate treatment, causal relationships between calcium and vitamin D intake, serum parathyroid hormone levels, and BP have not been seen consistently in large populations.[208] As a result, most current guidelines recommend a serum calcium level (and not parathyroid hormone) determination during blood testing for patients with an initial diagnosis of hypertension.

COARCTATION OF THE AORTA

Although most discrete isthmic constrictions of the aorta occur in or near the ductus arteriosus, there has been growing awareness that this fifth most common form of congenital cardiovascular disorders constitutes a spectrum of aortic and vasculopathic disorders and is not always cured by surgical procedures that relieve the obstruction. Most patients with the condition are hypertensive[209,210] and are diagnosed in infancy or childhood, but some escape detection until adulthood. Many cases are identified by suggestive physical findings (e.g., murmur, BP lower in the legs than the arms, radial-femoral pulse delay), some after imaging studies done for other reasons (e.g., rib notching or a "3" sign on chest radiograph; the latter results from indentation of the aorta, with prestenotic and poststenotic dilation), and others during investigation of associated abnormalities (e.g., bicuspid aortic valve).

Echocardiography is highly recommended for diagnosis and localization of the coarctation, although some patients (especially adults and those with associated anomalies) may require cardiac catheterization. Most pediatric patients undergo percutaneous catheter balloon dilation with stent placement; this can be followed by definitive surgical correction later, if needed. In a systematic review, 25% to 68% of patients with a coarctation had persistent hypertension despite satisfactory procedure results, with age at the time of surgery, age at follow-up, and the type of intervention being strong predictors of persistent hypertension.[211] A beta-blocker can lower BP and is often used in patients with coarctation, but this is not an indication approved by the US Food and Drug Administration (FDA).

ACROMEGALY

Hypertension occurs in more than 40% of patients with excessive growth hormone release causing acromegaly, and it can be exacerbated by concomitant sleep apnea.[212] Most such patients are easily identified by symptoms or signs of acral bony overgrowth, particularly in children or adolescents before epiphyseal closure, although some patients ignore or tolerate these changes for a decade or more. The vast majority (98%) of cases are caused by a pituitary adenoma; serum insulin-like growth factor-1 is the most useful initial laboratory screening test, although other tests, including the response of plasma growth hormone levels to an oral 75-g glucose load and prolactin levels, are often performed.

As with coarctation, successful treatment of acromegaly usually lowers BP, but hypertension often persists, especially in older and overweight patients.[213] No specific antihypertensive drug therapy seems to be more effective than others but, because acromegaly is a very unusual cause of resistant hypertension, antihypertensive drug therapy is usually effective.

SLEEP DEPRIVATION AND SLEEP-DISORDERED BREATHING

Poor sleep quality, if chronic, yields the same symptomology of paroxysmal hypertension and elevated BP, especially during the afternoon and evening hours. Poor sleep quality is not just the result of obstructive sleep apnea (OSA) but a host of sleep disorders, including restless legs syndrome and insomnia of a various causes.[214] Often, these different sleep disorders coexist and prevent the patient from achieving proper restful sleep.

The mechanism of poor sleep quality contributing to elevated BP and paroxysmal bouts of very high BP relates to activation of both the sympathetic and RAAS.[215,216] Sympathetic activity is also increased in sleep deprivation, restless legs syndrome, and OSA.[215,216]

Patients without OSA who suffer from sleep deprivation, defined as less than a minimum of 6 hours of uninterrupted sleep, also have increased sympathetic activity. In this case, it is a consequence of reduced time in non–rapid eye movement (NREM) or slow wave sleep that also affects the nocturnal dip in BP.[216] This supports the hypothesis that disturbed NREM sleep quantity or quality is a mechanism whereby sleep deprivation or restless legs syndrome leads to an increase in sympathetic tone.

OSA should be suspected in patients (men more so than women) who report severe snoring, daytime somnolence, witnessed nocturnal choking or gasping, and have a "crowded oropharynx" (limited or no visualization of the soft palate) on physical examination. Formal diagnosis is based on an ambulatory sleep study or in-center polysomnography.[217]

In contrast to reduced sleep time and quality, the increase in sympathetic activity associated with OSA is a function of intermittent hypoxia because the acute rise in BP parallels the severity of oxygen desaturation at night.[218] Indeed, increased sympathetic activity is seen in animal models subjected exclusively to intermittent hypoxia.[219] It is also important to note that OSA associated hypertension is only slightly reduced by continuous positive airway pressure (CPAP) treatment.[220] Nevertheless, addressing sleep disorders or sleep habits is relevant when considering the risk of developing or controlling preexistent hypertension.

Treatment of OSA with CPAP has a modest BP lowering effect (~1-mm Hg/hour use). However, patients who have more severe hypertension, more severe OSA, higher daytime sleepiness scores, and greater adherence to CPAP (average use >4 hours/night) tend to have greater responses.[221]

DRUG-INDUCED HYPERTENSIOM

Patients presenting with hypertension should be questioned about exposure to substances that can raise BP (Box 46.3).

Box 46.3 Drugs Commonly Associated With Hypertension

Oral contraceptives
Nonsteroidal antiinflammatory drugs (NSAIDs; selective and nonselective)
Sympathomimetics: pseudoephedrine, phenylpropanolamine, phentermine, cocaine, amphetamines (prescription or illegal); yohimbine (alpha-2 antagonist)
Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs)
Monoamine oxidase inhibitors (MAOIs)
Cyclosporine and tacrolimus
Erythropoietin and darbepoetin
Corticosteroids, mineralocorticoids (fludrocortisone)
Anti-VEGF antibodies (e.g., bevacizumab, ramucirumab) and certain tyrosine kinase inhibitors with anti-VEGF activity (e.g., sorafenib, sunitinib)
Licorice
Ethanol

VEGF, Vascular endothelial growth factor.
From Peixoto AJ. Secondary hypertension. In: Gilbert S, Weiner D, eds. *Primer in kidney diseases.* Philadelphia: Elsevier, 2018.

These include drugs of abuse and over-the-counter and prescription medications. Oral contraceptive pills (OCPS) can cause hypertension, although modern low-estrogen pills do so at rates much lower than with older preparations. Stopping the OCP cures the hypertension after several weeks to months in most but not all women. NSAIDs result in a modest average hypertensive effect (up to ~5 mm Hg), but some patients can have larger, clinically significant BP elevation. NSAID-induced hypertension may also present as loss of BP control in patients taking a diuretic or a blocker of the RAAS, whereas CCBs tend to be less affected in NSAID users.

Sympathomimetic amines (legal or illegal) usually cause hypertension acutely following their ingestion. Alcohol has an acute hypotensive effect, but chronic use in large amounts (more than four or five drink equivalents/day) is associated with increased BP. Glucocorticoids and mineralocorticoids can produce a dose-dependent rise in BP. Glucocorticoids with low mineralocorticoid activity (e.g., dexamethasone, budesonide) induce lesser pressor responses.

Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) can increase BP modestly, but some patients receiving SNRIs can have a severe hypertensive response. Interestingly, when used for hypertensive patients with depression, BP often improves as depressive symptoms improve. Angiogenesis inhibitors, such as anti–vascular endothelial growth factor (VEGF) antibodies (e.g., bevacizumab, ramucirumab) and tyrosine kinase inhibitors (e.g., sorafenib, sunitinib) can produce hypertension that often persists, despite discontinuation. Because hypertension during the use of these drugs correlates with better oncologic outcomes (likely a reflection of a successful antiangiogenic effect), treatment is usually continued unless reasonable BP control is not achievable, or if severe kidney injury develops.[222]

HYPERTENSIVE URGENCY AND EMERGENCY

A hypertensive emergency is the combination of elevated BP levels (with no specific diagnostic BP level) and signs or symptoms of acute, ongoing, target organ damage. Such patients are traditionally admitted to an intensive care unit and given parenteral infusions of short-acting antihypertensive agents to restore autoregulation in vascular beds. This is done because historical data from the 1920 to 1940 era (antedating effective antihypertensive drug therapy) showed a very poor prognosis, similar to that of many cancers. Traditionally, patients who presented with very elevated BP levels, but no acute, ongoing, target organ damage, were diagnosed with a "hypertensive urgency," observed for a few hours after treatment with one or more oral antihypertensive agents, and then discharged to a site of ongoing care for their hypertension. This practice is now undertaken primarily for medicolegal reasons. Two retrospective studies, one involving 1016 patients seen in an emergency department[223] and another involving 58,535 outpatients,[224] demonstrated very low rates of morbidity and mortality and no significant differences in prognosis for those treated acutely, compared with those who were discharged with rapid follow-up.

The initial evaluation of a severely hypertensive patient includes a thorough inspection of the optic fundi (looking for acute hemorrhages, exudates, or papilledema), a mental status assessment, a careful cardiac, pulmonary, and neurologic examination, a quick search for clues that might indicate secondary hypertension (e.g., abdominal bruit, striae, radial-femoral delay), and laboratory studies to assess renal function—dipstick and microscopic urinalysis, determination of serum creatinine level.

Several options for intravenous drug treatment exist, but nitroprusside is the least expensive and most widely available. It must be kept in the dark and is metabolized to cyanide and/or thiocyanate, particularly during long-term infusions. Fenoldopam mesylate, a dopamine-1 agonist, is very effective and acutely improves several parameters of renal function. Clevidipine is a dihydropyridine (DHP) calcium antagonist that is hydrolyzed within minutes by ubiquitous serum esterases; it is administered in an emulsion containing soy and egg proteins, either of which can cause immunologic reactions in allergic patients. Its elimination is not importantly affected by hepatic or renal functional impairment. Clevidipine and its older, longer acting cousin, nicardipine, are often used for patients with coronary disease because the reflex tachycardia is usually offset by coronary vasodilation. Nimodipine is typically used only for subarachnoid hemorrhage.

The quickest therapeutic response to a hypertensive emergency is recommended for an acute aortic dissection. In this condition, the BP should be lowered within 20 minutes to a systolic BP below 120 mm Hg (neither of which is supported by a strong evidence base), typically with a beta-blocker to reduce shear stress on the dissection and a vasodilator. Controversy exists about if and when BP lowering should be attempted in the setting of an acute ischemic stroke. If the patient is a candidate for acute thrombolytic therapy and the BP is higher than 180/110 mm Hg, acute BP lowering is recommended. Most US authorities suggest attempting slow and gradual BP lowering only if the BP is very high (e.g., ≥180/110 mm Hg) with a short-acting, rapidly titratable drug. However, two large, randomized trials done outside the United States have suggested that BP lowering in this setting is safe, but does not produce significant outcome benefits in ischemic[225] or hemorrhagic stroke.[226]

All other types of hypertensive emergencies can be handled with a gradual lowering of BP, typically 10% to 15% during the first hour and a further 10% to 20% during the next hour, for a total of approximately 25%. (See Table 46.8.)

Frequent monitoring of the patient's clinical status is important because not all patients can reestablish the normal autoregulatory capacity of the circulation in important vascular beds during the same short time period. Because hypertensive encephalopathy is a diagnosis of exclusion, it is often very rewarding to monitor these patients closely because their mental status improves markedly (and usually rather quickly) as the BP is carefully lowered.

Patients who present with hypertensive crises involving cardiac ischemia or infarction or pulmonary edema can be managed with nitroglycerin, clevidipine, nicardipine, or nitroprusside, although typically a combination of drugs (including an ACE inhibitor for heart failure or left ventricular dysfunction) is used in these settings. Efforts to preserve myocardium and open the obstructed coronary artery by thrombolysis, angioplasty, or surgery also are indicated.

Hypertensive emergency involving the kidney is commonly followed by a further deterioration in renal function, even when the BP is lowered properly. The most important predictor of the need for acute dialysis is not the BP level, but the degree of renal dysfunction (both eGFR and degree of albuminuria). Some physicians prefer fenoldopam to nicardipine or nitroprusside in this setting because of its lack of toxic metabolites and specific renal vasodilating effects. The need for acute dialysis often is precipitated by BP reduction in patients with preexisting stage 3 to 5 CKD, but many patients can avoid dialysis (and a remarkable few can even discontinue it) in the long term if BP is carefully well controlled during follow-up.

Hypertensive crises resulting from catecholamine excess states (e.g., pheochromocytoma, monoamine oxidase inhibitor crisis, cocaine intoxication) are most appropriately managed with an intravenous alpha-blocker (e.g., phentolamine), with a beta-blocker added later, if needed. Many patients with severe hypertension caused by sudden withdrawal of antihypertensive agents (e.g., clonidine) are easily managed by giving one acute dose of the missed drug.

Hypertensive crises during pregnancy must be managed in a more careful and conservative manner because of the

Table 46.8 Hypertensive Emergencies, Therapies, and Target Blood Pressures

Type of Emergency	Drug of Choice	Blood Pressure Target
Aortic dissection	Beta-blocker plus nitroprusside[a]	120 mm Hg systolic in 20 min (if possible)
Cardiac		
Ischemia, infarction	Nitroglycerin, nitroprusside,[a] nicardipine, or clevidipine	Cessation of ischemia
Heart failure (or pulmonary edema)	Nitroprusside[a] and/or nitroglycerin	Improvement in failure (typically only a 10%–15% decrease is required)
Hemorrhagic		
Epistaxis, gross hematuria, or threatened suture lines	Any (perhaps with anxiolytic agent)	To decrease bleeding rate (typically only 10%–15% reduction over 1–2 hours is required)
Obstetric		
Eclampsia or preeclampsia	$MgSO_4$, hydralazine, methyldopa	Typically <90 mm Hg diastolic, but often lower
Catecholamine Excess States		
Pheochromocytoma	Phentolamine	To control paroxysms
Drug withdrawal	Drug withdrawn	Typically only one dose necessary
Cocaine (and similar drugs)	Phentolamine	Typically only 10%–15% reduction over 1–2 hours
Renal		
Major hematuria or acute kidney injury	Nitroprusside,[a] fenoldopam	0%–25% reduction in mean arterial pressure over 1–12 hours
Neurologic		
Hypertensive encephalopathy	Nitroprusside[a]	25% reduction over 2–3 hours
Acute head injury, trauma	Nitroprusside[a]	0%–25% reduction over 2–3 hours (controversial)

[a]Many physicians prefer an intravenous infusion of clevidipine, fenoldopam, or nicardipine, none of which has potentially toxic metabolites, over nitroprusside, especially if a long duration of treatment is planned. Acute improvements in renal function occur during therapy with fenoldopam, but not with nitroprusside.

presence of the fetus. Magnesium sulfate, methyldopa, and hydralazine are the drugs of choice, with oral labetalol and nifedipine being drugs of second choice in the United States; nitroprusside, ACE inhibitors, and ARBs are contraindicated.[227] Delivery of the infant is often hastened by the obstetrician to assist in managing the hypertension in pregnancy.

Whether hypertensive urgencies (e.g., elevated BP, but without acute ongoing target organ damage) ought to be treated acutely is controversial because there is no evidence that such treatment improves prognosis. The BP in many such patients spontaneously falls during a 30-minute period of quiet rest. Conversely, immediate-release nifedipine capsules can cause precipitous hypotension, stroke, myocardial infarction, and death. According to the FDA, they "should be used with great caution, if at all." In such cases, true "hypotension" (e.g., systolic BP <90 mm Hg) may not be observed, yet the BP may fall below the autoregulatory threshold. This is likely different for every patient and may be unknown to the treating physician until it is surpassed, precipitating ischemia.

Clonidine, captopril, labetalol, several other short-acting antihypertensive drugs, and even amlodipine, have been used in this setting, but none has a clear advantage over the others, and each is usually effective in most patients. The most important aspect of managing a hypertensive urgency is to refer the patient to a good source of ongoing care for hypertension, where adherence to antihypertensive therapy during long-term follow-up will be more likely.

In short, patients presenting with a hypertensive emergency should be diagnosed quickly and started promptly on effective parenteral therapy (often, nitroprusside 0.5 μg/kg per minute) in an intensive care unit. BP should be gradually reduced by approximately 25% over 2 to 3 hours. Oral antihypertensive therapy should be instituted, usually after approximately 8 to 24 hours of parenteral therapy; evaluation for secondary causes of hypertension may be considered after transfer from the intensive care unit. Because of advances in antihypertensive therapy and management, the term *malignant hypertension* should be relegated to the dustbin of history (and used only by billers and coders) because the prognosis of patients with this condition has improved greatly since the term was introduced in 1927.

GOALS OF ANTIHYPERTENSIVE THERAPY

There are hundreds of clinical studies that have evaluated the efficacy and safety of the eight different classes of antihypertensive medications. All BP-lowering agents need to have at least two appropriately powered, placebo-controlled studies to meet specific FDA criteria for approval as antihypertensive agents. This section will not focus on the details of these studies but rather on data supporting BP reductions with certain BP-lowering classes and the impact on CKD progression, as well as trials evaluating CV outcomes in patients with kidney disease.

Meta-analyses of all commonly used antihypertensive drug classes have demonstrated that regardless of the agent used, reduction in BP corresponds to reduction in CV events if BP reduction is achieved.[228–230] This reduction in CV risk, however, is predominantly seen in people with stage 2 hypertension (systolic, 140–159 mm Hg, or diastolic, 90–99 mm Hg) with much less outcome data to support risk reduction in stage 1 hypertension (systolic, 130–139 mm Hg, or diastolic, 80–89 mm Hg).

Events that drive the risk reduction are derived predominantly from a reduced incidence of stroke, myocardial infarction, and heart failure. In all trials to date it is the group with the best overall BP control that has the best outcomes. An exception to this generalization is Avoiding Cardiovascular Events Through Combination Therapy in Patients Living with Systolic Hypertension (ACCOMPLISH), a CV outcome trial that included over 11,000 people.[231] In this trial, both groups had similar BP control, and both were randomized to the same ACE inhibitor (benazepril), yet the group initially randomized to a single-pill combination of benazepril with a calcium antagonist had a 20% CV risk reduction compared with the ACE inhibitor plus diuretic group (Fig. 46.13). The observed benefit for the benazepril-amlodipine combination also extended to slowing CKD progression.[232]

Almost all people with an eGFR of less than 60 mL/min/1.73 m^2 and hypertension will require two or more medications to achieve a BP goal of less than 140/90 mm Hg. Single-pill combinations, including the combination of an RAAS blocker with a calcium antagonist or diuretic, are preferred agents.[7,233] These combinations, when given generally in an additive fashion, reduce CV events and CKD progression. Other combinations that are efficacious for reducing BP but have not been tested in clinical trials include beta-blockers with a dihydropyridine (DHP) calcium antagonist and DHP calcium antagonists with diuretics.[233]

There have been a number of trials assessing both CV outcome and changes in CKD progression. All these trials have assumed adherence with antihypertensive medications. However, according to one report, only 71% of subjects with hypertension in the United States were on treatment, and only 48% had their BP under adequate control (<140/90 mm Hg). Moreover, two separate studies, one in the United Kingdom and the other in Germany, evaluating medication adherence, have shown that only approximately 45% of patients who claimed to be taking BP-lowering medication actually were, as assessed by urinalysis of drug metabolites.[234,235] Although there has been significant reduction in the age-adjusted death rate for stroke and coronary artery disease since the early 1980s as a result of better BP control (and better treatment of other risk factors, such as hyperlipidemia), heart disease and stroke remain the first and third leading causes of death in Western countries, which emphasizes the importance of identifying and treating patients with hypertension. This section will discuss outcome trials focused on BP reduction that evaluated CKD progression as well as CV outcomes in people with CKD.

BLOOD PRESSURE CONTROL AND CHRONIC KIDNEY DISEASE PROGRESSION

NONDIABETIC CHRONIC KIDNEY DISEASE

The rationale for antihypertensive therapy in nondiabetic CKD is discussed in detail in Chapter 59. There is clear evidence that partial blockade of the RAAS slows nephropathy progression in those with stage 3 or higher proteinuric kidney disease.[236–238] Although there is no evidence as to whether an ACE inhibitor versus ARB yields a better CKD

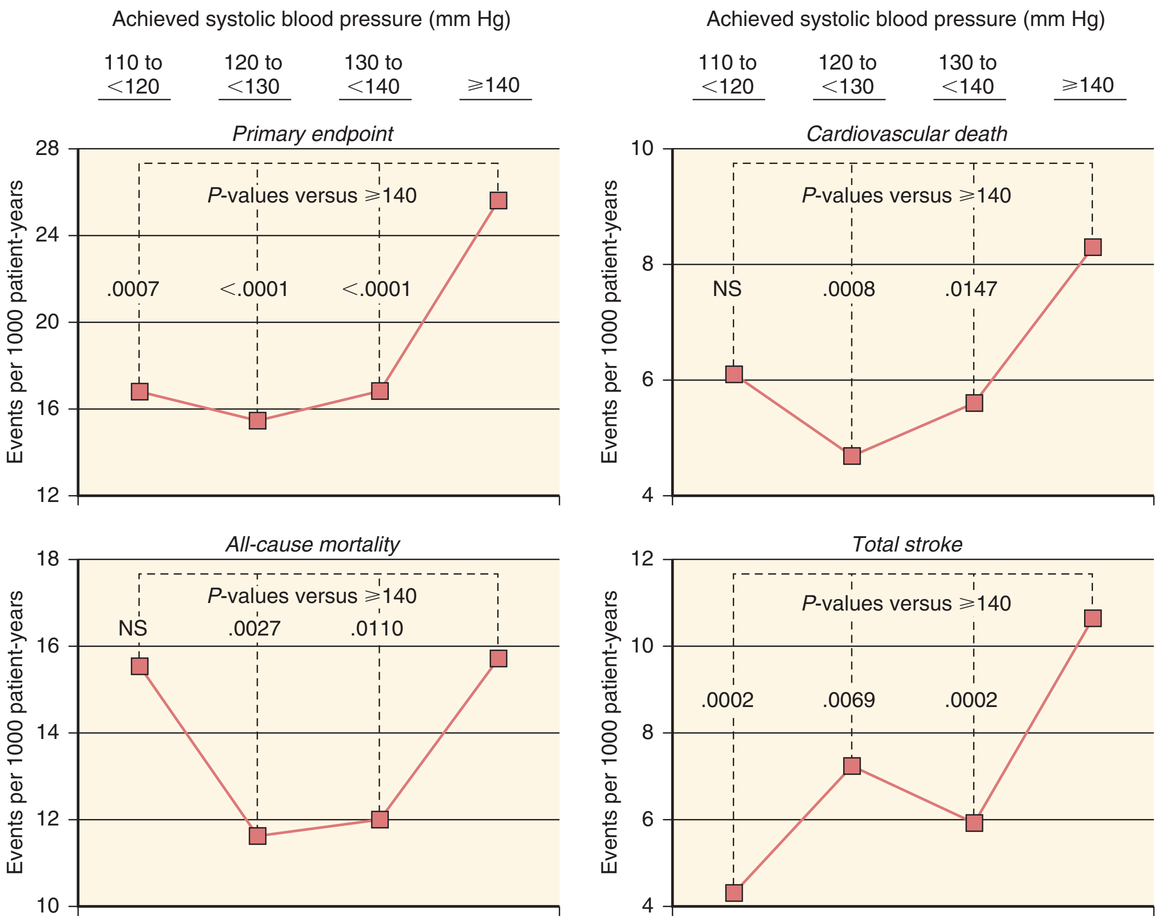


Fig. 46.13 Event rates (per 1000 patient-years) for major clinical outcomes in the Avoiding Cardiovascular Events Through Combination Therapy in Patients Living with Systolic Hypertension (ACCOMPLISH) trial categorized by systolic pressure. *NS,* Not significant. (From Weber MA, Bloch M, Bakris GL, et al. Cardiovascular outcomes according to systolic blood pressure in patients with and without diabetes: an ACCOMPLISH substudy. *J Clin Hypertens*. 2016;18:299–307.)

outcome, it is clear that both classes have similar benefits between trials.

In contrast to the protection afforded by RAAS blockade, it is also clear that a lower level of BP does not slow nephropathy progression in hypertensive patients per se. Four prospective, randomized, long-term CKD outcome trials in nondiabetic kidney disease (discussed in the following section) have failed to show a benefit on slowing nephropathy progression among the groups with the lower BP.[200,239–241] Hence, the updated KDIGO BP guidelines recommend a BP goal of less than 140/90 mm Hg in those with CKD and is backed by the highest level of evidence. The previous goal of less than 130/80 mm Hg has a much lower level of evidence and is endorsed in the presence of a very high urine albumin level (>300 mg/day). A lower level of blood pressure further reduces cardiovascular events in people with CKD, even though it does not slow CKD progression, and should be sought for that reason.[241]

Modification of Diet in Renal Disease Study

This was the first appropriately powered study to test whether a lower BP goal was associated with a slower progression of CKD. A low BP goal (target mean arterial pressure ≤92 mm Hg for patients ≤60 years old or 98 mm Hg for patients ≥61 years old) was associated with a significant reduction in proteinuria and to a slower subsequent decline in GFR. However, this was not significant compared with the higher pressure group with a mean arterial pressure of 102 to 107 mm Hg.

African American Study of Kidney Disease and Hypertension

The effect of antihypertensive therapy on the progression of CKD secondary to hypertension is more controversial. In the Multiple Risk Factor Intervention Trial (MRFIT), in which thiazide diuretics and beta-blockers were primarily used to control BP, slowing or stabilization of kidney function was not seen in black men but was seen in all other racial groups studied.[33]

In the African American Study of Kidney Disease and Hypertension (AASK), the use of an ACE inhibitor (ramipril) was found to be more effective at slowing CKD progression compared with either amlodipine (a dihydropyridine CCB) or metoprolol[31] (Fig. 46.14). This trial, in over 1000 blacks, failed to show superior protection with BP reduced to levels below 130/80 mm Hg compared with conventional BP targets of 140/90 mm Hg in subjects with hypertensive nephrosclerosis, even with a total of a 10-year follow-up.[240] Masked hypertension may be a confounder to these outcomes. A subanalysis of over 50% of the AASK participants who had

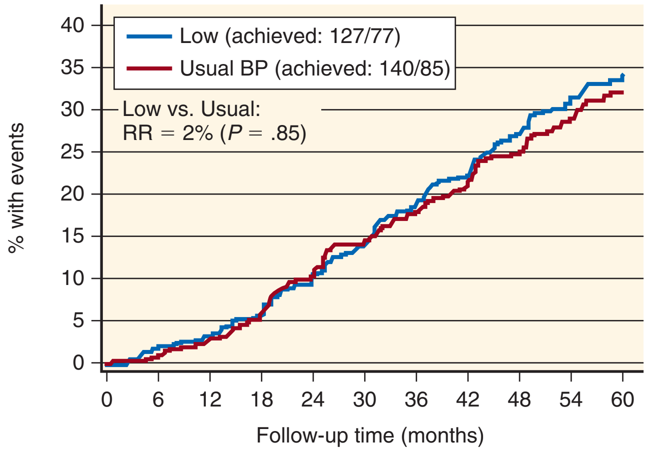


Fig. 46.14 Composite primary clinical endpoint decline in glomerular filtration rate, end-stage kidney disease, or death in the African American Study of Kidney Disease. *BP,* Blood pressure; *RR,* relative risk. (From Bakris GL, Sarafidis PA, Weir MR, et al. Renal outcomes with different fixed-dose combination therapies in patients with hypertension at high risk for cardiovascular events (ACCOMPLISH): a prespecified secondary analysis of a randomised controlled trial. *Lancet.* 2010;375(9721): 1173–1181.)

24-hour ABPM showed inadequate 24-hour BP control in approximately 36% of the cohort.[242] Masked hypertension and failure of nocturnal dipping were the two most common reasons for poor out-of-office BP control.

Further studies were performed to assess whether changes in antihypertensive dose timing corrected the failure of nocturnal dipping, but the results were negative.[243] In contrast, studies performed in Spain in white patients with CKD stages 2 to 3b have demonstrated an improvement in dipping status when dosing BP medications at night.[243]

Ramipril Efficacy in Nephropathy Trial (REIN-2)

This multicenter, randomized controlled trial was conducted in Italy in patients with proteinuria associated with nondiabetic kidney disease who were receiving background treatment with the ACE inhibitor ramipril (2.5–5 mg/day).[239] The aim was to assess the effect of intensified versus conventional BP control on progression to ESKD. Subjects were randomly assigned to conventional (diastolic <90 mm Hg; $n = 169$) or intensified (systolic/diastolic <130/80 mm Hg; $n = 169$) BP control. To achieve the intensified BP level, patients received add-on therapy with the dihydropyridine calcium channel blocker (DCCB) felodipine (5–10 mg/day). The primary outcome measure was time to ESKD over 36 months' follow-up. The authors found that over a median follow-up of 19 months, 38 of 167 patients (23%) assigned to intensified BP control and 34 of 168 patients (20%) who were allocated to conventional control progressed to ESKD (HR, 1.00; 95% confidence interval [CI], 0.61–1.64; $P = .99$). Hence, there was no benefit of aggressive BP lowering in slowing this nondiabetic nephropathy.

Systolic Blood Pressure Intervention Trial

The Systolic Blood Pressure Intervention Trial (SPRINT) randomized participants to intensive BP lowering (systolic BP [SBP] <120 mm Hg) versus standard BP control (SBP <140 mm Hg) on clinical outcomes. In 2646 patients who had CKD at baseline, the composite kidney outcome of a 50% decrease or more in eGFR from baseline or ESKD, was no different between the groups (HR, 0.90 for the intensive group; 95% CI, 0.44–1.83).[241] However, the intensive group had a lower rate of cardiovascular outcomes and all-cause death.

DIABETIC KIDNEY DISEASE

There have been no randomized trials of BP level and CKD outcome among patients with diabetes mellitus. The recommendation for a lower BP goal in diabetes has resulted from post hoc analyses of trials that evaluated CV outcomes in subjects with diabetic kidney disease. All these studies showed a benefit on CV risk reduction and some benefit in slowing CKD progression in the group randomized to the lower pressures.

The Hypertension Optimal Treatment (HOT) trial was not powered for a blood pressure diabetes outcome but was the first CV outcome trial that evaluated different levels of diastolic BP (80 mm Hg, 85 mm Hg, and 90 mm Hg) in 18,790 patients with hypertension, from 26 countries, mean age 61.5 years. In the subgroup of patients with diabetes mellitus, a post hoc analysis showed a 51% reduction in major CV events in the target group of 80 mm Hg or less compared with the target group of 90 mm Hg or less (P for trend = .005).[244] Nevertheless, because the primary endpoint of the trial failed to reach statistical significance on the CV outcome at the lowest randomized BP level, the diabetes subgroup could not be considered positive, even if it was statistically significant; the hypothesis tested was not specifically predefined for this subgroup.

The Ongoing Telmisartan Alone and in combination with Ramipril Global Endpoint Trial (ONTARGET) and the International Verapamil-Trandolapril Study (INVEST), like the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial, have failed to show a benefit on CV outcomes.[245–247] Taken together, these three studies demonstrated no additional benefit of BP lowering below 130/80 mm Hg on CV risk reduction compared with 130 to 139/80 to 85 mm Hg (Table 46.9).

It should be noted that there have been four recent systematic reviews of these trials and many others in diabetes that were summarized in the American Diabetes Association (ADA) Position Paper of Blood Pressure Goals in Diabetes. The committee supported a BP goal of less than 140/90 mm Hg, but acknowledged that most patients will achieve a greater CV risk reduction at blood pressure levels of less than 130/80 mm Hg. Thus, this suggests that most patients with diabetes strive for this goal.[157] Moreover, there is general agreement between the AHA/ACC BP guidelines and the ADA guidelines with regard to a BP approach in diabetes.[248]

Although there is no evidence of additional slowing of nephropathy progression at BP levels below 130/80 mm Hg, CV risk reduction did occur among those with CKD, as seen in the ACCORD trial.[249] The only prospective outcome trials that randomized groups to different BP levels and were powered statistically for CV outcomes were the United Kingdom Prospective Diabetes Study (UKPDS) and ACCORD. However, only the intensive treatment group in the ACCORD study attained a BP of less than 130/80 mm Hg.[246,250]

Table 46.9 Achieved Blood Pressure Levels in Cardiovascular Outcome Trials Involving Patients with Diabetes

Clinical Outcome Trial	Achieved Level of Systolic Blood Pressure (mm Hg)
ACCORD (primary)	119 (intensive); 133 (conventional)
UKPDS (primary)	144 (intensive); 154 (conventional)
ACCOMPLISH (secondary)	Overall mean, 133
INVEST (secondary)	144 (tight control); 149 (conventional)
ONTARGET (secondary)	Averaging approximately 140
VADT (secondary)	127 (intensive); 125 (conventional)
ADVANCE (secondary)	145 (in both intensive and conventional glucose control)

ACCOMPLISH, Avoiding Cardiovascular Events Through Combination Therapy in Patients Living with Systolic Hypertension; *ACCORD,* Action to Control Cardiovascular Risk in Diabetes; *VADT,* Veterans Affairs Diabetes Trial; *ADVANCE,* Action in Diabetes and Vascular Disease: Preterax and Diamicron Modified Release Controlled Evaluation; *INVEST,* International Verapamil-Trandolapril Study; *ONTARGET,* Ongoing Telmisartan Alone and in combination with Ramipril Global EndpoinT trial; *UKPDS,* United Kingdom Prospective Diabetes Study.

In ACCORD, there was no significant difference in the primary endpoint, all-cause and CV mortality, between the standard and intensive BP groups. However, the long-term follow-up ACCORDION study demonstrated a major interaction between the intensive glycemic control group and intensive blood pressure goal, such that the outcomes were confounded. When reevaluating the ACCORD data in the context of these statistical limitations, a benefit was seen trending for further CV risk reduction in a lower blood pressure group, albeit not significant.[251] Two important studies using ACCORD data are important. One study used SPRINT criteria applied to the ACCORD database and compared CV outcomes; there was a significant reduction in the intensive BP group in the primary endpoint.[251a] A second equally important study evaluated change in serum creatinine levels and CV outcome in ACCORD and demonstrated that up to a 30% increase in the serum creatinine level during the trial was associated with a reduced CV outcome.[251b] This was not seen among those with a greater than 30% increase in their serum creatinine level.

The UKPDS did not show a benefit from the lower BP group, which averaged above 140/90 mm Hg. Additional findings from post hoc analyses of diabetes subgroups of other trials have also failed to show benefit of BP levels below 130/80 mm Hg, demonstrating an additional CV outcome benefit.

A number of studies have demonstrated that some classes of antihypertensive drugs should be used preferentially in patients with diabetes who have nephropathy. The effect of blockers of the RAAS system in diabetic nephropathy to retard hard kidney endpoints, such as ESKD, has now been well documented in multiple studies.[252–254] However, this class of agents does not possess any specific advantages over other antihypertensive classes in diabetics who do not have nephropathy. Moreover, there is no evidence that blockers of the RAAS benefit people with normotension, with or without microalbuminuria, from developing declines in kidney function.[157,255,256] See Chapter 39 for an in-depth discussion of therapy in diabetic kidney disease.

ANTIHYPERTENSIVE THERAPY IN OLDER ADULTS

Clinical Trials in Older Adults

All major clinical trials involving persons older than 65 years are summarized in Table 46.10. Most patients recruited in these trials prior to the Hypertension in the Very Elderly Trial (HYVET) were younger than 80 years, limiting information about octogenarians.[24] Results in these trials have shown a reduction in the incidence of stroke and CV morbidity. Previous clinical trials have shown that lowering BP in older adults has no effect on overall mortality or on the incidence of fatal or nonfatal myocardial infarction.[210,257] The recent results of the older subgroup of the SPRINT, with a mean age of 79 years, however, has shown a clear benefit on all CV outcomes, especially heart failure risk, among those treated to a lower blood pressure.[25] Thus, clinical practice guidelines prior to HYVET, and subsequently the ACC/AHA, have recommended that in subjects aged 80 years and older, evidence for benefits of antihypertensive treatment is as yet inconclusive.[8]

The release of the HYVET,[23] and the subsequent publication of the "ACCF/AHA 2011 Expert Consensus Document on Hypertension in the Elderly" study,[211] changed the management of hypertension, particularly in patients older than 80 years. Before SPRINT, there were five trials performed in older people, only two of which randomized to different BP levels that were below a systolic BP of 140 mm Hg. The trials were SHEP—Systolic Hypertension in Europe (Syst-Eur)—HYVET, Japanese Trial to Assess Optimal Systolic Blood Pressure in Elderly Hypertensive Patients (JATOS), and Valsartan in Elderly Isolated Systolic Hypertension (VALISH; see Table 46.10).[212] Both randomized BP trials with the lower BP levels were carried out in Japan. The results from these two studies have shown no additional benefit of achieving a BP of less than 140/90 mm Hg compared with less than 150/90 mm Hg.

A concern in patients with isolated systolic hypertension is the reduction of diastolic BP after the initiation of antihypertensive therapy. A diastolic BP that is too low may interfere with coronary perfusion and possibly increase CV risk. The relationship between diastolic BP and CV death, particularly myocardial infarction, is like a J-shaped curve; thus, excessive reduction in diastolic pressures should be avoided in patients with coronary artery disease who are being treated for hypertension.[258,259] A BP of 119/84 mm Hg was identified as a nadir when treating patients with coronary artery disease[260]; however, among patients with CKD, a nadir of 131/71 mm Hg was identified as a level below which higher CV mortality was present[260] (Fig. 46.15).

When treating patients with isolated systolic hypertension, the JNC 7 has suggested a minimum posttreatment diastolic BP of 60 mm Hg overall, or perhaps 65 mm Hg in patients with known coronary artery disease, unless symptoms that could be attributed to hypoperfusion occur at higher pressures. This was seen in the findings of the SHEP trial,[261]

Table 46.10 Clinical Trials in Older Adults

Clinical Trials	Mean Age	No. of Subjects	Drugs Used	Achieved BP (Control)	Achieved BP (Rx)	Outcome
EWHPE, 1985	72	840	HCTZ + triamterene, ± methyldopa	159/85	155/84	+ CV risk reduction
Coope and Warrender, 1986	68	884	Atenolol ± bendroflumethiazide	180/89	178/87	Stroke reduction
SHEP, 1991	72	4736	Chlorthalidone ± atenolol or reserpine	155/72	143/68	Stroke reduction
STOP, 1991	72	1627	Atenolol ± HCTZ or amiloride	161/97	159/81	Stroke and MI reduction
MRC, 1992	70	3496	HCTZ ± amiloride vs. atenolol	≈169/79	≈150/80	Stroke, MI, and CHD reduction
CASTEL, 1994	83	665	Clonidine, nifedipine and atenolol + chlorthalidone	181/97	165.2/85.6	Reduced mortality
STONE, 1996	67	1632	Nifedipine	155/87	147/85	Stroke and CV event reduction
Syst-Eur, 1997	70	4695	Nitrendipine + enalapril or HCTZ	161/94	151/79	Stroke and CVD reduction
Syst-China, 2000	67	2394	Nitrendipine + captopril or HCTZ	178/93	151/76	Stroke, CVD, and HF reduction
HYVET, 2008	84	3845	Indapamide + perindopril	158.5/84	143/78	Stroke and HF reduction
JATOS, 2008	75	4418	Efonidipine hydrochloride	145.6/78.1	136/74	No difference between aggressive BP on renal and CV events

BP, Blood pressure; *CASTEL,* Cardiovascular Study in the Elderly; *CHD,* coronary heart disease; *CV,* cardiovascular; *CVD,* cardiovascular disease; *EWHPE,* European Working Party on High Blood Pressure in the Elderly; *HCTZ,* hydrochlorothiazide; *HF,* heart failure; *HYVET,* Hypertension in the Very Elderly Trial; *JATOS,* Japanese Trial to Assess Optimal Systolic Blood Pressure in Elderly Hypertensive Patients; *MI,* myocardial infarction; *MRC,* Medical Research Council; *Rx,* medication; *SHEP,* Systolic Hypertension in the Elderly Program; *STONE,* Shanghai Trial of Nifedipine in the Elderly; *STOP,* Hypertension, Swedish Trial in Old Patients with Hypertension; *Syst-China,* Systolic Hypertension in China; *Syst-Eur,* Systolic Hypertension in Europe.

where older patients with lower diastolic BP had higher CV event rates.

Additionally, in some trials involving older individuals, there is concern about thiazide diuretic use, hyponatremia, and worsening of kidney function in subjects with hypertension. In the European Working Party on High Blood Pressure in the Elderly trial, a significantly higher incidence of impaired kidney function was found in those receiving diuretics compared with placebo.[262] In the SHEP trial, the serum creatinine level increased significantly in subjects treated with thiazide diuretics compared with placebo. In ALLHAT, the chlorthalidone-treated group showed worse kidney function than the amlodipine- or lisinopril-treated group at both the 2- and 4-year endpoints.[263] Although this could likely be accounted for by volume depletion in many cases, diuretics have been shown to induce mild renal injury in various animal models, possibly because of hypokalemia, hyperuricemia, and stimulation of the RAAS related to the reduction in renal perfusion pressure.

The HYVET trial, in 2008, changed how hypertension in older adults is managed. It randomly assigned almost 4000 patients who were 80 years of age and older and had systolic BP above 160 mm Hg to indapamide or placebo. Results of HYVET have provided clear evidence that BP lowering with treatment using antihypertensive medications is associated with definite CV benefits. The use of indapamide supplemented by perindopril showed reductions in the incidence of stroke, congestive heart failure, and CV fatal events.

Stricter BP goals have shown no benefit in CV morbidity and mortality in older populations. The JATOS study has shown that systolic BP goals of less than 140 mm Hg had no statistical significance in the CV mortality of patients older than 65 years.[264] The results of the Japanese trials can be contrasted with the results of SPRINT, which demonstrated significantly lower CV event rates in the group randomized to a lower BP well below 130 mm Hg.[25]

There have been no specific trials powered to assess CKD progression in older people. There are prespecified and post hoc analyses of trials where the mean age is above 65 years that examined changes in CKD progression. A notable example is the prespecified analysis of the ACCOMPLISH trial, where more than 40% of the people with CKD were 70 years of age or older. This trial showed a benefit of the RAAS blocker–CCB combination over the RAAS blocker–diuretic for slowing progression and time to dialysis.[232] A post hoc analysis of the ALLHAT study in people older than 65 years also has shown that BP control slows progression.[263] However, in this trial, we had no information about albuminuria.

Nonpharmacologic Intervention in Older Adults

In the Trial of Nonpharmacologic Interventions in the Elderly (TONE),[265] the combination of weight loss and sodium restriction showed a drop of 5.3 ± 1.2 mm Hg in the systolic BP and 3.4 ± 0.8 mm Hg diastolic BP in obese older patients with hypertension. The goal of sodium restriction was 1.8 g/24

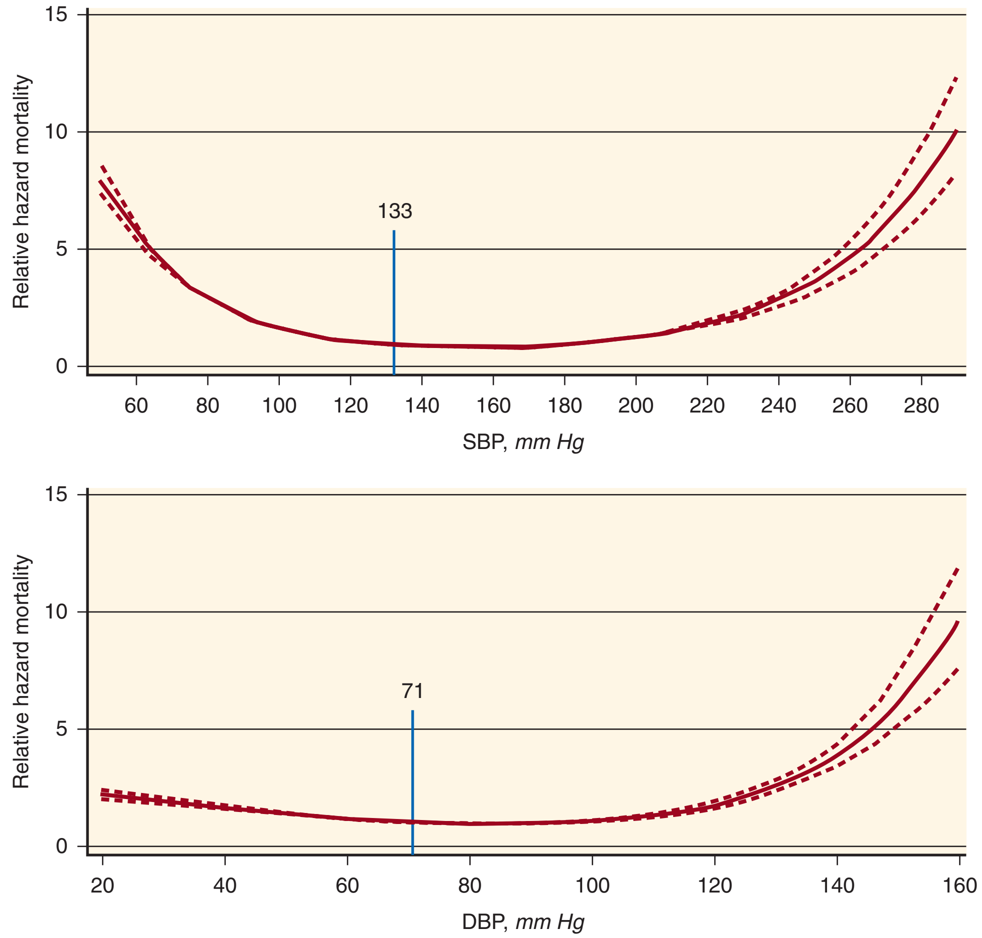


Fig. 46.15 Blood pressure and mortality in US veterans with chronic kidney disease. Shown are multivariable-adjusted relative hazards (hazard ratios [95% confidence intervals]) of all-cause mortality associated with systolic blood pressure *(SBP)* and diastolic blood pressure *(DBP)* relative to a hypothetical patient with the mean time-varying SBP (133 mm Hg) and DBP (71 mm Hg).

hours, and the goal for weight reduction was 10 lb. Other lifestyle changes recommended in the older patient include watching the potassium intake to keep the level of serum potassium in a safe range. Patient education about high-potassium foods is important. Intake of NSAIDs should also be decreased to a minimum because these patients are more likely to take NSAIDs for arthritis and pain. These drugs are known to cause elevations in BP by inhibiting the production of vasodilatory prostaglandins and may increase BP by as much as 6 mm Hg.[266,267] Their BP-raising effects can be blunted by calcium antagonists and, to a lesser extent, diuretics.[268]

Pharmacologic Intervention in Older Adults

The following is a brief summary of the approach to antihypertensive drug therapy in older adults. A more detailed discussion of the pharmacology and use of antihypertensive medications may be found in Chapter 49. The primary agents used in the treatment of hypertension include thiazide-type diuretics, ACE inhibitors, ARBs, and CCBs. Although many other drugs and drug classes are available, confirmation that these agents decrease clinical outcomes to a similar extent as the primary agents is lacking, or safety and tolerability may relegate their role to use as secondary agents.[13] In particular, there is inadequate evidence to support the initial use of beta-blockers for hypertension in the absence of specific cardiovascular comorbidities.

In considering the initial drug treatment of someone with a high BP, several different strategies may be contemplated. Many patients can be started on a single agent, but consideration should be given to starting with two drugs for those with BP higher than 20/10 mm Hg above the goal of 130/80 mm Hg. Many patients started on a single agent will subsequently require two or more drugs from different pharmacologic classes to reach their BP goals.

Knowledge of the pharmacologic mechanisms of action of each agent is important. Drug regimens with complementary activity, where a second antihypertensive is used to block compensatory responses to the initial agent or effect a different pressor mechanism, can result in additive lowering of BP. The use of combination therapy may also allow for treatment with lower doses of individual agents, which serves to minimize adverse effects and improve adherence. Several two and three fixed-dose drug combinations of antihypertensive drug therapy are available in generic form, with complementary mechanisms of action among the components.

The full dose of the drug is the highest pharmacologic dose of drug available; the maximum dose is the highest dose that the person can tolerate without side effects. If the person is having no therapeutic response or has had significant side effects, a drug from another class should be substituted. In fact, most older patients require two or more drugs to achieve the recommended goals in clinical trials.

Either a diuretic or calcium antagonist may be an initial drug, or a diuretic should be one of the first two agents when starting combination drugs (Box 46.4).[13] When the BP is more than 20/10 mm Hg above the goal, the recommendation is to initiate with two antihypertensive medications, with one of the choices being a diuretic. Single-pill combinations that incorporate logical doses of two agents may enhance convenience and compliance in older patients.[211] However, there is still a need for individualization of treatment; thus, treatment options should be carefully considered for older adults. The benefits of lowering BP must be weighed against the risks of side effects and the concomitant morbidity of the patient.

Thiazide diuretics such as hydrochlorothiazide, chlorthalidone, indapamide, and bendrofluazide, as well as calcium antagonists, are recommended for initiating therapy.[8,13] Diuretics cause an initial reduction of intravascular volume, peripheral vascular resistance, and BP in more than 50% of patients and are well tolerated and inexpensive.[269,270] However, they can cause hypokalemia, hypomagnesemia, and hyponatremia and are therefore not recommended for patients with baseline electrolyte abnormalities or those with a history of hyponatremia. The serum potassium level should be monitored, and supplementation should be given if needed.

Calcium antagonists are well suited for older patients whose hypertensive profile is based on increasing arterial dysfunction secondary to decreased atrial and ventricular compliance. This class of drugs dilates coronary and peripheral arteries in doses that do not severely affect myocardial contractility.[271,272] Most adverse effects relate to vasodilation, such as ankle edema, headache, and postural hypotension. Ankle edema is not secondary to sodium retention because calcium antagonists are natriuretic when given initially, but the profound vasodilation with poor venous return in older people is the major contributor.

First-generation drugs, such as nifedipine, verapamil, and diltiazem, should be avoided in patients with left ventricular dysfunction. Nondihydropyridines can precipitate heart block in older adults with underlying conduction defects.

RAAS blockers, such as ACE inhibitors, ARBs, and direct renin inhibitors, may be used in older adults.[273] Theoretically, as aging occurs, there is a reduction in angiotensin levels; thus, ACE inhibitors may not be an effective medication for hypertension in older adults. However, several clinical trials have shown otherwise. The use of ACE inhibitors is beneficial in the reduction of morbidity and mortality in patients with myocardial infarction, reduced systolic function, heart failure, and reduction in the progression of diabetic renal disease and hypertensive nephrosclerosis.[274,275] Finally, it appears that use of an RAAS blocker may provide greater benefit for CV and renal risk reduction than the use of a diuretic, based on data from ACCOMPLISH, a large outcome trial of 11,506 persons with a mean age of 68 years. Although there are very few data on kidney disease in older adults, ACCOMPLISH did provide some evidence worthy of being tested in a prospective trial—that is, a calcium antagonist–ACE inhibitor combination led to fewer people going on dialysis than a diuretic–ACE inhibitor combination.[232,276] Note that this outcome could not be explained by differences in BP of 1.2 mm Hg systolic. It must be noted that those older than 70 years tend to drink small amounts of fluid, and hence this makes them more vulnerable to decline in kidney function by RAAS blockade, as already discussed. Thus it is recommended that they increase their fluid intake to prevent volume depletion.

The clinical benefits of beta-blockers as monotherapy in uncomplicated older patients are poorly documented. They may have a role in combination therapy, especially with diuretics. Beta-blockers have established roles in patients with hypertension complicated by certain arrhythmias, migraine headaches, senile tremors, coronary artery disease, or heart failure.[277,278] Nebivolol, a selective beta-1 blocker with NO properties, does not show any associated symptoms of depression, sexual dysfunction, dyslipidemia, and hyperglycemia in older adults, unlike earlier generations of beta-blockers.[279]

Potassium-sparing diuretics are useful when combined with other agents. Aldosterone-blocking agents such as spironolactone and eplerenone reduce vascular stiffness and systolic BP.[280,281] They are also useful for hypertensive patients with heart failure or primary hyperaldosteronism. Gynecomastia and sexual dysfunction are the limiting adverse events, reactions that occur in men using spironolactone but are less frequent with eplerenone. The epithelial sodium transport

Box 46.4 American Society of Hypertension Evidence-Based, Fixed Dose Antihypertensive Combinations

Preferred

- ACE inhibitor, diuretic[a]
- ARB, diuretic[a]
- ACE inhibitor, CCB[a]
- ARB, CCB[a]

Acceptable

- Beta-blocker, diuretic[a]
- CCB (dihydropyridine), beta-blocker
- CCB, diuretic
- Renin inhibitor, diuretic[a]
- Renin inhibitor, ARB[a]
- Thiazide diuretics, K^+-sparing diuretics[a]

Less Effective

- ACE inhibitor, ARB
- ACE inhibitor, beta-blocker
- ARB, beta blocker
- CCB (nondihydropyridine), beta-blocker
- Centrally acting agent, beta-blocker

[a]Single-pill combination available in the United States.
ACE, Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *CCB,* calcium channel blocker.
From Gradman AH, Basile JN, Carter BL, Bakris GL; American Society of Hypertension Writing Group. Combination therapy in hypertension. *J Clin Hypertens (Greenwich).* 2011;13(3):146–154.

antagonists (e.g., amiloride, triamterene) are most useful when combined with another diuretic.

Other agents, such as alpha-blockers, centrally acting drugs (e.g., clonidine), and nonspecific vasodilators (e.g., minoxidil) should not be used as first- or second-line agents in an older adult with hypertension. Instead, they are reserved as part of a combination regimen to maximize BP control after other agents have been deployed.

BLOOD PRESSURE MANAGEMENT IN PATIENTS UNDERGOING DIALYSIS

Elevations in BP in dialysis patients are almost exclusively due to excessive volume. Hence, hypertension control related directly to volume management is a more common problem in hemodialysis than in peritoneal dialysis. There have been no large, multicenter, randomized trials in patients undergoing dialysis to evaluate different levels of BP on CV outcomes. However, there is a prospective randomized trial that has evaluated the effects of an ACE inhibitor versus a beta-blocker on LVH and as a secondary endpoint of CV mortality in patients undergoing dialysis.[282]

The purpose of the study was to determine, in 200 hypertensive patients undergoing hemodialysis with echocardiographic LVH, whether a beta-blocker or an ACE inhibitor used for BP lowering results in a greater regression of LVH. Subjects were randomly assigned to open-label lisinopril ($n = 100$) or atenolol ($n = 100$), each administered three times/week after dialysis. Monthly monitored home BP was controlled to less than 140/90 mm Hg with medications, dry weight adjustment, and sodium restriction. The primary outcome was the change in left ventricular mass index from baseline to 12 months. The results demonstrated that there was no between-group difference in 44-hour ambulatory BP. However, the monthly measured home BP was consistently higher in the lisinopril group, despite the need for a greater number of antihypertensive agents and a greater reduction in dry weight. An independent data safety monitoring board recommended termination because of CV safety. Serious CV events were more prominent in the lisinopril group (20 events; atenolol vs. 43 events, lisinopril; $P = .001$). Combined serious adverse events of myocardial infarction, stroke, and hospitalization for heart failure or CV death were much lower in the atenolol group ($P = .021$). Thus, it appears that beta-blocker therapy is superior to RAAS blockade in patients undergoing hemodialysis to reduce CV morbidity and all-cause hospitalizations.

Retrospective analyses of patients on hemodialysis have also supported other beta-blockers such as carvedilol for lowering CV events in hypertensive patients who were undergoing hemodialysis and Japanese investigators have further corroborated the results supporting beta-blocker use in patients undergoing dialysis. The effect of beta-blocker use on mortality in a cohort of patients undergoing hemodialysis was evaluated in a database analysis from the Dialysis Outcomes and Practice Patterns Study phase II of 2286 randomly selected patients on hemodialysis in Japan.[283] The main outcome measure was all-cause mortality. The authors found beta-blocker use was low (i.e., only 247 patients [10.8%] were administered beta-blockers, and 1828 patients [80%] were not). A Kaplan-Meier analysis revealed that all-cause mortality rates were significantly decreased in patients treated with beta-blockers ($P < .007$) compared with the group not on beta-blockers. In multivariable, fully adjusted models, treatment with beta-blockers was also independently associated with reduced all-cause mortality (HR, 0.48; $P = .02$).[236]

Data from the US Renal Data System have demonstrated a U-shaped relationship between systolic BP goals in patients undergoing dialysis and CV outcomes, suggesting that BP should be consistently above 120 mm Hg and below 150 mm Hg.[284] A meta-analysis by Agarwal and Sinha of antihypertensive medications in patients undergoing dialysis has demonstrated that regardless of class, reducing BP reduces CV events.[285] Data from epidemiologic studies have consistently shown that in patients undergoing maintenance dialysis, low BP values are associated with higher death rates when compared with normal to moderately high values.

Although there is no generalizable approach to manage BP in those undergoing dialysis, the following points are vital to have an accurate assessment of BP: (1) the most representive BP is the one taken the morning after dialysis; and (2) there should be a minimum of two and ideally three readings obtained 1 or 2 minutes apart during those morning readings and then averaged.[285,286] Given that heart failure and sudden death are the most common causes of death in dialysis patients, beta-blockers have an important role in the BP-lowering armamentarium, unlike in the general population. Ensuring euvolemia is the key, however. The clinical examination is important, but newer techniques of bioimpedance, blood volume monitoring, and inferior vena cava ultrasound are effective in improving the definition of dry weight and could be considered, particularly in patients with uncontrolled BP, despite conventional dry weight probing using the clinical examination.[190,287]

MANAGEMENT OF HYPERTENSION IN CHRONIC KIDNEY DISEASE

There are many guidelines written on the management of hypertension in the general population, and those have been summarized at the beginning of this chapter. Both the Kidney Disease Outcomes Quality Initiative (KDOQI) and KDIGO guidelines focus on the evidence for certain BP levels and classes of medication in patients with CKD. These guidelines have also been discussed in this chapter.

To summarize the key findings, the strongest level of evidence supports a BP goal below 140/90 mm Hg and the use of RAAS-blocking agents in people with stage 3 or higher CKD who have very high albuminuria. Nevertheless, there is weaker evidence to support RAAS blocker use in people with CKD in general, as well as very weak evidence supporting a BP of less than 130/80 mm Hg for those who have a urine albumin level of 1 g or more and an eGFR less than 60 mL/min/1.73 m^2.[288,289] These BP levels and recommendations for RAAS blockers are focused on the primary outcome of slowing CKD progression. A summary of all trials and effects on eGFR decline are shown in Fig. 46.16.

The most recent AHA/ACC BP guidelines have modified the approach slightly, with a focus on CV risk reduction rather than a focus on renal preservation. This is because the BP range of 125 to 130 mm Hg has not been shown to harm the kidneys yet reduce further CV events.[241] An algorithm proposed for BP management in CKD is shown in Fig. 46.17.

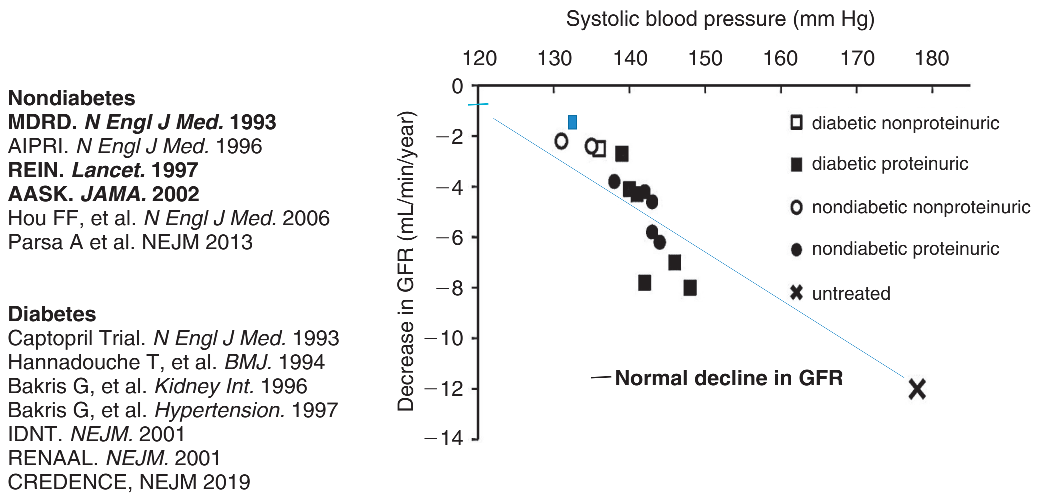


Fig. 46.16 Relationship between achieved blood pressure and decline in kidney function from primary renal endpoint trials. *GFR,* Glomerular filtration rate. (From Kalaitzidis R, Bakris GL. Update on reducing the development of diabetic kidney disease and cardiovascular death in diabetes. In Daugirdas JT, editor. *Handbook of chronic kidney disease management*. Philadelphia: Elsevier, 2018.)

Nuances in the management of BP in the patient with CKD are necessary because these patients have problems that are not seen in the general population. First, hyperkalemia is a risk in certain subgroups of patients. A review of clinical trials where hyperkalemia developed when managing hypertension in CKD has found three risk predictors. If a patient was already receiving an appropriate diuretic for her or his level of kidney function, then these were reliable risk predictors for hyperkalemia (i.e., $[K^+]_s > 5.5$ mEq/L): (1) eGFR of less than 45 mL/min/1.73 m^2; (2) serum potassium level above 4.5 mEq/L; and (3) body mass index less than 25.[290] Hyperkalemia has limited our ability to assess whether RAAS blockers are effective in slowing CKD progression in stage 3b and higher CKD. Newer agents to manage hyperkalemia will open the door to safer management and expanded research.[291,292]

Another nuance in BP management in the CKD patient is the critical importance of sodium restriction to less than 2400 mg/day and reduced alcohol consumption, as well as aerobic, not isometric, exercise.[293,294] To demonstrate the importance of exercise, a recent study has randomized 296 dialysis patients to normal physical activity (control, $n = 145$) or walking exercise ($n = 151$) over a 6-month period. Those who were in the active group using a simple, personalized, home-based, low-intensity exercise program managed by dialysis staff improved physical performance and quality of life.[294]

Studies evaluating the effect of sodium intake on BP control in people with stage 4 CKD have shown that approximately every 400 mg above a sodium intake base of 3000 mg/day requires an additional BP medication to maintain BP control.[293] Moreover, failure to reduce sodium intake suppresses the RAAS system and hence reduces the efficacy of RAAS blockers. Thus, failure to reduce sodium intake is a cause of resistant hypertension, as noted earlier in the chapter. Additionally, BP targets need to be clearly defined. Finally, the appropriate antihypertensive agents indicated as initial therapy by evidence-based guidelines should be used.

The role for initial combination therapy in patients with CKD with a BP that is 20/10 mm Hg above the goal has been championed for almost 2 decades to enhance adherence and efficacy.[8,233] Studies evaluating initial monotherapy versus single-pill combination among those with an eGFR greater than 60 mL/min/1.73 m^2 have uniformly shown an advantage of achieving the BP goal more quickly and with better tolerability.[295,296]

These new guidelines and concepts have been integrated into an algorithm that was in the original National Kidney Foundation consensus report.[297] It is meant to provide an approach to pharmacologically meaningful agents that when combined, actually further reduce BP. Box 46.4 provides an approved list of combination agents, noting which are meaningful for optimal BP reduction and which are not. Fig. 46.17 presents an evidence-based algorithm to help achieve BP control in the patient with advanced CKD.

RESISTANT HYPERTENSION

Resistant hypertension is currently defined as the failure to achieve a goal BP of less than 140/90 mm Hg in patients who are adherent with maximal tolerated doses of a minimum of three antihypertensive drugs, one of which must be a diuretic appropriate for kidney function.[152] The increasing prevalence of obesity and hypertension in the general population has resulted in this disorder gaining attention in the past decade. Large-scale population-based studies such as NHANES have specifically examined the prevalence and incidence of resistant hypertension and associated risk factors. These findings have suggested that the prevalence of resistant hypertension is approximately 8% to 12% of adult patients with hypertension (6–9 million people).[298] The increasing prevalence of resistant hypertension has contrasted with the improvement in BP control rates during the same period. Studies have also shown that patients with resistant hypertension who are older than 55 years, who are black, with a high

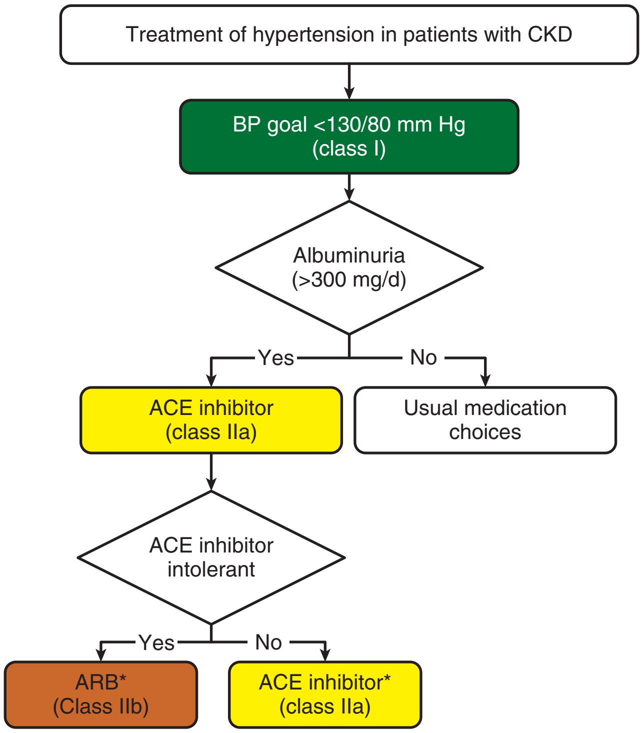


Fig. 46.17 Management of hypertension in patients with chronic kidney disease *(CKD)* according to American Heart Association (ACC)/American Heart Association (AHA) guidelines. Colors correspond to class of recommendation: *green,* class I (strong); *yellow,* class IIa (moderate); *orange,* class IIIb (weak). *CKD stage 3 or higher or stage 1 or 2 with albuminuria ≥300 mg/day or ≥300 mg/g creatinine. It should be noted that treating blood pressure *(BP)* in CKD patients to <140/90 mm Hg with an angiotensin receptor blocker or ACE inhibitor as first-line therapy is recommended for slowing progression of CKD by KDIGO guidelines. To further reduce CV risk, levels of <130/80 mm Hg need to be achieved. The use of combination therapy with diltiazem in the subgroup with very high albuminuria has an additive reduction, in addition, to lower BP further compared with other calcium channel blockers. *ACE,* Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker. (From Whelton PK, Carey RM, Aronow WS, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association and Task Force on Clinical Practice Guidelines. *Hypertension.* 2018;71(6):e13–e115.)

body mass index, diabetes, or CKD have an increased risk for CV events compared with patients with nonresistant hypertension. However, the effects of white coat hypertension and pseudoresistant hypertension have not been factored into many of the prevalence studies, and hence the true prevalence is not known. The white coat effect contributes greatly to the high perceived incidence of resistant hypertension, as was evidenced by the over 60% screen failure in the Renal Denervation in Patients with Uncontrolled Hypertension (SYMPLICITY HTN-3) trial of renal denervation due to WCH.[299,300]

Thus this is a diagnosis frequently seen by nephrologists and is usually the result of nonadherence with medication, as well as volume overload secondary to poor kidney function and nonadherence to a low-sodium diet. Once a diagnosis of resistant hypertension has been made using a 24-hour ABPM, and a fourth drug is needed after the use of a calcium antagonist, diuretic, and RAAS blocker, a mineralocorticoid inhibitor has demonstrated significant benefit in controlling BP in more recent studies.[301–304] A detailed discussion of this topic is beyond the scope of this chapter, but the reader is referred to the most recent consensus report by the AHA/ACC.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Which of the following is the current blood pressure guideline to slow kidney disease progression and reduce cardiovascular complications?
   a. <145/90 mm Hg
   b. <140/90 mm Hg
   c. <135/85 mm Hg
   d. <130/80 mm Hg
   e. <120/80 mm Hg
   Answer: d

2. Which of the following is the most common secondary cause of hypertension?
   a. Pheochromocytoma
   b. Primary hyperaldosteronism
   c. Obstructive sleep apnea
   d. Illicit drugs
   e. Cushing's syndrome
   Answer: c

3. All of the following combinations have been shown to have added blood pressure reductions EXCEPT:
   a. Renin angiotensin system blockers + calcium blockers
   b. Renin angiotensin system blockers + diuretics
   c. Calcium blockers + beta blockers
   d. Beta Blockers + diuretics
   e. Beta blockers + clonidine
   Answer: e

4. A 63-year-old African American male presents with a headache and history of poorly controlled blood pressure for over 10 years. His exam reveals 1+pedal edema and labs show a serum creatinine is 1.3 mg/dL (eGFR 51 ml/min/1.73 m^2) and has a normal bicarbonate, albuminuria of 430 mg/d and potassium of 4.2 mEq/L. He has taken clonidine 0.2 mg and amlodipine 10 mg within the past 2 hours and his BP is 160/90 mm Hg and heart rate 80. You educate him about low sodium diet and exercise and change his regimen to include chlorthalidone 25 mg/d continue amlodipine and add olmesartan 40 mg/d and stop the clonidine. He returns in a week with BP 132/80 mm Hg and feels tired. Physical exam shows resolution of edema. Repeat labs demonstrate-K-4.5 mEq/L, and creatinine 1.7 mg/dL. Which of the following would be the best approach based on guidelines?
   a. Stop olmesartan and re-evaluate in a week.
   b. Stop both diuretic olmesartan and re-evaluate in a week.
   c. Continue present management for another 2 to 3 weeks and re-evaluate.
   d. Continue management and add a beta blocker to further lower blood pressure.
   e. Change diuretic to loop diuretic now that GFR is lower and still has elevated blood pressure.
   Answer: c

5. Which of the following statements best summarizes the rate of decline in kidney function today in someone with an eGFR of 50 mL/min/1.73 m^2 based on optimal standard of care per guidelines in people with diabetic kidney disease?
   a. −6 mL/min/year
   b. −2 mL/min/year
   c. −4 mL/min/year
   d. −1 mL/min/year
   e. Unclear
   Answer: b